ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Communication and Information Systems Officer
(CISO)

National Qual - NWO 00207 – LT(N)

(AEEC - 119154)

National Qual – TDL 100

(ALDF)

National Qual – TDL 200

(ALDG)

RECORD OF AMENDMENTS

1. This Qualification Standard and Plan (QSP) is developed under the authority of the Royal Canadian Navy (RCN) through the Training Authority (TA), Naval Personnel and Training Group (NPTG) and the Training Development Center Atlantic (NTDC (A)). The approval or revision date should appear in the footer of every page as an additional means of verification. Details on the changes must be amplified in Annex J “Course History”.

Revision	Approval date	Authority
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FOREWORD

1. This training document is developed under the authority of C/NPTG and in accordance with (IAW) the following references:

a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification dated 23 April 2014;

c. A-PD-055-002/PP-001 Maritime Surface and Sub-Surface (NWO) dated 01 March 2015;

d. A-PD-055-003/PQ-001 Communication and Information Systems Officer Specialty Specification dated 25 June 14;

f. NAVORD 4500-0; and

g. NPTORD 4500-22.

2. This QSP consists of specific Performance Objectives (POs) in Chapter 2 and Lesson Specifications and Enabling Objectives (EOs) in Chapter 4. A trainee will achieve the qualification when the standards listed in the POs have been met.

3. This QSP is effective upon approval and supersedes previous versions.

4. Suggestions for amendments to this QSP may be initiated by anyone and shall be forwarded in writing to NTDC (A), who will forward to NPTG for approval, as appropriate.

5. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A)	Mod 1 through 4		AEEC - 119154
NFS(P)	Mod 1 through 4		AEEC - 119154
CFCSS Halifax	Mod 5	Nil.	COMSEC Custodian
CFCSS Esquimalt	Mod 5	Nil.	COMSEC Custodian

CHAPTER 1: TRAINING MANAGEMENT DETAILS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (C/NPTG)

AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Communication and Information System Officer. This QSP is designed to assist NTDC (A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

Course duration is calculated in accordance with (IAW) NPTORD 4500-22, para 5.5. For this course, a period of instruction is 50 minutes in duration.

Course Duration Calculation	
Actual Training Time (ATT)	**Periods**
Sum of all EOs	448
Sum of all ECs	52
Sum of all PCs	38
Sub-total	538 periods **77 trg days**
BAT, TAT, TTT and PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	16
Training Administrative Time (TAT)	16
Training Travel Time (TTT)	0
Physical Fitness (PT)	45
Sub-total	77 periods **11 trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT)/8 periods	615 periods **88 trg days**

RESOURCES

Identify key personnel and facilities required to conduct the training. Additional TE resources can be listed in Annex H, if required.

Required Course Staffing				
Position	Rank	Qualifications Required	Qty	PO/EO
Course Officer	Lt(N)	CISO (AEEC), Post-Director Level Sea Tour	1	1 through 12
Senior Instructor	Lt(N)	CISO (AEEC), MCCIS, CAFNetOPS	1	1 through 12
Invigilator	Lt(N)	CISO (AEEC), or IWO	1	1 through 12
Guest Lecturers	IAW Annex H			

Critical Training Resources		
CFB Esquimalt	**CFB Halifax**	**No of Days**
Multimedia Classroom	Multimedia Classroom	
Classroom with Unclassified VTC	Classroom with Unclassified VTC	
SECRET References	SECRET Research LAB	
DMOC Lab	DMOC Lab	
SEACOT	SEACOT	
HMC Ship with CMS flashed up	HMC Ship with CMS flashed up	
DWAN / GPNet workstation	DWAN / GPNet workstation	

COURSE CAPACITY

Planning for the conduct of training is based on the guidelines in NPTORD 4500-40.

a. Maximum – 12;

b. Minimum – 1; and

c. Optimum – 6.

MILITARY INDIVIDUAL TRAINING AND EDUCATION (MITE)

Loading. The authority to load members onto this course lies with the BTL manager at NPTG HQ. TE, including coastal CFCSSs, is to be included as an INFO addressee on all loading messages.

Scheduling. Each session of this course shall be scheduled, in MITE, IAW NPTORD 4500-22 and Campus Orders by the TE conducting the training, and NPTG through the Quantity Control section. The course length in MITE is to be based on the course duration, planned annual leave, training breaks, weekends and training system holidays.

Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Naval Officer Professional Qualification (NOPQ);

b. Top Secret Security Clearance; and

c. Valid fitness test.

Qualification. Successful completion of all modules constitutes the primary means of attaining the Communication and Information Systems Officer (AEEC). Additionally, the following MITE Course IDs will be granted:

a. NWO Communication and Information System Officer – 119154 (AEEC);

b. NWO Communication and Information Systems Officer (CISO) Mod 1: NEW MITE

c. NWO Communication and Information Systems Officer (CISO) Mod 2: NEW MITE

d. NWO Communication and Information Systems Officer (CISO) Mod 3: NEW MITE

e. NWO Communication and Information Systems Officer (CISO) Mod 4: NEW MITE

f. NWO Communication and Information Systems Officer (CISO) Mod 5: NEW MITE

COURSE REPORTING

Upon passing this course, trainees shall receive a Course Report (CF 377) and/or Certificate of Military Achievement (CF 289) completed IAW CMP Instruction 01/06 and NPTORD 4500-51.

EVALUATION

Program evaluation shall be conducted IAW the guidance provided in CFITES Vol. 11.

VALIDATION

This training will be validated every five years IAW CFITES Vol. 8 and based on the task list in Annex B.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

Module 4 – shall be scheduled in concert with the NAV COMM CISN Manager course (QL6B) in order to capitalize on training integration and save instructor and SME resources. Additionally, this will achieve the aim of creating a professional cadre of Senior NAV COMMs and Communication and Information Systems Officers who will work alongside one another similar to the other warfare directors.

CHAPTER 2: PERFORMANCE OBJECTIVES

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDER NAVAL PERSONNEL AND TRAINING GROUP (C/NPTG)

PO 001 Lead NAV COMM Section

1. **Performance**: Lead NAV COMM section (N0005).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access; and

      (3) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall lead the NAV COMM section (N0005), IAW <A14, A68, A89>, by:

   g. Managing NAV COMM personnel (N0006), IAW <A68, A89> by;

      (1) Identifying ship's NAV COMM team responsibilities and functions;

      (2) Verifying manning requirements during shipboard evolutions; and

      (3) Identifying NAV COMM occupation progression;

   h. Coordinating Ship's CIS training (N0007), IAW <A14> by:

      (1) Directing ship's CIS Combat Readiness Requirements (CRRs);

      (2) Managing section refresher training requirements;

      (3) Directing new CIS equipment training; and

      (4) Explaining additional training requirements.

4. **Training Limitations**: Nil.

PO 002 Draft Communications Improvement Documents

1. **Performance**: Draft communications improvement documents (N0008).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access;

      (3) Message drafting software;

      (4) Message distribution software;

      (5) Message traffic; and

      (6) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall draft communications improvement documents (N0008), IAW Assystnet, A20, A71, A74, C2, to include:

   a. Recommending CIS Improvements (N0009), by:

      (1) Identifying systems' performance deficiencies, malfunctions, or failures (B0540);

      (2) Submitting an IMR / Assyst Ticket IAW <assystnet>;

      (3) Drafting a SOCD IAW <A74>; and

      (4) Drafting a UCR IAW <C2>;

   b. Preparing operational evaluation reports (B0140), IAW <A71>, by:

      (1) Drafting CIS input to evaluation reports IAW <A20>;

      (2) Drafting CIS input routine reports IAW <A20>; and

      (3) Facilitating the testing of CIS resources by external agencies IAW <A20>.

4. **Training Limitations**: Nil.

PO 003 Advise Command on CIS Security

1. **Performance**: Advise Command on CIS security (N0010).

1. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access;

      (3) Message drafting software;

      (4) Message distribution software;

      (5) Message traffic;

      (6) Enforcement tools; and

      (7) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

2. **Standard**: The CISO shall advise Command on CIS security (N0010), IAW A57, A67, A69, by:

   a. Directing enforcement of communication and information security (T0004), IAW <A57>, by:

      (1) Ensuring IS accreditation IAW <A69>;

      (2) Directing information security procedures IAW <A67>;

      (3) Drafting unit IS security regulations IAW <A67>;

      (4) Coordinating physical protection of IS assets IAW <A67>; and

      (5) Managing IS security incidents IAW <A67>;

3. **Training Limitations**: Nil.

PO 004 Direct Operational Messaging

1. **Performance**: Direct operational messaging (N0012).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access;

      (3) Message drafting software;

      (4) Message distribution software;

      (5) Message traffic; and

      (6) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall direct operational messaging (N0012), IAW A12, A28, B6, B18, B19, B42, by:

   a. Preparing Operational Messages (B0130), IAW <B18, B19, B42> by:

      (1) Overseeing message preparation;

      (2) Verifying message addressing;

      (3) Reviewing Communication Information Bulletins (CIBs); and

      (4) Drafting CIS messages IAW <A28>; and

   b. Directing Operational Message Traffic (N0013), IAW <A12, B6, B19, B42> by:

      (1) Directing message processing and handling systems; and

      (2) Coordinating the distribution of operational message traffic, IAW <A12, B6>.

4. **Training Limitations**: Nil.

PO 005 Advise Command on Electromagnetic Spectrum Warfare

1. **Performance**: Advise Command on electromagnetic spectrum warfare (N0014).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access;

      (3) Prediction software;

      (4) Electromagnetic footprint software;

      (5) Message traffic; and

      (6) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall advise command on electromagnetic spectrum warfare (N0014), IAW <reference>, by:

   a. Manipulating RF prediction software (N0015), IAW <reference>;

   b. Correlating RF information from operational sources (N0016), IAW <reference>, and;

   c. Explaining RF theory (N0017), IAW <reference>.

4. **Training Limitations**: Electromagnetic footprint software is not yet available at the training establishments as of 3 October 2017.

PO 006 Manage LF/MF/HF/VHF/UHF Communications Equipment

1. **Performance**: Manage LF/MF/HF/VHF/UHF Communications Equipment (T0002)

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified Network Access;

      (3) Cryptographic support;

      (4) Ships RF Equipment;

      (5) FMF support;

      (6) NCS support;

      (7) Message Traffic; and

      (8) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall manage LF/MF/HF/VHF/UHF communications equipment (T0002), IAW references, by:

   a. Identifying communications monitoring faults (N0018), IAW <reference>, by:

      (1) Configuring Communications Monitoring System (CMS), IAW <reference>;

      (2) Configuring RF systems, IAW <reference>;

      (3) Initializing RF systems, IAW <reference>;

      (4) Troubleshooting RF systems, IAW <reference>;

   b. Configuring ships personal radio communicators (PRCs) (N0019), IAW <reference>;

   c. Maintaining ships PRCs (N0020), IAW <reference>; and

   d. Maintaining Sub Net Relay (SNR) (N0021), IAW <reference>, by troubleshooting SNR, IAW <reference>.

   e. Managing PRC 117 equipment (N0022), IAW <reference>.

4. **Training Limitations**:

a. Training facilities are not available as of 24 Apr 2017 and training support will need to be provided by HMCS Ships for SNR.

b. CMS trainer is degraded and the system is expected to be replaced in future ships.

PO 007 Manage Satellite Communications Equipment

1. **Performance**: Manage Satellite Communications (SATCOM) equipment (N0023).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access;

      (3) Cryptographic support;

      (4) Ships SATCOM equipment;

      (5) NCS support;

      (6) Message traffic; and

      (7) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall maintain satellite communications equipment (N0023), IAW <Space Applications Course Manual>, by:

   a. Coordinating Naval SATCOM (N0043), IAW <reference>

   b. Maintain Satellite Communications (N0024), IAW <reference>, by:

      (1) Initializing SATCOM equipment, IAW <reference>;

      (2) Configuring SATCOM equipment, IAW <reference>; and

      (3) Troubleshooting SATCOM equipment, IAW <reference>.

4. **Training Limitations**: Navy Multi-band Terminal (NMT) trainer is not available at NFS (P) therefore HMC Ships will need to be utilized locally for this phase.

PO 008 Manage Cryptographic Equipment

1. **Performance**: Manage cryptographic equipment (N0025).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access;

      (3) Cryptographic support;

      (4) Ship's cryptographic equipment;

      (5) Message traffic; and

      (6) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall maintain cryptographic equipment (N0025), IAW A45, by:

   a. Maintain cryptographic equipment (N0026), IAW <reference>, by:

      (1) Troubleshooting cryptographic equipment IAW <reference>;

      (2) Configuring cryptographic equipment IAW <reference>; and

      (3) Initializing cryptographic equipment IAW <reference>.

   b. Configuring KIV-7M, IAW KIV-7M QSP (AJKP)

   c. Configuring TACLANE, IAW TACLANE QSP (AIVC)

4. **Training Limitations**: Nil.

PO 009 Manage Network Operations

1. **Performance**: Manage network operations (N0030).

Conditions:

a. Given:

(1) Network Credentials;

(2) DWAN enabled workstation;

(3) Classified Network Access;

(4) NavIS Hardware;

(5) NavIS Software

(6) Network Management Tools;

(7) Telephone - landline; and

(8) References.

b. Denied: Nil.

c. Environment: All naval environmental conditions at sea, alongside or ashore.

Standard: The CISO shall manage network operations (N0030), IAW <reference>, by:

a. Managing ship's IS/IT configuration (B0470), IAW <reference>, to include:

(1) Maintaining ship's network architecture;

(2) Facilitating changes to network architecture;

b. Managing network administrator functions IAW <reference> (N0032);

c. Brief Command on network performance (N0033), IAW <reference>, to include:

a. Maintaining information security;

b. Advising on network applications;

c. Configuring bandwidth management hardware; and

d. Configuring bandwidth management software.

Training Limitations:

a. CISOs are not entitled to an administrator account for DWAN IAW NavIS security orders, however they may be available for other networks.

b. Administrator access to Steelhead appliances, HSDC Routers, and other NavIS hardware are restricted by Network Operations Center staff and are only provided as required in exigent circumstances to Ship's staff.

PO 010 Coordinate Defensive Cyber Operations – Internal Defensive Measures

1. **Performance**: Coordinate defensive cyber operations – internal defensive measures (N0034).

2. **Conditions**:

   a. Given:

      (1) Network Credentials;

      (2) DWAN enabled workstation;

      (3) Classified Network Access;

      (4) NavIS Hardware;

      (5) NavIS Software

      (6) Network Monitoring Tools;

      (7) Telephone - landline;

      (8) Cyber Threat Brief; and

      (9) References.

   b. Denied: Nil.

   c. Environment: All naval cyberspace conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall coordinate defensive cyber operations – internal defensive measures (N0034), IAW <references>, by:

   a. Briefing Command on cyber impact to operations (N0035), IAW <reference> by:

      (1) Identifying a cyber-attack;

      (2) Interpreting a cyber threat assessment;

      (3) Developing a cyber security plan; and

      (4) Implementing a cyber security plan;

      (5) Briefing personnel on cyber threat(s) (N0036), IAW <reference>.

4. **Training Limitations**:

   a. The CISO is not responsible for, nor will they have the capability to, conduct any Offensive Cyber Operations (OCO), or any Defensive Cyber Operation - Response Attacks (RA).

PO 011 Coordinate a Communications and Information Systems Plan

1. **Performance**: Coordinate a Communications and Information Systems plan (B0350).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified Network Access;

      (3) Intelligence Brief;

      (4) Threat Assessment;

      (5) Message Drafting Software;

      (6) Message Traffic;

      (7) Implementation Plan; and

      (8) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall coordinate a communications and information systems plan (B0350), IAW references, by:

   a. Coordinating the installation of CIS resources (N0037), IAW references by:

      (1) Arranging Formation CIS support, IAW references;

      (2) Arranging Fleet CIS support, IAW references; and

      (3) Arranging external agency support, IAW references;

   b. Preparing a communications plan (N0038), IAW references by:

      (1) Identifying shipwide C4 resource requirements (T0003), IAW references;

      (2) Extracting operational CIS information, IAW references;

      (3) Managing CIS operational correspondence, IAW references;

      (4) Executing the Operational Planning Process (OPP), IAW references; and

      (5) Conducting a maritime tactical estimate, IAW references.

4. **Training Limitations**: Nil.

PO 012 Direct Shipboard Communications Tactics

1. **Performance**: Direct shipboard communications tactics (T0001).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified Network Access;

      (3) Intelligence Brief;

      (4) Threat briefs; and

      (5) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall direct shipboard communications tactics (T0001), IAW references, by:

   a. Managing the configuration of CIS resources (N0039), IAW references, by;

      (1) Advising Command on tactical employment of communications equipment, IAW references;

      (2) Drafting communications battle orders, IAW references;

      (3) Managing the implementation of EMCON states, IAW references;

   b. Developing communications tactics (N0040), IAW references;

      (1) Interpreting communication threat analysis, IAW references,

      (2) Interpreting ELINT analysis, IAW references; and

      (3) Coordinating communications intercept responses, IAW references.

   c. Advising Command on Command and Control in a Denied or Degraded Environment (C2D2E) (N0041), IAW references by:

      (1) Assessing communications jamming, IAW references;

      (2) Coordinating communications anti-jamming procedures;

4. **Training Limitations**: Communications tactics phase will be based on allied communications publications, CFCDs and theoretical academic publications due to the burgeoning recognition of IW as a warfare domain within the naval battlespace.

PO 013 Coordinate all aspects of ship's COMSEC

1. **Performance**: Coordinate all aspects of ship's COMSEC (B0480).

2. **Conditions**:

   a. Given:

      (1) References;

      (2) Equipment; and

      (3) Applications.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

Standard: The CISO shall coordinate all aspects of ship's COMSEC (B0480), IAW references by completing the COMSEC course.

3. **Training Limitations**: Nil.

CHAPTER 3: ASSESSMENT OF TRAINEE

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDING OFFICER TRAINING DEVELOPMENT CENTER ATLANTIC (NTDC (A))

GENERAL

1. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this Qualification Standard and Plan (QSP) and the standard listed must be met to achieve this qualification. PCs will be administered in the form of a performance-based assessment and the trainee must obtain a pass in all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

Enabling Checks (ECs). ECs are formative tests that help the trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructors and trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the trainee's final mark.

Performance Checks (PCs). PCs are summative tests used to determine if the trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

Assessment of trainees is to be done in accordance with (IAW) CFITES Vol. 7.

The Training Establishments (TE) provide quality control by administering and scoring all Performance Checks. The Training Development Center provides quality control through designing course content, PCs and ECs. Any PC not administered or invigilated by the instructor will be administered by a non-involved instructor designated by the senior instructor. Enabling Checks are the responsibility of the instructional staff.

PROGRESS MONITORING

Monitoring learning is the responsibility of all instructional staff and instructional supervisors. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol. 6, and TE SOs or SOPs.

Continuous monitoring of candidate progress is required on this course in order to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the Training Review Board (TRB) or Progress Review Board (PRB) (if required); and

d. Information to assist in completion of the CF 377 Course Report.

Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs or SOPs.

PC failures and supplemental PC failures are to be handled IAW NPTORD 4500-54 or TE SOs/SOPs. Supplemental ECs and PCs shall not be given prior to 24 hours having elapsed since trainee was informed of their EC or PC failure.

TRAINING REVIEW BOARD

TRBs shall be convened and conducted IAW with NPTORD 4500-1.

COURSE REPORTS

Course reports (CF 377) shall be prepared and distributed IAW CFAO 26-12 and NPTORD 4500-51.

Trainee will be graded pass or fail for individual modules and the NQual overall. Students will not be ranked IAW NPTORD 4500-54.

Where necessary for Rank Qualifications (RQ), the narrative will indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager. Training limitations and/or incomplete training must also be included in the course report. Record of trainee counselling is to be maintained by the TE.

CHAPTER 4: LESSON SPECIFICATIONS

AUTHORIZATION FOR THIS CHAPTER AND AMENDMENTS MUST BE GRANTED BY COMMANDING OFFICER OF THE TRAINING DEVELOPMENT CENTER ATLANTIC (NTDC (A))

AIM

1. The aim of the training resulting from this Training Plan (TP) is to prepare personnel who can perform the duties of a Communication Information Systems Officer**.**

OUTLINE OF TRAINING

2. This TP contains Lesson Specifications with associated Enabling Objectives (EOs), Enabling Checks (ECs) and Performance Checks (PCs) that the candidate shall achieve.

TRAINING STRATEGY

3. The training requirement will be achieved through the following modules:

Module	Delivery	Reg/P Res	MITE ID	NQual
Mod 1	DL	Reg	New MITE	Nil.
Mod 2	In-house	Reg	New MITE	Nil.
Mod 3	In-house	Reg	New MITE	Nil.
Mod 4	In-house	Reg	New MITE	Nil.
Mod 5	CFCSS	Reg	New MITE	Nil.

4. Further details can be found at Annex G – Modularization and Suggested Timings.

OCCUPATION REQUIREMENT ORGANIZATION

5. Occupation requirements were organized under the following categories:

PO	EO	Maintain Readiness	Plan	Train	Execute
001	**01**	Manage NAV COMM Personnel			
	02	Coordinate Ship's CIS Training			
002	**01**	Recommend CIS Improvements			
	02	Prepare Evaluation Operational Reports			
003	**01**	Direct Information Security Procedures			
	02	Direct Enforcement of Communication and Information Security			

004	**01**	Prepare operational messages			
	02	Direct operational message traffic			
005	**01**	Manipulate RF prediction software			
006	**01**	Identify communications monitoring faults			
	02	Maintain ship's PRCs			
	03	Maintain SubNet relay			
007	**01**	Coordinate naval SATCOM			
	02	Maintain satellite communication			
008	**01**	Maintain cryptographic equipment			
	02	KIV-7M (AJKP)			
	03	TACLANE (AIVC)			
009	**01**	Manage Ship's IS/IT Configuration			
	02	Manage Network Administrator Functions			
010	**01**	Brief Command on Cyber Impact to Operations			
011	**01**	Coordinate the Installation of CIS Resources			
012	**01**	Manage the Configuration of CIS Resources			
	02	Develop Communications Tactics			
	03	Advise Command on C2D2E			
013	**01**	Coordinate All Aspects of Ships COMSEC			

PO 001 Lead NAV COMM Section

EO 001.01 Manage NAV COMM Personnel

1. **Performance**: Lead NAV COMM section (N0005).

2. **Conditions**:

   d. Given:

      (1) DWAN enabled workstation;

      (2) Classified network access; and

      (3) References.

   e. Denied: Nil.

   f. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard**: The CISO shall lead the NAV COMM section (N0005), IAW <A14, A68, A89>, by:

   a. Managing NAV COMM personnel (N0006), IAW <A68, A89> by;

      (1) Identifying ship's NAV COMM team responsibilities and functions;

      (2) Verifying manning requirements during shipboard evolutions; and

      (3) Identifying NAV COMM occupation progression;

4. **Time Required**:
   a. EO – 5 x 50 min periods;

   b. EC – 1 x 50 min period; and

   c. PC – 1 x 50 min period.

5. **Instructional Methods**:
   a. Study assignment; and
   b. Guided discussion.

6. **Training Aids**:

   a. Watch and Station Bill;

   b. NAV COMM Career Progression; and

   c. CCR Functional Org. Chart.

7. **Test Details**: EO 001.01 will be tested in EC 001.01 and PC 001.

8. **Remarks**: The CISO Pre-Instruction Package will include assigned readings for the below teaching points followed by the Instructor leading a guided discussion in class.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Identifying ship's NAV COMM team responsibilities and functions (TASK) IAW ref.						
1. Course Scenario Introduction a. Situation: Trainees are CISOs on an HMC Ship which is en route to a Specified Operation / Mission	**50**			Course Scenario Pack – Part 1	Briefing	Instructor will use the Course Scenario Pack and read in the students on the situation they are in.
2. Ship's NAV COMM team responsibilities and functions: a. CISO b. SNC (ISSO) c. ISA d. Watch Supervisor e. A/ISA f. Bridge Senior Hand off the Watch (SHOW) g. CCR Senior Hand of the Watch (SHOW) h. Bridge Junior Hand of the Watch (JHOW) i. CCR Junior Hand of the Watch (JHOW) j. Message Center /Main Signalling Officer k. Expectations of each positions from SNC/CISO perspective	**50**				S1: In-Class discussion based on PIP S2: Distributed Learning (DLN) S3: Nil.	Instructor to discuss PIP assigned readings
Verifying manning requirements and actions taken during various evolutions (TASK) IAW ref.						
3. Personnel requirements and action taken during the following evolutions:	**50**				S1: In-Class discussion based on PIP	Instructor to discuss PIP

a. Normal b. Minimum Manning (CCR/Bridge) c. Action Stations d. Emergency Stations e. Emergency Flying Stations f. RAS g. Special Parties h. Enter/Leave Harbour					S2: Distributed Learning (DLN) S3: Nil.	assigned readings
Advise subordinates on occupation progression IAW ref.						
4. Identify NAV COMM occupation progression a. Junior CISN OP (QL3) b. OJPR on Ship (QL4) c. CISN OP (QL5A) d. CISN ADMIN (QL5B) e. CISN Supp (QL6A) f. CISN Mgr. (QL6B)	**50**				S1: In-Class discussion based on PIP S2: Distributed Learning (DLN) S3: Nil.	Instructor to discuss PIP assigned readings
5. MOC advisor perspective on strategic / operational NAV COMM considerations.	**50**			Guest Lecturer	S1: In-Class discussion based on PIP S2: Distributed Learning (DLN) S3: Nil.	NAV COMM MOC Advisor NTDC(A) CPO2 or MARPAC ISSO

Enabling Check – EO 001.01 Manage NAV COMM Personnel

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.01 will be a written assessment based on content from EO 001.01.
3. To pass this EC, the trainee must achieve 5/8 SATs or 70%.
4. Time required: 1 x 50 min period for all trainees.
5. References will not be required.

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a written assessment based on content from EO 001.01.
2. References will not be provided.
3. Pass mark is 70%.
4. You will have 1 x 50 min period(s) to complete this assessment.
5. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Ensure candidates have a good understanding of the career progression and roles of their subordinates.

PART IV – ENABLING CHECK LIST

Item	EC 001.01 Manage NAV COMM Personnel	SAT	UNSAT	Remarks
1. Understand Career Progression				
List all QL's – 3, 4, 5A, 5B, 6A, 6B		☐	☐	
Briefly describe each		☐	☐	
2. Roles and Responsibilities				
Role of SNC		☐	☐	
Role of ISSO		☐	☐	
Role of ISA		☐	☐	
Role of Watch Supervisor		☐	☐	
Role of SHOW		☐	☐	
Role of JHOW		☐	☐	

Score: ***/***

Trainee must achieve 5/8 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 001.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 001.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 001.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 001.02 Coordinate Ship's CIS Training

1. **Performance**: Coordinate ship's CIS training (N0007).

2. **Conditions**:

   a. Given:

      (1) DWAN Access;

      (2) Secret Network Access;

      (3) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall coordinate ship's CIS training (N0007) IAW A14 by:

   b. Coordinating Ship's CIS training (N0007), IAW <A14> by:

      (1) Directing ship's CIS Combat Readiness Requirements (CRRs);

      (2) Managing section refresher training requirements;

      (3) Directing new CIS equipment training; and

      (4) Explaining additional training requirements.

4. **Time Required**:

   a. EO – 4 x 50 min periods; and

   b. EC – 1 x 50 min periods.

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Self-study; and

   c. Field trip.

6. **Training Aids**:

   a. Voice Trainer

   b. Flag Trainer

c. Flashing Light Trainer

d. Crypto Trainer

e. KIV-7 / TACLANE Trainer

f. NMT Trainer

g. PRC 117 Trainer

h. Comms Equipment Trainer

7. **Test Details**: EO 001.02 will be tested in EC 001.02 and PC 001.

8. **Remarks**:

a. The CISO Pre-Instruction Package will include assigned readings discussion in class.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Directing ship's CIS Combat Readiness Requirements (CRR) IAW ref:						
1. Use ARMOR a. Purpose b. Reporting c. Governing authority d. DWAN vs CSNI versions	**25**				S1: In-class lecture. S2: Distributed Learning (DLN) S3: Nil.	
2. Directing ship's CIS Combat Readiness Requirements by: a. Explaining the CRR process; b. Identifying CIS CRRs; c. Scheduling CRRs; d. CRR Delivery e. CRR at sea or ashore	**50**				S1: In-class lecture. S2: Distributed Learning (DLN) S3: Nil.	
Managing Section Refresher Training requirements IAW ref :						
3. Managing section refresher training requirements:	**100**				S1: In-class lecture. S2: Distributed	

a. Explaining section training requirements; b. Explaining Training PO, SNC, and CISO relationship; c. Trade skills refresher; d. .50 cal trainer e. Pre-sailing refresher plans; f. Readiness requirements; g. Describing external CIS trainers to include: (1) Voice Trainer (2) Flag Trainer (3) Flashing Light (4) Crypto trainer (5) KIV-7 (6) TACLANE (7) NMT (8) PRC 117 (9) SKL (10) Comms Equipment Trainer (11) Sea Training Resources					Learning (DLN) S3: May include virtual equipment trainer.	
Directing new CIS equipment training IAW ref :						
4. Directing new CIS equipment training by: a. Liaising with Training Establishments, Project Managers b. CFCSS for Crypto fill devices	**30**				S1: In-class lecture. S2: Distributed Learning (DLN) S3: Nil.	

Enabling Check – EO 001.02 Coordinate Ship's CIS Training.

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 001.02 will be a practical assessment based on content from EO 001.02.
3. To pass this EC, the trainee must achieve 3/4 SATs.
4. Time required: 1 x 50 min period.
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a practical assessment based on content from EO 001.01.
2. References will not be provided.
3. Pass mark is 3/4 SATs.
4. You will have 1 x 50 min period(s) to complete this assessment.
5. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Candidates provided with FLEX based on scenario. They are to identify the serials which NAV COMMs will participate in, and be provided a training plan by the PO2 for each of those watches. They will need to compare the Training Plan, FLEX and current status of comms CRRs (handout) to determine what is missing and what items need to be added or amended in the FLEX.

PART IV – ENABLING CHECK LIST

Item	EC 001.02 Performance Statement	SAT	UNSAT	Remarks
1. Combat Readiness Requirements and Training				
	Identified the CRRs that were not in the FLEX but had expired?	☐	☐	
	Identified personnel shortages for serials	☐	☐	
	Identified that the Training PO had not scheduled refresher training for the Watch for WoD serial.	☐	☐	
	Recognized there was no sequence of manoeuvres for upcoming TABORD for OOW Manoeuvres with their ship as OCS	☐	☐	

Score: ***/***

Trainee must achieve 3/4 SATs

Positive Feedback	Improvement Feedback

EC 001.02 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 001.02 with the candidate.	**Candidate** I have been debriefed on the result of EC 001.02.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 001 Lead NAV COMM Section

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. PC 001 will be a practical assessment based on content from PO 001.

3. To pass this PC, the trainee must achieve 7/9 SATs.

4. Time required: 1 x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 001 will be a practical assessment based on content from PO 001.

2. During this practical assessment, you will prepare the NAV COMM section for work-ups. You must pass this PC to pass PO 001.

3. References provided: FLEX, W&S bill, CFCD 102, ARMOR Example

4. Pass mark is 7/9 SATs IAW PCL 001.

5. You will have 1 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Threaded Scenario: Ship is on its way or has already arrived on a training exercise like Joint Warrior, or RIMPAC.

Prepare the NAV COMM section for work-ups by:
Given flex, identify serials your section will need to prepare for;
Compare W&S bill to see if there are any gaps;
Compare CRRs to Sea Training Guide for requirements;
Create a training plan to get section trained for WUPs.

PART IV – PERFORMANCE CHECK LIST

Item	PC 001 Lead NAV COMM Section	MET	NOT MET	Remarks
1. FLEX Comparison				
Identify CRRs in FLEX		☐	☐	
Identify NAV COMM serials		☐	☐	
Assessment of readiness level based on CRRs		☐	☐	
Identified 2/3 deficiencies in FLEX		☐	☐	
2. W&S Bill Assessment				
Identified 2/3 personnel shortages for serials		☐	☐	
WoD Serial Assessment		☐	☐	
3. Workups Plan				
Created plan to include missing CRRs		☐	☐	
Identified OPI from section for each serial		☐	☐	
Assessment of time for each serial based on CFCD 102		☐	☐	

Score: /

The trainee must achieve 7/9 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 001 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 001 with the candidate.	**Candidate** I have been debriefed on the result of PC 001.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 002 Draft Communications Improvement Documents

EO 002.01 Recommend CIS Improvements

1. **Performance**: Recommend CIS improvements (N0009).

2. **Conditions**:

   a. Given:

      (1) DWAN Access:

      (2) Classified Network Access;

      (3) Message Traffic; and

      (4) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall recommend CIS improvements (N0009) IAW references by:

   a. Identifying systems performance deficiencies, malfunctions, or failures (B0540) IAW references;

   b. Submitting an IMR IAW references by:

      (1) Drafting an Assyst ticket;

      (2) Completing the Request for Support (RFS) process;

      (3) Completing Request for Change (RFC) process; and

      (4) Monitoring the Engineering Change process.

   c. Drafting a Statement of Operational Capability Deficiency IAW references by:

      (1) Reviewing the SOCD database;

      (2) Liaising with SMEs;

      (3) Applying the SOCD and UCR formats; and

      (4) Submitting lessons learned via KMS IAW references to include reviewing the KMS database.

   d. Drafting an Unsatisfactory Condition Report (UCR) IAW references.

4. **Time Required**:

   a. EO – 5 x 50 min periods;

   b. EC – 1 x 50 min period; and

   c. PC – 2 x 50 min period.

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Demonstration - performance;

   c. Study assignment;

   d. Self-study;

6. **Training Aids**:

   a. Assyst Portal / IMR database

   b. KMS program (Unclassified and Classified)

   c. SOCD Database

   d. UCR Message

7. **Test Details**: EO 002.01 will be tested in EC 002.01 and PC 002.

8. **Remarks**:

   a. For TP 2 there is an assignment that the students are required to complete as a part of the PIP in preparation for this lesson. For TP 3, the instructor will need to liaise with their coastal Fleet SNC or CISO to determine if there are any outstanding deficiencies that require an SOCD to be drafted. The CISO instructor may also approach Ship's CISOs, BIS, N6, and CFCSS to determine if there are outstanding deficiencies.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Identify systems performance deficiencies, malfunctions or failures IAW ref.						
1. Identifying systems performance deficiencies,	**50**				S1: in-class interactive lecture S2: DLN course	Instructor to lead discussion on the ways that deficiencies

malfunctions, or failures to include: a. Identification by personnel to include: i. STOW; ii. CSEO; iii. ORS; iv. OOW; v. NAV COMMS; b. Equipment error chain					S3: Nil.	can be identified.
Submit an IMR						
2. Submitting an IMR Assyst ticket process Local manager (ISA) Approval process IMR process to include : a. RFS b. RFC IMR considerations: Base Information Services Shared Services Canada	**50**				S1: in-class interactive lecture with DWAN assyst portal as training aid S2: DLN course with imbedded assignment to use the assyst portal S3: Nil.	Trainee will submit a screenshot of their ticket IAW the PIP to the Instructor which will be discussed in class upon arrival, or via DLN forums.
Draft a Statement of Operational Capability Deficiency and Unsatisfactory Condition Report						
3. Drafting a Statement of Operational Capability Deficiency and Unsatisfactory Condition Report to include: a. Reviewing the SOCD database; b. Liaising with SMEs; c. Applying the SOCD and UCR formats;	**25**		**75**		S1: in-class interactive lecture with group assignment S2: DLN course with individual assignment S3: Nil.	Instructor to liaise with Fleet SNC/CISO to determine possible SOCD to draft on behalf of Fleet. Trainees to work together to draft SOCD in-class for submission.
Submit lessons learned via Knowledge Management System						
4. Submitting lessons learned via Knowledge Management System (KMS) to include: a. Purpose	**15**	**25**	**15**		S1: in-class explanation and demonstration with student lab	Provide information for students to generate CIS input to KMS

b. Reporting Requirements c. Location d. How to provide input / make changes					S2: DLN course with imbedded labs. S3: Nil.	

Enabling Check – EO 002.01 Recommend CIS Improvements

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 002.01 will be a practical assessment based on content from EO 002.01.
3. To pass this EC, the trainee must achieve 8/11 SATs.
4. Time required: 1 x 50 min period(s).
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.01 will be a practical assessment based on content from EO 002.01.
2. During this practical assessment, you will prepare evaluation operational reports. You must pass this EC to pass EO 002.01.
3. References will not be provided.
4. Pass mark is 8/11 SATs IAW ECL 002.01.
5. You will have 1 x 50 min period(s) to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

EC shall not have SOCD or KMS Submission Information. Those were covered via practical assignment. EC shall be comprised of a scenario where the trainee must explain the escalation of the troubleshooting process. Part 2 of the EC shall be explaining the Assyst, RFC, IMR process from start to finish

PART IV – ENABLING CHECK LIST

Item	EC 002.02 Prepare Evaluation Operational Reports	SAT	UNSAT	Remarks
1. Troubleshooting Process				
Explanation of NAV COMM responsibilities		☐	☐	
Explanation of ORS's role		☐	☐	
STOW's role and actions		☐	☐	
CSE department actions		☐	☐	
CISO's responsibilities		☐	☐	
2. IMR Process				
Explain Assyst Ticket portal/process		☐	☐	
IMR Manager's Role		☐	☐	
RFC Process		☐	☐	
Explained that an SOCD is required		☐	☐	
Explanation of when a SA&A is req.		☐	☐	
Explanation of when a TCI is req.		☐	☐	

Score: /

Trainee must achieve 8/11 SATs.

Positive Feedback	Improvement Feedback

EC 002.02 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 002.02 with the candidate.	**Candidate** I have been debriefed on the result of EC 002.02.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 002.02 Prepare Evaluation Operational Reports

1. **Performance**: Prepare evaluation operational reports (B0140).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified Network Access;

      (3) Message Traffic; and

      (4) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall prepare evaluation operational reports (B0140) IAW references by:

   a. Drafting CIS input to evaluation operational reports IAW references by:

      (1) Draft input for OPEVAL;

      (2) Draft input for OPANAL;

      (3) Draft input to Post Deployment Reports; and

      (4) Drafting RCN Observation Report.

   b. Drafting CIS input to routine reports IAW references; and

   c. Facilitate the testing of shipboard CIS resources by external agencies IAW references.

4. **Time Required**:

   a. EO – 2 x 50 min periods;

   b. EC – 1 x 50 min periods; and

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Study assignment;

c. Self-study;

d. Case study;

6. **Training Aids**:

a. OPEVAL Format

b. OPANAL Format

c. PDR Format

d. RCN Observation Form

7. **Test Details**: EO 002.02 will be tested in EC 002.02 and PC 002.

8. **Remarks**:

a. TP 2 does not require guest lectures by Subject Matter Experts (SME).

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Draft CIS input to evaluation and routine reports (Task number) IAW ref.						
1. Drafting CIS input to evaluations and routine reports a. Draft input for OPEVAL; b. Draft input for OPANAL; c. Post Deployment Reports; and d. RCN Observation Report.	**50**		**25**		S1: in-class lecture with practical group assignment S2: DLN Case study with assignment for PDR S3: Nil.	Instructor to provide group assignment to draft parts of an OPEVAL or PDR. Different scenario from EC.
Facilitate the testing of CIS resources by external agencies						
2. Facilitate the testing of CIS resources by external agencies to include: a. DMSS b. FMF/N37 c. NFR d. N6/BIS	**25**				S1: in class interactive lecture S2: DL content S3: nil.	Discussion lead by instructor on the resources each of these units require

e. Fleet Staff f. Ship Staff g. CFMWC h. NETE						when testing equipment.

Enabling Check – EO 002.02 Prepare Evaluation Operational Reports

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 002.02 will be a practical assessment based on content from EO 002.02.
3. To pass this EC, the trainee must achieve 2/3 SATs.
4. Time required: 1 x 50 min period(s).
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 002.02 will be a practical assessment based on content from EO 002.02.
2. During this practical assessment, you will prepare evaluation operational reports. You must pass this EC to pass EO 002.02.
3. References will not be provided.
4. Pass mark is 2/3 SATs IAW ECL 002.02.
5. You will have 1 x 50 min period(s) to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Three questions where the trainee must identify what report fits the scenario, and provide one reason why.

PART IV – ENABLING CHECK LIST

Item	EC 002.02 Prepare Evaluation Operational Reports	SAT	UNSAT	Remarks
1. Evaluation Reports				
Determined OPEVAL for Q1		☐	☐	
Determined OPANAL for Q2		☐	☐	
Determined PDR for Q3		☐	☐	

Score: /

The trainee must achieve 2/3 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 002.02 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 002.02 with the candidate.		**Candidate** I have been debriefed on the result of EC 002.02.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 002 Draft Communications Improvement Documents

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. PC 002 will be a practical assessment based on content from PO 002.

3. To pass this PC, the trainee must achieve 5/6 SATs.

4. Time required: 2 x 50 min period(s).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 002 will be a practical assessment based on content from PO 002.

2. During this practical assessment, you will draft communications improvement documents. You must pass this PC to pass PO 002.

3. References will not be provided.

4. Pass mark is 5/6 SATs IAW PCL 002.

5. You will have 2 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Scenario: While on exercise the Student will be informed of an issue with regard to IS/IT equipment and provided a statement from the ISA. They will take the proposed solution from the ISA's statement and draft an SOCD, UCR, or submit a LL depending on the scenario and draft one paragraph for the Post Deployment Report on the issue.

PART IV – PERFORMANCE CHECK LIST

Item	PC 002 Drift Communications Improvement Documents	MET	NOT MET	Remarks
1. SOCD/ UCR/ Lesson Learned				
Submission captured issue at hand		☐	☐	
SOCD Format Used		☐	☐	**Use only if SOCD was scenario**
UCR Format Used		☐	☐	**Use only if UCR was required**
Lesson Learnt submitted through KMS		☐	☐	**Use only if LL was submitted**
2. Post Deployment Report				
Included Issue		☐	☐	
Included Troubleshooting measures		☐	☐	
Included probable cause		☐	☐	
Included recommendation for repair or mitigating measures		☐	☐	

Score: /

Based on Exam, mark only 1 of the SATs for SOCD, UCR, or LL.

The trainee must achieve 5/6 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 002 – Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of PC 002 with the candidate.	**Candidate** I have been debriefed on the result of PC 002.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 003 Advise Command on CIS Security

EO 003.01 Direct Enforcement of Communications and Information Security

1. **Performance**: Direct enforcement of communications and information security (T0004).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation; and

      (2) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall direct enforcement of communications and information security (T0004) IAW reference by:

   a. Ensuring IS Accreditation IAW references by:

      (1) Facilitating the Security Analysis & Accreditation Process;

      (2) Facilitating the Technical COMSEC Inspection Process;

      (3) Developing Risk Management Plan; and

      (4) Drafting Operating Security Orders.

   b. Directing information security procedures IAW references by:

      (1) Explaining ISS IAW NDSODs;

      (2) Differentiate between Operating Authorities vs Technical Authorities;

      (3) Explaining data transfer workstation;

      (4) Utilize the JIIFC; and

      (5) Maintaining COMPUSEC.

   c. Drafting unit IS security regulations;

   d. Coordinating physical protection of IS assets by:

      (1) Explaining physical protection policies; and

      (2) Applying PSS process.

e. Managing IS security incidents by:

(1) Assessing operational impact of IS Security infractions and breaches;

(2) Conducting a Threat Risk Analysis (TRA);

(3) Reviewing ISSIRs; and

(4) Applying corrective measures for ISSIs.

4. **Time Required**:

a. EO – 11 x 50 min periods;

b. EC – 3 x 50 min periods; and

c. PC – 2 x 50 min periods.

5. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration - performance;

c. Study assignment; and

d. Self-study;

6. **Training Aids**:

a. SA&A Checklist;

b. TCI Checklist;

c. PSS Checklist;

d. ISSIR Form;

e. Threat Evaluation Matrix;

f. Ship's Unit Security Orders

g. MLPs

7. **Test Details**: EO 003.01 will be tested in EC 003.01 and PC 003.

8. **Remarks**:

a. NDSODs will be assigned as pre-course reading material.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Ensure IS accreditation IAW ref.						
1. Ensure IS Accreditation a. SA&A Process b. TCI Process c. Risk Management Plan d. Operating Security Orders	**100**			NDSODs	S1:in-class interactive lecture S2: DLN S3: Nil.	
Direct information security procedures IAW ref.						
2. Directing information security procedures to include: a. Explaining ISS IAW NDSODs b. Operating Authorities vs Technical Authorities c. Explaining data transfer workstation; d. Capabilities of JIIFC e. Maintaining COMPUSEC by: (a) Conducting IT/IS Security education; (b) Assessing IS Security threats; (c) Assessing IS Security posture; and (d) Implementing IS Security procedures	**100**	**15**	**15**		S1:in-class interactive lecture S2: DLN with submission using JIIFC S3: Nil.	Instructor to lead interactive lecture on material and demonstrate the JIIFC data transfer tool. Trainees will utilize tool for practical time.
Draft unit IS security regulations IAW ref.						

3. Drafting unit IS security regulations a. IAW NDSODs b. IAW Fleet and MARLANT or MARPAC Orders c. IAW local BIS rules d. IAW CIBs e. More Restrictive vs. National Standard	**50**		**50**		S1:in-class interactive lecture S2: DLN with submitted assignment S3: Nil.	Instructor to discuss Ship's Security orders and have students work together to improve them.
Coordinate physical protection of IS assets IAW ref.						
4. Coordinating physical protection of IS assets f. Explaining physical protection policies g. Explain PSS process	**50**				S1:in-class interactive lecture S2: DLN that ties into the DL that is created for the Nav Comm QL3. S3: Nil.	
Manage IS security incidents IAW ref.						
5. Managing IS security incidents (TASK) by: a. Assessing operational impact of IS Security infractions and breaches b. Conducting a Threat Risk Analysis (TRA) c. Reviewing ISSIRs d. Corrective measures for ISSIs	**100**		**25**	Guest Lecturer	S1:in-class interactive lecture S2: DLN S3: Nil.	Formation ISSO to come and discuss TP and ISSO responsibilities, requirements, etc.

Enabling Check – EO 003.01 Direct Enforcement of Communications and Information Security

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 003.01 will be a practical assessment based on content from EO 003.01.

3. To pass this EC, the trainee must achieve x/y SATs

4. Time required: 3 x 50 min period(s)

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a practical assessment based on content from EO 003.01.

2. During this practical assessment, you will direct the enforcement of communications and information security. You must pass this EC to pass EO 003.01.

3. References will not be provided.

4. Pass mark is x/y SATs IAW ECL 003.01.

5. You will have 3 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Trainees will be shown to a room where they must identify the physical and IT/IS security issues and achieve 7 of 10 within 15 minutes.

PART IV – ENABLING CHECK LIST

Item	EC 003.01 Direct Enforcement of Communications and Information Security	SAT	UNSAT	Remarks
1. Physical Security				
Unlocked safe		☐	☐	
Combination to safe written on back of lock		☐	☐	
Window open / door unlocked		☐	☐	
PROTECTED B information left unattended		☐	☐	
Fake Confidential documents left on top of shredder		☐	☐	
2. IT/IS Security				
PROTECTED B USB stick left out		☐	☐	
Incorrectly labeled media (CD, USB, Disk)		☐	☐	
SECRET stick in DWAN workstation		☐	☐	
Computer left logged in and unlocked		☐	☐	
Password written on a sticky note		☐	☐	

Score: /

The trainee must achieve 7/10 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 003.01 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 003.01 with the candidate.		**Candidate** I have been debriefed on the result of EC 003.01.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 003 Advise Command on CIS Security

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom/Ship/Submarine.

Arrangements and Procedures

2. PC 003 will be a practical assessment based on content from PO 003.

3. To pass this PC, the trainee must achieve 4/5 SATs.

4. Time required: 2 x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 003 will be a practical assessment based on content from PO 003.

2. During this practical assessment, you will advise command on CIS security. You must pass this PC to pass PO 003.

3. References will be provided.
5
4. Pass mark is 4/5 SATs IAW PCL 003.

5. You will have 2 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Part 1: The CISO trainee will amend a COTM pertaining to a new system onboard their ship. He/she must research regulations pertaining to the system to include NDSODs, DAODs, and ITSGs.

Part 2: The CISO trainee will review an ISSIR drafted by the ISSO regarding an incident involving the new system and conduct quality control IAW the scenario.

PART IV – PERFORMANCE CHECK LIST

Item	PC 003 Advise Command on CIS Security	MET	NOT MET	Remarks
1. COTM for new system				
Applied NDSODs		☐	☐	
Applied ITSGs		☐	☐	
Applied NAVORD		☐	☐	
2. Review ISSIR				
Identified 2/3 errors on ISSIR		☐	☐	
Identified 2/3 errors on Threat Matrix		☐	☐	

Score: **/**

The trainee must achieve 4/5 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 003 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 003 with the candidate.		**Candidate** I have been debriefed on the result of PC 003.	
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 004 Direct Operational Messaging Traffic

EO 004.01 Prepare Operational Messages

1. **Performance**: Prepare operational messages (B0130).

2. **Conditions**:

   a. Given:

      (1) DWAN Access;

      (2) Classified Network Access:

      (3) Messaging Drafting Software;

      (4) Message Traffic; and

      (5) References.

   b. Denied: Ability to transmit on HF.

   c. Environment: Classroom, Trainer, NCS.

3. **Standard**: The trainee shall prepare operational messages (B0130) IAW references by:

   a. Overseeing message preparation;

   b. Verifying message addressing;

   c. Apply Communications Information Bulletins (CIBs); and

   d. Drafting CIS messages.

4. **Time Required**:

   a. EO – 7 x 50 min periods;

   b. EC – 1 x 50 min periods; and

   c. PC – 2 x 50 min periods.

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Demonstration - performance;

   c. Study assignment;

   d. Self-study;

6. **Training Aids**:

   a. APP 11 Message Format Handout

   b. Classified and Unclassified CIB webpages

   c. CAF Message Addressing Manual (online – DWAN)

   d. Example messages for all messages in TP 3

7. **Test Details**: EO 004.01 will be tested in EC 004.01 and PC 004.

8. **Remarks**:

   a. Nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Oversee Message Preparation (Task number) IAW ref.						
1. Explaining the various parts of messages: a. Message Design (1) Basic Message Format (2) Parts: Heading, Text, Ending (3) Schematic Diagram b. Message Preparation: (1) OPSIGs (2) Prosigns c. Service Messages	**50**				S1: in-class interactive lecture S2: DLN S3: Interactive application that teaches the trainee how to write a generic message.	Instructor must use the APP 11 message format handout.
Verify Message Addressing (task) IAW ref						
2. Verify message addressing to include: a. Plain Language Address (PLAs) b. National Registry c. Allied Registry d. Routing Indicators (RI)	**25**				S1: in-class interactive lecture S2: DLN S3: Interactive application that teaches the trainee how to write a	APP 11 Message Format handout and CAF Address Manual (online)

e. Formulation of RI (1st, 2nd, 3rd, 4th, and 5th letters) f. RI Assignment Authority & Management g. RI Supplements h. Collective RIs i. UNCLAS / CLASSIFIED RIs j. RI Change Promulgation k. AIGs l. Callsigns					generic message.	
3. Message Comparison Assignment: a. ACP 124, b. ACP 127, c. ACP 128, d. ACP 126 Modified, ACP 176	**15**	**25**	**15**		S1: in-class interactive lecture and format study S2: DLN comparison assignment S3: addition of NATO standard messaging software	
Review Communications Information Bulletins (task) IAW refs						
4. Review Communications Information Bulletins (CIBs)	**25**				S1: In-class interactive lecture S2: DLN Link to material (unclassified) S3: Nil.	Instructor to review UNC and Classified CIBs
Draft CIS messages (task) IAW refs						
5. Drafting CIS messages IAW references to include: a. TABORDs b. CIS Service Request c. Daily Sitrep / OPSUM d. COMSPOT e. COMSTAT f. Network Outage Report g. OPGEN h. OPORD	**150**		**50**		S1: In-class interactive lecture with demonstration by instructor and practical phase where trainees draft a TABORD. S2: DLN for unclassified content. S3: Classified content DLN.	Instructor to cover each message briefly and discuss the assigned readings where the trainees would have read each message in detail. Assign an exercise to

i. OPTASK messages j. OPSTAT messages k. OPSTAT CIS Supp l. Tactical Frequency Clearance Request m. Non-tactical Frequency Clearance Request n. Satellite Access Request o. Satellite Allocation Message p. C4I Requirements Message q. Crypto Callout Message r. COMSEC Incident Report s. CANCOMSECGEN t. CANCOMSECADMINGEN u. ROUTE STAT REP v. Common Source Route File (CSRF)						each trainee and require a TABORD to be drafted and handed into the instructor.

Enabling Check – EO 004.01 Prepare Operational Messages

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom/Ship/Submarine.

Arrangements and Procedures

2. EC 004.01 will be a practical assessment based on content from EO 004.01.

3. To pass this EC, the trainee must achieve 7/10 SATs.

4. Time required: 1 x 50 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.01 will be a practical assessment based on content from EO 004.01.

2. During this practical assessment, you will verify the formatting and content of operational messages. You must pass this EC to pass EO 004.01.

3. References will be provided.

4. Pass mark is 7/10 SATs IAW ECL 004.01.

5. You will have 1 x 50 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Instructor shall give the trainees three messages and the trainees will identify the errors in the messages, and annotate the corrections onto it.

PART IV – ENABLING CHECK LIST

Item	EC 004.01 Prepare Operational Messages	SAT	UNSAT	Remarks
1. ACP 176 Format				
Error 1		☐	☐	
Error 2		☐	☐	
Error 3		☐	☐	
Error 4		☐	☐	
2. Message Text Format (MTF) APP11 OPGEN/SUPP/TASK etc.				
Error 1		☐	☐	
Error 2		☐	☐	
Error 3		☐	☐	
Error 4		☐	☐	
3. TABORD				
Error 1		☐	☐	
Error 2		☐	☐	

Errors types will be identified on the Master Copy only.

Score: /

Trainee must achieve 7/10 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 004.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 004.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 004.02 Direct Operational Message Traffic

1. **Performance**: Direct operational message traffic (N0013).

2. **Conditions**:

   a. Given:

      (1) DWAN Access;

      (2) Classified Network Access:

      (3) Messaging Drafting Software;

      (4) Message Traffic; and

      (5) References.

   b. Denied: Ability to transmit on HF.

   c. Environment: Classroom, Trainer, NCS.

3. **Standard**: The trainee shall direct operational message traffic (N0012) IAW references by:

   a. Directing Message Processing and Handling Systems

      (1) Applying E-Mail Type Messaging (ACP 123 Environment)

      (2) Managing Military Messaging Handling Instructions

      (3) Directing method of message distribution throughout the ship

   b. Coordinating the distribution of operation message traffic IAW references by:

      (1) Configuring RATT Communications

      (2) Configuring Ship / Shore – National, NATO, US, ALLIED

      (3) Explaining Strategic Message Switching System (SMSS)

4. **Time Required**:

   a. EO – 11 x 50 min periods;

   b. EC – 1 x 50 min periods; and

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Demonstration - performance;

c. Guided discussion;

d. Tutorial;

6. **Training Aids**:

a. MHDS Trainer;

b. MCOIN Messaging Account;

c. Strategic Messaging System, System Diagram (Handout); and

d. MLPs.

7. **Test Details**: EO 004.02 will be tested in EC 004.02 and PC 004.

8. **Remarks**:

a. This portion of the course should be taught by a MS or above NAV COMM under direct supervision by the CISO instructor to ensure that the QSP is adhered to and the trainees are provided the information that is relevant to the QSP, and their position as CISO.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Direct the handling of operational message traffic IAW ref.						
1. Message Processing and Handling Systems: a. Message Handling System (MHS) b. Message Processing System (MPS) c. MHDS d. MCOIN Messaging e. Air Gap message distribution process	**25**	**15**	**15**		S1: In class interactive lecture and MHDS Lab S2: DLN with in-class lab S3: Virtual MHS trainer	Compare MSO duties to regulations, and SOP on MSO
2. E-Mail Type Messaging (ACP 123 Environment): a. Principles and Theory b. X.400 message exchange c. X.500 directory services	**50**				S1: Guest Lecture or CCR Tour of ship S2: DLN S3: Nil.	Overview with details on how they are used.

3. Manage Military Messaging Handling Instructions: a. OPSIGS b. MINIMIZED c. CONSIDERED d. LIMITED DISTRIBUTION e. NO DISTRIBUTION f. SEND TO xxxx g. ALARM REQUIRED h. NIGHT ACTION	**25**				S1: Guest Lecture or CCR Tour of ship S2: DLN S3: Nil.	Importance of MSO and proper handling by NAV COMMs shall be emphasized here.
4. Direct method of message distribution throughout the ship a. Paper b. Electronic c. Emailing Messages to entire Ship's COY d. Implementing a shared folder for messages e. Managing the CO's File	**25**		**50**		S1: Guest Lecture or CCR Tour of ship S2: DLN with case study assignment. S3: Lasers.	
Coordinate the distribution of operational message traffic (TASK) IAW ref						
5. Ship / Shore – National, NATO, US, ALLIED: a. Basic Principles b. Categories of Stations (Primary, Secondary, Local) c. Methods: Simplex / Half- Duplex / Duplex d. Types of ship shore procedures e. Ship shore Tables	**50**				S1: In-class lecture based on assigned readings S2: DLN S3: Nil.	
6. RATT Communications: a. Broadcasts – NATO, ALLIED, National b. Broadcast Authorities and Responsibilities: (1) BCOA (2) BCA (3) BCS c. Message Delivery Facilities: (1)NRS	**150**				S1: In-class lecture based on assigned readings S2: DLN S3: Virtual NRS/NCS.	

(2) Maritime Air Fixed land communications site (AGA) (3) COMFAC / COMCEN d. Broadcast Types: (1) Area (2) Sub – Area (3) Local (4) Special e. Broadcast Designators: (1) Geographic Boundary (2) NATO Regions (3) ITU Regions (4) Service Type (5) On-line / Off-line Encrypted (6) RATT / CW / Voice (7) METOC / Submarine / Small Vessel / Intelligence (8) Geographic Area (9) Area / Sub Area / Special f. Broadcast Procedure Requirements: (1) Broadcast Tables (2) Multi-Channeling (3) Setting/Ceasing Watch Rules (4) CHOP Procedures (5) Guard Duties (6) Dual Routing (7) Operator periods (8) Crypto periods (9) Service Status Reports (10) Loss of Service Procedures						

7. Explain Strategic Message Switching System (SMSS)	**60**				S1: In-class interactive lecture S2: interactive DLN. S3: Nil.	
8. Tour of Message Handling Facilities, NRS/COMMCEN	**120**				N/A	

Enabling Check – EO 004.02 Direct Operational Message Traffic

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom/Ship/Submarine.

Arrangements and Procedures

2. EC 004.02 will be a practical assessment based on content from EO 004.02.

3. To pass this EC, the trainee must achieve x/y SATs or %.

4. Time required: 1 x 60 min period(s) per trainee (# total (time per trainee x number of trainees)).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 004.02 will be a practical assessment based on content from EO 004.02.

2. During this practical assessment, you will be required to direct operational message traffic. You must pass this EC to pass EO 004.02.

3. References will be provided.

4. Pass mark is x/y SATs IAW ECL 004.02.

5. You will have 1 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The trainee must be able to explain the following systems, how they interact, which ones are on ship, ashore, and what level they operate (Regional, National, etc.)

PART IV – ENABLING CHECK LIST

Item	EC 004.02 Direct Operational Message Traffic	SAT	UNSAT	Remarks
1. Message Handling Systems				
MHDS		☐	☐	
CSNI Messaging		☐	☐	
DEMS II		☐	☐	
2. Message Distribution Systems				
Ship-Shore		☐	☐	
HF Broadcast		☐	☐	
Types		☐	☐	
Areas		☐	☐	
National		☐	☐	
Allied		☐	☐	
SMSS Infrastructure		☐	☐	

Score: **/**

The trainee must achieve 7/10 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 004.02 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 004.02 with the candidate.		**Candidate** I have been debriefed on the result of EC 004.02.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 004 Direct Operational Messaging

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom/Ship/Submarine.

Arrangements and Procedures

2. PC 004 will be a practical assessment based on content from PO 004.

3. To pass this PC, the trainee must achieve 5/6 SATs.

4. Time required: 1 x 60 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 004 will be a practical assessment based on content from PO 004.

2. During this practical assessment, you will be required to direct operational messaging. You must pass this PC to pass PO 004.

3. References will be provided.

4. Pass mark is 5/6 SATs IAW PCL 004.

5. You will have 1 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Part 1: The CISO student will proof a prepared message for formatting and addressing errors and correct the errors found IAW SSO's Vol2 and ship's COTM.

Part 2: The CISO student will trace the routing of a lost message from their ship to a unit in Northwood, UK, or Hawaii, USA by creating a diagram outlining the path is has taken.

PART IV – PERFORMANCE CHECK LIST

Item	PC 004 Direct Operational Messaging	MET	NOT MET	Remarks
1. Proofing Message				
Assessment of Routing Indicators (AIGs)		☐	☐	
Formatting Assessment		☐	☐	
Information within the message		☐	☐	
2. Routing Trace				
Explanation of route the message would take		☐	☐	
Explanation how the message could be "lost"		☐	☐	
Proposed resolution (how to trace lost message)		☐	☐	

Score: **/**

The trainee must achieve 5/6 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 004 – Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of PC 004 with the candidate.	**Candidate** I have been debriefed on the result of PC 004.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 005 Advise Command on Electromagnetic Spectrum Warfare

EO 005.01 Manipulate RF prediction software

1. **Performance**: Manipulate RF prediction software (N0015).

2. **Conditions**:

   a. Given:

      (1) DWAN Access;

      (2) Classified Network Access:

      (3) Secret CEO Workstation;

      (4) RF prediction software;

      (5) EM footprint software;

      (6) Message traffic; and

      (7) References.

   b. Denied: Nil.

   c. Environment: Classroom. Battle Lab.

6. **Standard**: The trainee shall manipulate RF prediction software (N0015) IAW references by:

   a. Explaining RF Theory to include:

      (1) RF Properties and Variables

      (2) Radio communication system equipment:

      (3) Antenna Theory:

      (4) RF Modulation

   b. Correlating RF information from operational sources IAW references;

   c. Manipulate RF prediction software

3. **Time Required**:

   a. EO – 31 x periods;

   b. EC – 2 x periods; and

c. PC – 2 x periods.

4. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration - performance;

c. Tutorial;

d. Field trip;

5. **Training Aids**:

a. Open source animated sinusoidal wave;

b. Open source animation of change in frequency;

c. RF Properties of HCM Ships;

d. Lobe Diagrams;

e. HF Prediction Software;

f. EM configuration software (Spiral 3); and

g. MLPs.

6. **Test Details**: EO 005.01 will be tested in EC 003.01 and PC 005.

7. **Remarks**:

a. Majority of this portion of the course should be taught by a CSEO Instructor who is familiar with the NEI Course Material.

7. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Explaining RF Theory IAW ref.						
1. RF Properties and Variables: a) Wavelength VS Frequency	**120**		**120**		S1: In-class interactive lecture, video, or open source	Content to be given and discussed by CISO Instructor and

b) Frequency Bands and Groupings (Comm & Radar) c) ITU Military d) Wave theory: i. Direct Wave: i. Diffraction ii. Attenuation iii. Reflection iv. Refraction v. Sub-refraction/Super-refraction ii. Sky Wave: i. Affected frequencies ii. Layers (D, E, F1, F2) iii. Magnetosphere iv. Predictable variations v. Diurnal vi. Seasonal vii. Sun cycle viii. Sunspots ix. Flux measurement x. Shadow zones xi. Critical angle					S2: Interactive tools and lessons S3: Tools distributed via DLN	CSEO Instructor to facilitate learning. Practical aspect shall be given in the COMM TECH labs where students can assemble equipment and apply in-class knowledge to technology.

iii. Ground Wave: i. Cause ii. Frequency VS Power relationship iii. Seamless coverage e) Frequency properties: iv. Bandwidth v. Water/Dust penetration vi. Ranges vii. Typical military applications						
2. Radio communication system equipment: b) Transmitter and receiver: i. Block diagram/signal flow ii. Characteristics (Sensitivity and Selectivity) c) Block diagram and signal flow of an AM Superheterodyne receiver d) Transmission medium characteristics: e) Waveguides f) Line communications	**24 0**		**24 0**		S1: In-class interactive lecture, video, or open source S2: Interactive tools and lessons S3: Tools distributed via DLN	Content to be given and discussed by CISO Instructor and CSEO Instructor to facilitate learning Practical aspect shall be given in the COMM TECH labs where students can assemble equipment and apply in-class knowledge to technology.
3. Antenna Theory: Fundamental Types: g) Single Array: i. Dipole	**24 0**		**24 0**		S1: In-class interactive lecture, video, or open source	Content to be given and discussed by CISO Instructor and CSEO

ii. Marconi (Monopole or “Whip”) iii. Loop h) Multi-Array: i. End Fire Driven (Yagi) ii. Broadside Driven i) Element Arrangement Configurations: i. Mono-Static ii. Bi-Static iii. Focused (“Dish”) iv. Phased Array j) Principle Components and Purpose: i. Waveguides ii. Couplers k) Polarization: i. Horizontal ii. Vertical l) Field strength with respect to: i. Antenna patterns ii. Side lobes iii. Directivity iv. Directional gain m) Relationship of wave length to antenna size n) Half wave VS quarter wave dipole antenna o) Antenna and special configurations:					S2: Interactive tools and lessons S3: Tools distributed via DLN	Instructor to facilitate learning. Practical aspect shall be given in the COMM TECH labs where students can assemble equipment and apply in-class knowledge to technology.

i. Random wire ii. Tuned antenna iii. Microwave dish type iv. Phased systems (e.g.: Broadside array and end fire array)						
4. RF Modulation a) Methods of modulation: i. Analogue ii. Digital b) Types of modulation: i. Analogue: i. AM ii. FM iii. PM ii. Digital: i. CW ii. ASK iii. FSK iv. PSK v. QPSK c) Optimizing modulation: i. Suppressing: i. Carrier waves ii. Sidebands d) Mixing Superhetrodyne e) Multiplexing i. FDMA	**24 0**		**24 0**		S1: In-class interactive lecture, video, or open source S2: Interactive tools and lessons S3: Tools distributed via DLN	Content to be given and discussed by CISO Instructor and CSEO Instructor to facilitate learning. Practical aspect shall be given in the COMM TECH labs where students can assemble equipment and apply in-class knowledge to technology.

ii. TDMA iii. CDMA iv. DAMA f) Spread Spectrum						
Correlate RF information from operational sources IAW ref:						
5. Operational Sources a. OPTASK COMMS b. COMPLAN c. Space Weather Report d. METOC e. Lat and Long vs. time of Day	**30**		**30**		S1: In-class interactive lecture, video, or open source S2: Interactive tools and lessons S3: Tools distributed via DLN	CISO Instructor combined with PO1 SNC
6. Ship's Emitter Types a. Handout with list of ship's emitters b. Students determine what types of antennas they are and label them	**0**				S1: Handout S2: DLN interactive tool S3: Virtual Reality using Virtual Platforms where they can select an antenna and receive technical data on it	No time allocation. Expectation is that they will complete it on their own time and hand it in for marking.
Manipulate RF prediction software IAW ref:						
7. Use other HF prediction software to predict HF propagation based on operational sources	**15**	**15**	**30**		S1: Investigate other software S2: New HF predict software in Battle Lab S3: NavIS accepted HF predict software	Each session, NTDC will need to ensure MLP updated based on post-HCM EM Trials
8. Use EM footprint prediction software to predict the EM footprint of the ship as	**15**	**15**	**30**		S1: In-class discussion based on EM footprint	Each session, NTDC will need to ensure MLP

configured with a COMPLAN					reports from CFMWC S2: EM Software in Battle Lab S3: NavIS accepted EM Footprint software via Classified Network	updated based on post-HCM EM Trials

Enabling Check – EO 005.01 Manipulate RF Prediction Software

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer.

Arrangements and Procedures

2. EC 005.01 will be a practical assessment based on content from EO 005.01.

3. To pass this EC, the trainee must achieve 3 of 4 SATs.

4. Time required: 2 x 60 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 005.01 will be a practical assessment based on content from EO 005.01.

2. During this practical assessment, you will be required to manipulate RF prediction software. You must pass this EC to pass EO 005.01.

3. References will be provided.

4. Pass mark is 3 of 4 SATs IAW ECL 005.01.

5. You will have 2 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Spiral 1: The trainees shall use open source software (HF Predict, RF Modulation simulator) to manipulate broadcast and UHF voice frequencies at various power outputs to achieve scenario goals. There should be three different scenarios provided that revolve around the Exercise sail that the trainees are operating on.

Spiral 2: Incorporation of EM emission trial data from post-HCM ships with a goal of achieving ranges vice tactical EM awareness.

Spiral 3: EM Prediction and graphic visualization software.

PART IV – ENABLING CHECK LIST

Item	EC 005.01 Manipulate RF Prediction Software	SAT	UNSAT	Remarks
1. High Frequency				
Frequencies based on COMPLAN		☐	☐	
Power output achieved required ranges		☐	☐	
2. Ultra High Frequency				
Frequencies based on COMPLAN		☐	☐	
Power output achieved required ranges		☐	☐	

Score: **/**

The trainee must achieve 3/4 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 005.01 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 005.01 with the candidate.		**Candidate** I have been debriefed on the result of EC 005.01.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 005 Advise Command on Electromagnetic Spectrum Warfare

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. PC 005 will be a practical assessment based on content from PO 005.

3. To pass this PC, the trainee must achieve 5/7 SATs.

4. Time required: 2 x 60 min period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 005 will be a practical assessment based on content from PO 005.

2. During this practical assessment, you will advise command on electromagnetic spectrum warfare. You must pass this PC to pass PO 005.

3. References will be provided.

4. Pass mark is 5/7 SATs IAW PCL 005.

5. You will have 2 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – PERFORMANCE CHECK NOTES

Part 1: The CISO student will determine the optimum HF frequencies within the COMPLAN to use to reach Arleigh Burke Consort ## nm away by using HF predict software.

Part 2: The CISO student will create a top down profile of the ship's lobes when all COMPLAN circuits are up using EM footprint software. JW planning staff require EMI signature report for OP Areas in the Hebrides.

PART IV – PERFORMANCE CHECK LIST

Item	PC 005 Advise Command on Electromagnetic Spectrum Warfare	MET	NOT MET	Remarks
1. COMPLAN Assessment				
HF Predict Software Use		☐	☐	
Determined Frequency IAW COMPLAN		☐	☐	
Determined Power Output to reach Consort		☐	☐	
2. Communications Lobe Profiles				
HF Lobes		☐	☐	
VHF Lobes		☐	☐	
UHF Lobes		☐	☐	
Special Systems (if any)		☐	☐	

Score: **/**

The trainee must achieve 5/7 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 005 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 005 with the candidate.	**Candidate** I have been debriefed on the result of PC 005.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 006 Manage LF/MF/HF/VHF/UHF Communications Equipment

EO 006.01 Identify Communications Monitoring Faults

1. **Performance**: Identify communications monitoring faults (N0014).

2. **Conditions**:

   a. Given:

      (1) Communications Monitoring System;

      (2) Fitted Communications Equipment;

      (3) Communications Control System;

      (4) Message Traffic;

      (5) DWAN Access;

      (6) Confidential Network Access; and

      (7) References

   b. Denied: Permission to transmit on LF through UHF

   c. Environment: Classroom, HMC Ship, and trainers.

3. **Standard**: The trainee shall identify communications monitoring faults (N0018) by:

   a. Initializing RF systems, IAW references to include:

      (1) RF equipment architecture and components;

      (2) RF equipment capabilities and limitations; and

      (3) RF equipment initialize and shut down procedures; and

   b. Configuring RF systems, IAW references to include:

      (1) Operating procedures and commands;

      (2) Configuration settings; and

      (3) RF circuit and traffic management procedures; and

   c. Configuring Communications Monitoring Systems, IAW references to include:

      (1) Conducting CCS initialize/shut down procedures to include CCS architecture

      (2) Verifying CCS circuit/traffic management to include:

(a) CCS lists, tables, and databases

(b) CCS configuration settings

(3) Configuring CCS operating procedures and commands to include;

(a) Configuring voice circuits

(4) Troubleshooting Communications Control System (CCS), IAW references, to include:

(a) CCS flow diagrams;

(b) CCS capabilities and limitations;

(c) CCS fault finding procedures; and

(d) CCS failure procedures;

d. Troubleshoot RF systems (TASK), IAW ref to include:

(1) Troubleshooting LF/MF/HF communications, IAW references, to include:

(a) HF block diagrams; and

(b) Frequency selection.

(2) Troubleshooting UHF (LOS) communications, IAW ref, to include

(a) UHF block diagrams;

(b) Frequency selection.

(3) Troubleshooting Message Processing System (MPS), IAW ref, to include:

(a) MPS flow diagrams;

(b) MPS fault finding procedures; and

(c) MPS failure procedures;

(4) Troubleshooting VHF communications, IAW Ref to include:

(a) VHF block diagrams; and

(b) Equipment settings.

4. **Time Required**:

a. EO – 25 x 60 min periods;

b. EC – 2 x 60 min periods; and

c. PC – 8 x 60 min periods.

5. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration - performance;

c. Tutorial;

d. Field trip;

6. **Training Aids**:

a. Flashup checklist

b. COMPLAN

c. Architecture diagrams

d. B-Switch

e. MLPs

f. Ship's Emitters, Spaces and Equipment Diagrams.

7. **Test Details**: EO 006.01 will be tested in EC 006.01 and PC 006.

8. **Remarks**:

a. The aim of this PO is to teach the trainee how to initialize, configure, and troubleshoot all RF communications equipment, systems, monitoring systems, software, and configuration systems and software. Through manipulation and understanding of the systems, the trainee should also understand what the configuration of the ship's communications equipment is when baselined, not including mission fit equipment.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Initialize RF systems IAW ref.						

1. Initialize/shut down MPS equipment to include: a) MPS capabilities and limitations to include: i. Introduction to theory of operations. b) MPS architecture and components to include:; i. System summary. c) MPS initialize/shut down procedures to include: i. Start up and logon procedures.	**60**	**30**	**150**	A35 art 4.1 A35 art 3.3 A35 art 2.2, 3.17 A35= CTM2100424-01 MHDS User Manual	S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	
Configure RF systems IAW ref.						
2. Configuring circuits within MPS to include: a) MPS lists, tables, and databases to include: i. Preparation for use; b) MPS operating procedures and commands; i. Port monitoring and control. c) MPS configuration settings to include: i. Broadcast circuits; ii. Ship-shore circuits; and iii. Inter-ship circuits. d) MPS circuit/traffic management procedures.	**60**	**30**	**150**	A35 art 2.1, 2.6, 2.7, 2.8, 2.9, 2.14, 2.15, 2.16, 2.17, 2.18 A35 art 3.2, 3.4, 3.5, 3.6, 3.7, 3.8, 3.9, 3.10	S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	
Configure communications monitoring systems IAW ref						
3. CCS initialize/shut down procedures to include:	**60**	**30**	**150**	A31 arts. 3.1-3.4	S1: Communications Trainer	

a) CCS architecture and components to include: i. CCMS operator consoles; ii. Alarm Panels; iii. UHF distress frequency monitor panel; iv. AN/UYK 507 computer; v. CCMS processor select panel; vi. CMTU magnetic tape drive; vii. CCMS disk drive; viii. UGC-123BX printer; ix. Wideband audio and data switch (WADS); x. CCMS subsystems; and xi. Communications suite to include: 1. Power requirements; 2. Modems; 3. HF; 4. UHF; 5. LF; 6. VHF; and 7. Crypto equipment. b) Flash-up from disk; and c) Shut down.				A31= 101733100 CCMS User Manual	S2: Improved Trainer S3: Virtual Trainer	
4. CCS circuit/traffic management to include: a) CCS lists, tables, and databases (BK0080) to include: i. System tables:	**60**	**30**	**150**	A31 art 3.5 A31 Appendix B A31 Appendix E	S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	

1. Availability; 2. Circuit Configuration; 3. COMPLAN Name; 4. BSwitch ii. COMPLAN tables. b) CCS configuration settings (BK0065) to include: i. Unattended menu commands: 1. Modify; 2. COMPLAN; 3. Backup; 4. Halt. ii. Attended menu commands: 1. Setup; 2. Cleardown; 3. Alter; 4. Report (circuit status only); 5. Exit. c) CCS operating procedures and commands (BK0075) to include: i. Modes of operation; ii. Display terminal screen layout; iii. Special function keys; and iv. Circuit setup.						
Troubleshoot RF systems IAW ref.						
5. Troubleshooting LF/MF/HF communications a) HF system Block diagram b) Frequency selection	**30**	**15**	**60**		S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	Block diagrams required.

c) Fault Finding Procedures d) Failure process						
6. Troubleshooting UHF (LOS) communications a) UHF system Block diagram b) Frequency selection c) Fault Finding Procedures d) Failure process	**30**	**15**	**60**		S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	Block diagrams required.
7. Troubleshooting Message Processing System (MPS) a) MPS system Block diagram b) Fault Finding Procedures c) Failure process	**30**	**15**	**60**		S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	Block diagrams required.
8. Troubleshooting VHF communications a) VHF system Block diagram b) Frequency selection c) Fault Finding Procedures	**30**	**15**	**60**		S1: Communications Trainer S2: Improved Trainer S3: Virtual Trainer	Block diagrams required.
9. HMC Ship Tour CPF/AOPs/CSC Tour to include : a) Communications Control Room b) Communications Equipment c) Emergency Equipment d) Emitter/Antenna Locations e) (upper decks) f) Any space which includes comms equipment, and network equipment.			**120**		S1: CPF Tour S2: Virtual AOPs and CSC S3: Platform depends on posting of students (HCM/AOPS/CSC)	Instructor to assign communications work sheet to be completed at the ship. Spaces, Antennas, Equipment in each space, etc.

Enabling Check – EO 006.01 Identify Communications Monitoring Faults

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer.

Arrangements and Procedures

2. EC 006.01 will be a practical assessment based on content from EO 006.01.

3. To pass this EC, the trainee must achieve 2/3 SATs.

4. Time required: 10 minutes per trainee or (2 x 60 minute periods = total).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 001.01 will be a practical assessment based on content from EO 006.01.

2. During this practical assessment, you will be required to identify communications monitoring faults. You must pass this EC to pass EO 006.01.

3. References will be provided.

4. Pass mark is 2/3 SATs IAW ECL 006.01.

5. You will have 10 x minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

In this EC the trainee must use the CCMS to identify where the fault in the system lies and recommend a repair, or carry out the repair.

PART IV – ENABLING CHECK LIST

Item	EC 006.01 Identify Communications Monitoring Faults	SAT	UNSAT	Remarks
1. Troubleshooting				
Identification of Fault		☐	☐	
Followed the TS process		☐	☐	
Rectification of fault		☐	☐	

Score: /

The trainee must achieve 2/3 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 006.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 006.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 006.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 006.02 Maintain Ship's PRCs

1. **Performance**: Maintain Ship's PRCs (N0020).

2. **Conditions**:

   a. Given:

      (1) Personal Radio Communication (PRC) Equipment;

      (2) Configuration Software;

      (3) Accounting Logs; and

      (4) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall maintain ship's PRCs (N0020) IAW references by:

   a. Apply PRC accounting procedures IAW refs:

      (1) Identify Governing authorities

      (2) Carry out Life cycling

      (3) Verify Ship's allocation

   b. Configuring Ship's PRCs (N0019) IAW references to include:

      (1) Programming PRC frequencies;

      (2) Installing crypto sleeves;

   c. Adhering to DFSM international regulations to include:

      (1) Canadian requirements

      (2) American requirements

      (3) International requirements

4. **Time Required**:

   a. EO – 3 x 60 min periods; and

   b. EC – 2 x 60 min periods.

5. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration - performance; and

c. Tutorial.

6. **Training Aids**:

a. Example Supply Control Account (SCA);

b. Reconfigurable PRC radios;

c. Laptop with PRC configuration software;

d. Connector cabling;

e. Example Crypto sleeve;

f. Training Crypto; and

g. MLPs.

7. **Test Details**: EO 006.02 will be tested in EC 006.02 and PC 006.

8. **Remarks**:

a. The trainee shall account for all PRCs on their ships, explain the accounting process, and configuration constraints for the PRCs. They shall configure the PRC radios with the associated software and load crypto onto the sleeve.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Apply PRC accounting procedures IAW ref.						
1. Adhering to accounting procedures a) Governing authorities b) Life cycling c) Ship's allocation	**30**				S1: Reading Assignment S2: Graphical Handout	
Configure ship's PRCs IAW ref.						
2. Configuring Ship's PRCs IAW references to include:	**15**	**15**	**60**		Equipment Lab	Instructor shall demonstrate

a) Programming frequencies; b) Installing crypto sleeves;						how to load and delete crypto from PRC. Each student shall run through the process at least once.
Adhere to DFSM international regulations IAW ref						
3. Adhering to DFSM international regulations to include: a) Canadian requirements b) Canadian Frequencies c) Operating constraints d) American requirements e) Authorized frequencies for US waters and ports f) DFSM restraints and constraints g) International requirements	**60**				S1: Lecture S2: DL S3: regulations included in Battle Lab and EM prediction software	

Enabling Check – EO 006.02 Maintain Ship's PRCs

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

2. EC 006.02 will be a practical assessment based on content from EO 006.02.
3. To pass this EC, the trainee must achieve 4/6 SAT.
4. Time required: 10 minutes per trainee (2x 60 min periods total).
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 006.02 will be a practical assessment based on content from EO 006.02.
2. During this practical assessment, you will reconfigure a ship's PRC, load crypto and then unload it. You must pass this EC to pass EO 006.02.
3. References will not be provided.
4. Pass mark is 4/6 SATs IAW ECL 006.02.
5. You will have 10 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The trainees shall be given a blank un-programmed PRC, told to program it to a specific configuration, then load a crypto sleeve and then to unload the crypto sleeve and return the radio to a blank state.

PART IV – ENABLING CHECK LIST

Item	EC 006.02 Maintain Ship's PRCs	SAT	UNSAT	Remarks
1. Program the PRC				
Connected PRC to laptop		☐	☐	
Selected configuration profile		☐	☐	
Loaded profile		☐	☐	
Returned PRC to blank state		☐	☐	**Last action of EC**
2. Crypto Sleeve				
Installed Crypto Sleeve		☐	☐	
Uninstalled Crypto Sleeve		☐	☐	

Score: **/**

The trainee must achieve 4/6 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 006.02 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 006.02 with the candidate.	**Candidate** I have been debriefed on the result of EC 006.02.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 006.03 Maintain Subnet Relay (SNR)

1. **Performance**: Maintain Subnet Relay (N0021).
2. **Conditions**:
   a. Given:
      (1) SNR Equipment;
      (2) SNR Architecture;
      (3) Message Traffic; and
      (4) References.
   b. Denied: Permission to transmit.
   c. Environment: Classroom, and trainer.
3. **Standard**: The trainee shall maintain Subnet Relay (N0021) IAW references by:
   a. Configuring Subnet Relay IAW references by:
      (1) Capabilities and limitations;
      (2) System architecture;
         (a) Internal; and
         (b) External;
      (3) SNR considerations;
   b. Troubleshoot Subnet Relay IAW references by:
      (1) Executing troubleshooting flowchart;
      (2) Executing operating instructions;
      (3) Conferring with operating and troubleshooting authorities.
4. **Time Required**:
   a. EO – 6 x 60 min periods;
   b. EC – 2 x 60 min periods; and
5. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration – Performance; and

c. Simulation.

6. **Training Aids**:

a. SNR Theory Video;

b. DMOC SNR simulation;

c. SNR Trainer; and

d. MLPs

7. **Test Details**: EO 006.03 will be tested in EC 006.03 and PC 006.

8. **Remarks**:

a. The trainee shall have an understanding of how SNR equipment ties into the ship's RF systems and networks. They shall understand who the OA, TA, and how it is governed by the taskgroup, as well as who to contact in the event of issues beyond the scope of Ship's operators and technicians.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Configure Subnet Relay IAW ref.						
1. Configure Subnet Relay IAW references by: a) Capabilities and limitations; b) System architecture; i. Internal; ii. External; c) SNR considerations;	**120**	**30**	**60**		S1: interactive lecture on system S2: EM footprint software S3: SNR Trainer	SNR equipment is mission fit, and special liaison between Fleet and NFS will be required to get hands on the equipment
Troubleshoot Subnet Relay IAW ref.						
2. Troubleshoot Subnet Relay (TASK) IAW references by: a) Troubleshooting Flowchart;	**60**	**30**	**60**		S1: Interactive lecture on OA, technical aspects, and how to TS	SNR equipment is mission fit, and special liaison

b) Operating Instructions; c) Operating and Troubleshooting authorities.					S2: improved information as equipment made more available S3: SNR trainer	between Fleet and NFS will be required to get hands on the equipment

Enabling Check – EO 006.03 Maintain Subnet Relay

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Ship.

Arrangements and Procedures

2. EC 006.03 will be a practical assessment based on content from EO 006.03.
3. To pass this EC, the trainee must achieve 3/4 SATs.
4. Time required: 15 minutes per trainee (3x 60 minute periods total).
5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 006.03 will be a practical assessment based on content from EO 006.03.
2. During this practical assessment, you will maintain Subnet Relay. You must pass this EC to pass EO 006.03.
3. References will be provided.
4. Pass mark is 3/4 SATs IAW ECL 006.03.
5. You will have 15 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Each trainee shall flashup, configure and ensure the passing of data using the SNR system. HMC Ship may be required until SNR Equipment is installed at Campus for training.

PART IV – ENABLING CHECK LIST

Item	EC 006.03 Maintain Subnet Relay	SAT	UNSAT	Remarks
1. Subnet Relay				
Flashup		☐	☐	
Configuration		☐	☐	
Verification working		☐	☐	
Confirmed passing of data		☐	☐	

Score: **/**

The trainee must achieve 3/4 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 006.03 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 006.03 with the candidate.	**Candidate** I have been debriefed on the result of EC 006.03.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 006.04 Manage PRC 117 Equipment

1. **Performance**: Manage PRC 117 Equipment (N0022).

2. **Conditions**:

    a. Given: IAW PRC 117 QSP.

    b. Denied: IAW PRC 117 QSP.

    c. Environment: IAW PRC 117 QSP.

3. **Standard**: The trainee shall complete the PRC 117 course IAW the PRC 117 QSP.

4. **Time Required**:

    a. EO – 7 x 60 min periods;

    b. EC – 3 x 60 min periods IAW the PRC 117 PCL

5. **Instructional Methods**: IAW PRC 117 QSP

6. **Training Aids**:

    a. IAW PRC 117 QSP.

7. **Test Details**: EO 006.04 will be tested IAW the PRC 117 QSP.

8. **Remarks**:

    a. The trainees shall complete the PRC 117 course IAW the PRC 117 QSP and MLP. Campus scheduling shall schedule a PRC 117 course to match the times required in the CISO QSP. In the event that the PRC 117 course is unavailable, material on the initialization, configuration, and troubleshooting of the PRC 117 shall be used in consort with the trainer or on an HMC Ship to instruct the trainees on the configuration, use, and troubleshooting of the PRC 117. In this case, PCL 010 shall be used to assess students.

9. **Teaching Points & EC**: IAW the PRC 117 QSP.

Performance Check – PO 006 Manage LF/MF/HF/VHF/UHF Communications Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Ship.

Arrangements and Procedures

2. PC 006 will be a practical assessment based on content from PO 006.

3. To pass this PC, the trainee must achieve 5/7 SATs.

4. Time required: 1 x 60 min period(s) per student with a maximum of 8 periods.

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 006 will be a practical assessment based on content from PO 006.

2. During this practical assessment, you will be required to advise command on LF/MF/HF/VHF/UHF communications equipment. You must pass this PC to pass PO 006.

3. References will be provided.

4. Pass mark is 5/7 SATs IAW PCL 006.

5. You will have 45 minutes (8 x periods total) per student to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The CISO student will initialize and configure communications equipment in order to establish communications IAW a COMPLAN. The student will troubleshoot faults in the communications equipment and solicit assistance from the proper technicians.

IAW threaded scenario the CISO student is posted to a ship transiting to an exercise and has been tasked to test the COMPLAN.

PART IV – PERFORMANCE CHECK LIST

Item	PC 006 Manage LF/MF/HF/VHF/UHF Communications Equipment	MET	NOT MET	Remarks
1. COMPLAN				
Interpreted COMPLAN		☐	☐	
Input COMPLAN into CCMS or Black Patch Panel		☐	☐	
Printed B-Switch		☐	☐	
2. Fault Finding				
Identify Fault in system		☐	☐	
Determine equipment at fault		☐	☐	
Determine patch		☐	☐	
Apply Black Patch to system		☐	☐	

Score: /

The trainee must achieve 5/7 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 006 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 006 with the candidate.		**Candidate** I have been debriefed on the result of PC 006.	
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 007 Manage Satellite Communications Equipment

EO 007.01 Coordinate Naval SATCOM

1. **Performance**: Coordinate Naval SATCOM (TASK).

2. **Conditions**:

   a. Given:

      (1) Satellite Communications Equipment;

      (2) Configuration Software;

      (3) Message Traffic; and

      (4) References.

   b. Denied: Nil.

   c. Environment: Classroom, trainer.

3. **Standard**: The trainee shall coordinate naval SATCOM (N0043) IAW references by:

   a. Explaining Naval SATCOM considerations (TASK) IAW references to include:

      (1) Space weather;

      (2) Terran weather;

      (3) Ground stations;

      (4) Network topography;

      (5) Latency;

   b. Explaining communications satellites IAW references to include:

      (1) Payloads;

      (2) Orbits;

      (3) Inter-satellite communications

      (4) Hardening techniques;

   c. Explaining satellite constellations IAW references to include:

      (1) Military SATCOM (AUSCANZUKUS, NATO)

      (2) Commercial SATCOM

(3) Canadian constellations

(4) Global Positioning Systems (GPS)

d. Explaining geographical considerations for SATCOM IAW references to include:

(1) RF Coverage types and patterns

(2) Look angles

(3) Wooding

(4) Arctic considerations

4. **Time Required**:

a. EO – 8 x 60 min periods;

b. EC – 1 x 60 min periods; and

c. PC – 2 x 60 min periods.

5. **Instructional Methods**:

a. Interactive lecture; and

b. Self-study;

6. **Training Aids**:

a. CIBs;

b. Messages for SATCOM Allocation, Requests, and Reports;

c. NMT Trainer;

d. INMARAT Fleet Broad Band Sailor 500 Equipment;

e. Harris Terminal Equipment; and

f. MLPs.

7. **Test Details**: EO 007.01 will be tested in EC 007.01 and PC 007.

8. **Remarks**:

a. Nil.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Explaining Naval SATCOM considerations IAW ref.						
1. Explaining Naval SATCOM considerations: a) Space weather; i. SUNSPOTS ii. Ionosphere iii. Debris iv. Solar Flares b) Terran weather; i. In relation to the effects on a SGT c) SATCOM Antennas: i. types ii. Location d) Ground stations; i. Locations ii. Considerations e) Network topography; f) Latency;	**120**				S1: interactive lecture S2: DL S3: Space weather software	
Explaining communications satellites IAW ref.						
2. Explaining communications satellites: a) Payloads; i. Primary vs. secondary b) Orbits; i. LEO ii. MEO iii. GEO iv. GSO v. HEO vi. Molinya vii. Arctic viii. Sun Synchronous c) Inter-satellite communications i. Satellite links	**120**				S1: interactive lecture S2: DL S3: DL combined with simulations	

d) Hardening techniques; i. Spot beams, ii. Shielding iii. Crypto iv. Spread spectrum						
Explaining satellite constellations IAW ref						
3. Explaining satellite constellations: a) Military SATCOM (AUSCANZUKUS, NATO) i. AEHF ii. WGS iii. UFO III (DAMA) iv. Skynet b) Commercial SATCOM i. INNWOAT ii. Iridium iii. Others c) Canadian constellations i. Sapphire ii. RadarSAT d) Global Positioning Systems (GPS)	**120**				S1: interactive lecture S2: DL S3: DL combined with simulations	
Explaining geographical considerations for SATCOM IAW ref:						
4. Explaining geographical considerations for SATCOM: a) RF Coverage types and patterns i. Spot beams ii. Broadcast iii. Radiation footprints b) Look angles i. Complications ii. Latitude issues c) Wooding i. Own ship ii. Berths iii. Fjords d) Arctic considerations	**120**				S1: interactive lecture S2: DL S3: DL combined with simulations	

i. Coverage ii. Look angle iii. Wooding						

Enabling Check – EO 007.01 Coordinate Naval SATCOM

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 007.01 will be a practical assessment based on content from EO 007.01.
3. To pass this EC, the trainee must achieve 6/9 SATs.
4. Time required: 1 x 60 min period(s).
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 007.01 will be a practical assessment based on content from EO 007.01.
2. During this practical assessment, you will be required to coordinate naval SATCOM. You must pass this EC to pass EO 007.01.
3. References will not be provided.
4. Pass mark is 6/9 SATs IAW ECL 007.01.
5. You will have 1 x 60 min period(s) to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Group question and answer session where the instructor will use a pre-determined question bank with questions based on the characteristics of each SATCOM system with the expectation that the trainees will be able to answer the following IAW the ECL. System, Equipment Specifications, Modem & its Bands, Terminal characteristics, Constellations the system can connect to, Uplink/downlink bands. Weather/space/radiation considerations.

Questions will be asked in a round robin fashion. If they trainee fails to answer the same question passes to the next. Once answered, the next question is asked. If a trainee is consistently failing to answer questions they will be increasingly relied upon to answer questions on the first try.

For a course of 12, it is suggested that the EC be broken into two sessions.

PART IV – ENABLING CHECK LIST

Item	EC 007.01 Coordinate Naval SATCOM	SAT	UNSAT	Remarks
1. System Information				
NMT, Harris Terminal, FBB Sailor 500		☐	☐	
Equipment/terminal specifications		☐	☐	
Modem Characteristics (bands)		☐	☐	
Constellation		☐	☐	
Connection limitations		☐	☐	
Uplink and Downlink Bands		☐	☐	
2. Weather and Space Considerations				
Orbit Characteristics (LEO, MEO, GEO, GSO, Sun Synch)		☐	☐	
Precipitation considerations		☐	☐	
Rear-link/SGT/terrestrial considerations		☐	☐	

Score: ***/***

The trainee must achieve 6/9 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 007.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 007.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 007.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 007.02 Maintain Satellite Communications

1. **Performance**: Maintain satellite communications (N0024).

2. **Conditions**:

   a. Given:

      (1) Satellite Communications Equipment;

      (2) Configuration Software;

      (3) Message Traffic; and

      (4) References.

   b. Denied: Nil.

   c. Environment: Classroom, trainer.

3. **Standard**: The trainee shall maintain satellite communications (N0024) IAW references by:

   a. Adhering to SATCOM access requirements IAW references by:

      (1) Explaining National vs Elemental SATCOM assets

      (2) Identifying SATCOM priorities

      (3) Extracting information from SATCOM Availability and Allocation messages

      (4) Reporting requirements based on constellation

      (5) Reporting PIM requirements

      (6) Drafting Post Deployment Reports (PDRs) for SATCOM

      (7) Reviewing SATCOM Communication Information Bulletins (CIBs)

   b. Initializing SATCOM equipment IAW references by:

      (1) Initializing the Navy Multiband Terminal

      (2) Initializing the Fleet Broad Band Sailor 500

      (3) Initializing DAMA equipment

   c. Configuring SATCOM equipment IAW references by:

      (1) Configuring the Navy Multiband Terminal

(a) Loading Crypto into the NMT

(b) Pairing the NMT

(2) Configuring the Fleet Broad Band Sailor 500

(3) Configuring DAMA equipment

(a) Loading Crypto into the DAMA

d. Troubleshooting SATCOM equipment IAW references by:

(1) Troubleshooting the Navy Multiband Terminal

(2) Troubleshooting the Fleet Broad Band Sailor 500

(3) Troubleshooting DAMA equipment

4. **Time Required**:

a. EO – 20 x 60 min periods; and

b. EC – 6 x 60 min periods;

5. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration - performance; and

c. Tutorial;

6. **Training Aids**:

a. Navy Multiband Trainer;

b. DAMA Trainer (DAMA in a box);

c. Satellite Access Allocation message;

d. WGS/AEHF Reporting Messages;

e. SATCOM Post Deployment Report;

f. NMT Configuration SOP;

g. FBB Configuration SOP;

h. SATCOM CIBs; and

i. MPLs.

7. **Test Details**: EO 007.02 will be tested in EC 007.02 and PC 007.

8. **Remarks**:

a. When Initializing, Configuring and troubleshooting each system they should be scheduled to allow the instructor to teach one system at a time.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Adhere to SATCOM access requirements IAW ref.						
1. Adhering to SATCOM access requirements: a) Explaining National vs Elemental SATCOM assets b) Identifying SATCOM priorities c) Extracting information from SATCOM Availability and Allocation messages d) Reporting requirements based on constellation e) PIM requirements f) PDRs for SATCOM g) SATCOM CIBs	**90**	**30**			S1: Assigned readings and in-class discussion S2: Nil. S3: IAW SpoC, SOC and NCS training.	Assign students the readings (CIBs) and message requirements as dictated by NCS, NOC, and Constellation owners. 90min for readings
Initializing SATCOM equipment IAW references:						
2. Initializing the Navy Multiband Terminal	**30**	**30**	**180**		S1: NMT SOP S2: NMT Course S3: Virtual SATCOM trainer	Lesson IAW the NMT Couse
3. Initializing the Fleet Broad Band Sailor 500 a) Procedure b) SOP	**15**	**15**	**30**		S1: SOPs S2: DL	

c) Checklist					S3: Nil (expect to be replaced)	
4. Initializing DAMA equipment a) Procedure b) SOP c) Checklist	**15**	**15**	**30**		S1: SOPs S2: DL S3: Virtual SATCOM trainer	
Configure SATCOM equipment IAW references:						
5. Configuring the Navy Multiband Terminal a) Loading Crypto into the NMT b) NMT Endorsement process	**30**	**30**	**180**		S1: NMT SOP S2: NMT Course S3: Virtual SATCOM trainer	Lesson IAW the NMT Couse
6. Configuring the Fleet Broad Band Sailor 500		**30**	**30**		S1: SOPs S2: DL S3: Nil (expect to be replaced)	Intro Material included in TP3
7. Configuring DAMA equipment a) Loading Crypto into the DAMA		**30**	**30**		S1: SOPs S2: DL S3: Virtual SATCOM trainer	Intro material Included in TP4
Troubleshooting SATCOM equipment IAW references by:						
8. Troubleshooting the Navy Multiband Terminal a) Drawing NMT system block diagrams	**30**	**30**	**120**		S1: NMT SOP S2: NMT Course S3: Virtual SATCOM trainer	Lesson IAW the NMT Couse
9. Troubleshooting the Fleet Broad Band Sailor 500 a) Drawing FBB system block diagrams		**30**	**30**		S1: SOPs S2: DL S3: Nil (expect to be replaced)	Intro Material included in TP3
10. Troubleshooting DAMA equipment		**30**	**30**		S1: SOPs S2: DL S3: Virtual SATCOM trainer	Intro material Included in TP4

a) Drawing DAMA system block diagrams						

Enabling Check – EO 007.02 Maintain Satellite Communications

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Ship.

Arrangements and Procedures

2. EC 007.02 will be a practical assessment based on content from EO 007.02.
3. To pass this EC, the trainee must configure the NMT SATCOM system IAW the NMT Delta Course PCL
4. Time required: IAW the NMT Course PC.
5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 007.02 will be a practical assessment based on content from EO 007.02 and the NMT Delta Course PCL.
2. During this practical assessment, you will be required to configure the NMT satellite communications. You must pass this EC to pass EO 007.02.
3. References will be provided.
4. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The EC for EO 007.02 is the PC for the NMT Course. Trainees shall initialize, setup, configure and trial a connection on the NMT. Scheduling for this EC shall include a connection to WGS or AEHF to properly connect and pass data from the NMT Trainer, or a Ship Fitted with NMT equipment.

Performance Check – PO 007 Manage SATCOM Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. PC 007 will be a practical assessment based on content from PO 007.
3. To pass this PC, the trainee must achieve 5/7 SATs.
4. Time required: 2 x 60 min period(s).
5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 007 will be a practical assessment based on content from PO 007.
2. During this practical assessment, you will be required to manage SATCOM equipment. You must pass this PC to pass PO 007.
3. References will be provided.
4. Pass mark is 5/7 SATs IAW PCL 007.
5. You will have 2 x 60 min period(s) to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Part 1: The CISO student will troubleshoot SATCOM systems IOT maintain satellite communications for the given scenario. A lease issue has arisen in a shared lease between the students' ship and an MCDV consort (MCDV with STSCU and FBB)

Part 2: The CISO student will solve the PIM issue from the given scenario and determine what the reporting requirements are for your SATCOM connection. The Navigator and Bridge Watch Keepers have not been reporting the PIM properly to the NAV COMMs and your ship has lost SATCOM connection for WGS/AEHF. Consequently your section has stopped reporting with the required messages.

PART IV – PERFORMANCE CHECK LIST

Item	PC 007 Manage SATCOM equipment	MET	NOT MET	Remarks
1. Lease Issue				
Identify lease restrictions		☐	☐	
Explain process to request new AAG		☐	☐	
Explain Authority/permission to change AAG		☐	☐	
2. PIM Requirements for NMT Connection				
WGS vs AEHF PIM requirements		☐	☐	
Explain expectations of PIM reporting		☐	☐	
Identified where in PIM reporting process the issue arose		☐	☐	
Proposed resolution was IAW reporting ref.		☐	☐	

Score: /

The trainee must achieve 5/7 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 007 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 007 with the candidate.		**Candidate** I have been debriefed on the result of PC 007.	
PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 008 Manage Cryptographic Equipment

EO 008.01 Maintain Cryptographic Equipment

1. **Performance**: Maintain cryptographic equipment (N0025).

2. **Conditions**:

   a. Given:

      (1) DWAN Access;

      (2) Classified Network Access;

      (3) COMSEC Accounting System;

      (4) Secure Storage;

      (5) Cryptographic Equipment; and

      (6) References.

   b. Denied: Nil.

   c. Environment: Classroom, trainer.

3. **Standard**: The trainee shall maintain cryptographic equipment (N0026) IAW references by:

   a. Initialize cryptographic equipment IAW references to include:

      (1) Explaining capabilities and settings

      (2) Using cryptographic operating procedures

   b. Configure cryptographic equipment IAW references to include:

      (1) Conducting cryptographic loading procedures

      (2) Conducting over the air distribution

   c. Troubleshoot cryptographic equipment IAW references to include:

      (1) Conducting cryptographic troubleshooting procedures

      (2) Identifying Technical Authorities for Equipment

   d. Manage COMSEC Equipment IAW refs by:

      (1) Identify Controlled Cryptographic Items (CCI)

(2) Identify Cryptographic Equipment

(3) Identify NAVSETs

(4) Request Increase in Entitlements

(5) Submit COMSEC Material Request

(a) Timelines for requests

(6) Draft Equipment Failure Report Process (EFR)

(7) Coordinate the courier process

4. **Time Required**:

a. EO – 18 x 60 min periods;

b. EC – 1 x 60 min periods; and

c. PC – 8 x 60 min periods.

5. **Instructional Methods**:

a. Interactive lecture;

b. Demonstration - performance; and

c. Tutorial.

6. **Training Aids**:

a. Training Crypto;

b. MLPs;

c. Checklists;

d. Canadian Cryptographic Doctrine (CCD);

e. Equipment SOPs; and

f. Aide Memoirs.

7. **Test Details**: EO 008.01 will be tested in EC 008.01 and PC 008.

8. **Remarks**:

a. This EO is designed to supplement the COMSEC Course where information that is critical to being a COMSEC Custodian is lacking. It is not designed to replace the COMSEC Course.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Initialize cryptographic equipment (Task number) IAW ref.						
1. Initializing/shutting down cryptographic equipment to include: a) Capabilities and settings for: i. KY57/58; i. Purpose ii. KEYMAT ii. KG 84C; i. Purpose ii. KEYMAT iii. KWR 46; i. Purpose ii. Baud rate iii. KEYMAT iv. KYV-5/ANDVT i. Purpose ii. Baud rate iii. KEYMAT v. KIV 77/78 i. Purpose ii. Settings iii. Keymat b) Transfer and storage devices: i. KOI 18 ii. KYK 13 iii. KYX 15 iv. SKL c) Cryptographic operating procedures to include: i. KY 57/58; ii. KG 84C; iii. KWR 46; iv. KYV-5/ANDVT; v. KIV 77/78	**120**	**60**	**120**		S1: Crypto Equipment Lab and Lecture S2: No Change S3: Virtual Crypto Equipment Trainer	

vi. HARPOON d) AHWCS GPS vii. Initialization viii. Loading Keymat						
Configure cryptographic equipment (TASK)						
2. Crypto loading procedures to include: a) KY 57/58; b) KG 84C; c) KWR 46; d) KYV-5/ANDVT e) HARPOON	**30**	**30**	**120**		S1: Crypto Equipment Lab and Lecture S2: No Change S3: Virtual Crypto Equipment Trainer	
3. Conducting over the air distribution (OTAD) procedures: a) Definitions; i. OTAD ii. OTAR iii. OTAT iv. TEK v. KEK vi. TRKEK vii. Octal numbers viii. KW 46 Community variable (CV)	**15**	**15**	**60**		S1: Crypto Equipment Lab and Lecture S2: No Change S3: Virtual Crypto Equipment Trainer	
Troubleshoot cryptographic equipment						
4. Crypto Troubleshooting Procedures a) KY57/58; b) KG 84C; c) KWR 46; d) KYV-5/ANDVT; e) KIV 77/78; f) KOI 18; g) KYK 13; h) KYX 15; i) HARPOON	**15**	**15**	**60**		S1: Crypto Equipment Lab and Lecture S2: No Change S3: Virtual Crypto Equipment Trainer	
Manage COMSEC equipment						
5. Technical Authorities for Equipment a) FCA b) CFCSS	**60**				S1: Lecture S2: DL S3: Not Required	Purpose of this TP is to provide Trainees with

c) Help Desk d) LCMMs e) CSE (Lvl III)						contact information for these organizations and what they can help a COMSEC Custodian with.
6. Maintenance Requirements: a) Rainbow Tags b) Returning Equipment c) Cryptographic Material Requests d) Transfers between Accounts e)	**60**				S1: Tags, SOPs, and lecture S2: DL S3: nil.	
7. Secure Telephone a) STE i. initialization ii. Operating procedures iii. Troubleshooting b) vIPer i. initialization ii. Operating procedures iii. Troubleshooting	**15**	**30**	**60**		S1: Manual S2: Equipment in Crypto Lab S3: Virtual Crypto Equipment Trainer	
8. NMT AEHF Endorsement Process c) Liaison Considerations d) SOC's role e) N6's role f) Changing KGV-136R faceplates	**60**	**30**	**120**	PMSC NMT Job Aides	S1: NMT Trainer or HMC Ship S2: Nil. S3: Virtual SATCOM Trainer	

Enabling Check – EO 008.01 Maintain Cryptographic Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 008.01 will be a practical assessment.

3. To pass this EC, the trainee must achieve 5/7 SATs.

4. Time required: 1 x 60 minute period.

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 008.01 will be a practical assessment based on content from EO 008.01.

2. During this practical assessment, you will be required to maintain cryptographic equipment. You must pass this EC to pass EO 008.01.

3. References will be provided.

4. Pass mark is 5/7 SATs IAW ECL 008.01.

5. You will have 1 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Trainee is given a “broken” piece of COMSEC equipment with a sticker or piece of paper explaining what is wrong. They must then fill out a Crypto Material Request, Rainbow Tag, and any other required paperwork IAW Infosec 2 to return the piece of equipment to CFCSS. They must then explain the requirements to get the equipment to the CFCSS, and any special considerations. (Two Person Integrity, courier certificate, etc.)

PART IV – ENABLING CHECK LIST

Item	EC 008.01 Maintain Cryptographic Equipment	SAT	UNSAT	Remarks
1. Rainbow Tag				
Fields completed		☐	☐	
Tagged to Equipment		☐	☐	
Signed		☐	☐	**Yes or No**
2. Crypto Material Request				
Completed		☐	☐	
Description matches equipment issue		☐	☐	
3. Special Considerations				
Were the special considerations accurate?		☐	☐	
Were they IAW NDSOD/INFOSEC?		☐	☐	**<CRITICAL>**

Score: **/**

Trainee must achieve 5/7 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 008.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 008.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 008.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 008.02 Configure KIV-7M

1. **Performance**: Configure KIV-7M

2. **Conditions**: IAW KIV-7M QSP (AJKP)

3. **Standard**: The trainee shall complete the KIV-7 course, IAW KIV-7M QSP (AJKP)

4. **Time Required**: IAW KIV-7M QSP (AJKP)

   a. EO – 32 x 60 min periods;

   b. EC – 5 x 60 min periods; and

   c. PC – 4 x 60 min periods.

5. **Instructional Methods**: IAW KIV-7M QSP (AJKP)

6. **Training Aids**:

   a. IAW KIV-7M (AJKP) MLP

7. **Test Details**: EO 008.02 will be tested IAW KIV7-M QSP (AJKP).

8. **Remarks**:

   a. Schedule this part of the course IAW the TKIV-7M Admin QSP however the PC for the course will be scheduled as a part of PC 008

9. **Teaching Points**: IAW KIV-7M (AJKP) MLP.

EO 008.03 Configure TACLANE (KG-175D)

1. **Performance**: Configure TACLANE (KG-175D) (AIVC)

2. **Conditions**:

   a. Given: IAW TACLANE QSP

   b. Denied: IAW TACLANE QSP

   c. Environment: IAW TACLANE QSP

3. **Standard**: The trainee shall complete the TACLANE (KG-175D) course, IAW KG-175D (AIVC).

4. **Time Required**:

   a. EO – 28 x 60 min periods;

   b. EC – 4 x 60 min periods; and

   c. PC – 4 x 60 min periods.

5. **Instructional Methods**: IAW TACLANE QSP

6. **Training Aids**:

   a. IAW TACLANE MLP

7. **Test Details**: EO 008.03 will be tested IAW TACLANE QSP.

8. **Remarks**:

   a. Schedule this part of the course IAW the TACLANE Admin QSP however the PC for the course will be scheduled as a part of PC 008

9. **Teaching Points**: IAW KIV-7M (AIVC) QSP.

Performance Check – PO 008 Manage Cryptographic Equipment

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer.

Arrangements and Procedures

2. PC 008 will be a practical assessment based on content from PO 008.

3. To pass this PC, the trainee must complete the KIV-7M and TACLANE Administrator Courses.

4. Time required: 8 x 60 minute period(s).

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 008 will be a practical assessment based on content from PO 008.

2. During this practical assessment, you will be required to manage cryptographic equipment. You must pass this PC to pass PO 008.

3. References will be provided.

4. Pass mark is IAW the KIV-7M and TACLANE PCLs .

5. You will have 4 x 60 min period(s) per PC (4 for KIV-7M, and 4 for TACLANE) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

PART IV – PERFORMANCE CHECK LIST

Item	PC 008 Manage Cryptographic Equipment	MET	NOT MET	Remarks
1. KIV-7M Administrator Course				
Met the standard IAW course PCL		☐	☐	
2. TACLANE Administrator Course				
Met the standard IAW course PCL		☐	☐	

Score: **/**

The trainee must achieve 2/2 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 008 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 008 with the candidate.	**Candidate** I have been debriefed on the result of PC 008.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 009 Manage Network Operations

EO 009.01 Managing Ship's IS/IT Configuration

1. **Performance**: Manage ship's Information Systems and Information Technology configuration (B0470).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) NavIS Hardware;

      (3) Telephone - landline; and

      (4) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall manage Ship's IS/IT configuration (N0031) IAW <references> by;

   a. Maintaining ship's network architecture to include:

      (1) Computer Hardware;
      (2) Storage Devices and configurations;
      (3) Network Architecture;
      (4) Network Operations;
      (5) Network Security; and
      (6) Troubleshooting.

   b. Facilitating Changes to network architecture to include

      (1) Network Topologies;
      (2) Industry Standards;
      (3) WAN Theory
      (4) HSDC Architecture

4. **Time Required**:

   a. EO – 25 x 60 min periods;

   b. EC – 7 x 60 min periods; and

   c. PC – 4 x 60 min periods.

5. **Instructional Methods**:

   a. Self-study;

   b. Self-paced CBT (asynchronous); and

   c. Video Lectures.

6. **Training Aids**:

   a. CompTIA A+ Certification Video Course (YouTube Video);

   b. Professor Messer's CompTIA N10-006 Network + Course (YouTube Video);

   c. TP 2-6 Alternate Reference: CompTIA Network + Certification Video (YouTube Video);

   d. HSDC Architecture Network Topology Diagram;

   e. Data Flow Diagram (DWAN) to NOC and into INTRANET; and

   f. MLPs.

7. **Test Details**: EO 009.01 will be tested in EC 009.01 and PC 009.

8. **Remarks**:

   a. The videos are the primary source for instruction in this EO. Should the Instructor not be able to obtain a computer lab with GP Net access to enable YouTube access, then work around solutions shall be entertained, such as students given time at home for the duration of these TPs to watch the videos, or coordinating the students bringing a laptop, smart phone, or tablet to an internet café for the duration of the TP to enable YouTube access. Alternatively, the instructor may copy the videos from youtube and provide the students with a hard copy CD (Unclassified) for use in their issued DVPNI laptop.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Threaded Scenario Brief, IAW Course Scenario						
1) CJOC Frag Order a. TOCA to CJOC for Named Operation IAW Course Threaded Scenario b. MCC c. Mission Frag Order	**60**			Course Threaded Scenario Pack	Lecture/Brief on Situation, TOCA, and Named Mission	Instructor to relay to students that the Exercise stage of the course is over. All assessments

d. New reporting responsibilities e. Mission Orders f. All other portions of the Scenario Package						now pertain to the named CJOC Mission
Maintaining ship's network architecture IAW ref to include:						
2) A+ Standard a. Storage Devices b. Power Supply c. Motherboard Form Factor d. Motherboard I/O Interfaces e. Adapter/Expansion Cards f. RAM Slots g. Printers h. Cooling i. RAM j. CPU, Socket k. Chipset l. Bus Slots m. PATA Interface n. Networking Cables & Connectors o. TCP/IP Characteristics p. Well-known Ports q. TCP/IP Protocol Suites r. Wireless Networking s. Router Configuration t. Internet Access technologies u. Types of Networks v. Network Components w. Input Devices & Peripherals x. Monitors y. BIOS/CMOS z. Windows OS aa. Windows features, ISSUES, tools & installation bb. RAID	**210**	**15**	**30**	CompTIA A+ Certification Video Course (YouTube)	S1: Online Video Lecture S2: Video Hosted on DLN S3: Computer Hardware Lab for Hands on Practical	Classroom with GP Net Capability must be scheduled for this TP. YouTube access is required.
Facilitating Changes to network architecture IAW ref to include:						
3) Network Architecture a. Network Devices b. Remote Access c. Network Services	**340**			Professor Messer's CompTIA N10-006	S1: Online Video Lecture	Classroom with GP Net Capability must be

d. WAN Technology e. Cable Types f. Topologies g. Infrastructure h. Addressing i. Routing j. Cloud and Virtual Technologies k. Implementing Networks				Network + Course Section 1 (Youtube Video)	S2: Video Hosted on DLN S3: Computer Network Lab for Hands on Practical	scheduled for this TP. YouTube access is required.
4) Network Operations a. Monitoring The Network b. Metrics and Reporting c. Configuration Management d. Network Segmentation e. Patches and Updates f. Switch Configuration g. Wireless Networks	**130**			Professor Messer's CompTIA N10-006 Network + Course Section 2 (Youtube Video)	S1: Online Video Lecture S2: Video Hosted on DLN S3: Computer Network Lab for Hands on Practical	Classroom with GP Net Capability must be scheduled for this TP. YouTube access is required.
5) Network Security a. Risk b. Vulnerabilities and Threats c. Network Hardening d. Physical Security Controls e. Firewalls f. Network Access Control g. Forensics	**180**			Professor Messer's CompTIA N10-006 Network + Course Section 3 (Youtube Video)	S1: Online Video Lecture S2: Video Hosted on DLN S3: Computer Network Lab for Hands on Practical	Classroom with GP Net Capability must be scheduled for this TP. YouTube access is required.
6) Troubleshooting a. Methodology b. Tools c. Wireless Issues d. Copper Cable Issues e. Fiber Issues f. Network Issues g. Security Issues h. WAN Issues	**135**			Professor Messer's CompTIA N10-006 Network + Course Section 4 (Youtube Video)	S1: Online Video Lecture S2: Video Hosted on DLN S3: Computer Network Lab for Hands on Practical	Classroom with GP Net Capability must be scheduled for this TP. YouTube access is required.
7) Industry Standards a. The OSI Model	**100**			Professor Messer's CompTIA	S1: Online Video Lecture	Classroom with GP Net Capability

b. Theory and Concepts c. Wireless Network Standards d. Wired Network Standards e. Policies and Procedures f. Safety g. Installation and Configuration h. Change Management i. Network Ports and Protocols j. Application Ports and Protocols				N10-006 Network + Course Section 5 (Youtube Video)	S2: Video Hosted on DLN S3: Computer Network Lab for Hands on Practical	must be scheduled for this TP. YouTube access is required.
8) WAN Theory: a. SUBNET Relay b. MTWAN: c. Definition; d. Capability; e. Concepts of operation: i. Principles ii. Operational view iii. Line of sight networking iv. Multi bearer routing f. Systems view	**60**					
9) HSDC Architecture: a. Theory: b. 24/7 Connectivity c. Systems/services supported hardware Ship-side Architecture Identity and Functions: d. Terminals e. Black switch box f. Servers g. Routers h. Patch panel i. Crypto patch panel j. Modem patch panel	**90**		**30**			

k. KIV-7/KG-84 l. Riverbed; m. KG-175 n. Network services o. VOIP Shore-side Architecture: p. Modems q. Routers r. Switches s. PBX exchanges t. Location Troubleshooting: u. Responsibility v. Tools						
10) SECLAN Configuration a. Host Application Servers b. Network Application Servers c. Thin Clients vs. Thick Clients d. Configuration and Topology e. Administrative requirements	**60**		**60**		S1: Demonstration of SECLAN configuration on HMC Ship D2: Configuration Documents for OA and TA S3: SECLAN Trainer	

Enabling Check – EO 009.01 Managing Ship's IS/IT Configuration

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 009.01 will be a practical assessment based on content from EO 009.01.

3. To pass this EC, the trainee must achieve 3/4 SATs or 75%.

4. Time required: 7 x 60 min period(s).

5. Open source references will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 009.01 will be a practical assessment based on content from EO 009.01.

2. During this practical assessment, you will be required to setup a local area network and pass data across it. You must pass this EC to pass EO 009.01.

3. References will not be provided, however open source research for troubleshooting purposes is encouraged.

4. Pass mark is 3/4 SATs IAW ECL 009.01.

5. You will have 7 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The instructor will provide each student with the required parts to assemble a computer tower that is capable of booting, having an operating system installed on it, connecting to a network, network hardware (switch, cabling), monitors, mouse, keyboards, and STARCRAFT II software.

The EC is 7 periods and the students are expected to assemble their tower, install the operating system, connect to the network, install STARCRAFT II, connect to the LAN in the game, and commence a player vs. player match to pass.

PART IV – ENABLING CHECK LIST

Item	EC 009.01 Manage Ship's IS/IT Configuration	SAT	UNSAT	Remarks
1.				
Assemble Computer Tower		☐	☐	Must be able to boot
Install Operating System		☐	☐	Windows XP or greater
Connect to LAN		☐	☐	**<CRITICAL>**
Install provided program and pass data across the network		☐	☐	STARCRAFT II

Score: /

Trainee must achieve 3 / 4 SATs **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 009.01 – Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of EC 009.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 009.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 009.02 Manage Network Performance and Administration

1. **Performance**: Manage Network Performance and Administration (N0032).

2. **Conditions**:

   a. Given:

      (1) Network Credentials;

      (2) DWAN enabled workstation;

      (3) Classified Network Access;

      (4) NavIS Hardware;

      (5) NavIS Software;

      (6) Network Management Tools; and

      (7) References.

   b. Denied: Nil.

   c. Environment: Classroom.

8. **Standard**: The trainee shall Brief Command on network performance (N0033), IAW <reference>, to include:

   a. Maintaining Information Security;

   b. Describe the function of network management authorities;

   c. Describe IS Administrative Requirements;

   d. Advising on Network Applications;

   e. Configuring Bandwidth Management Hardware; and

   f. Configuring Bandwidth Management Software.

3. **Time Required**:

   a. EO – 17 x 60 min periods;

   b. EC – 3 x 60 min periods.

4. **Instructional Methods**:

   a. Interactive lecture;

b. Demonstration - performance;

c. Study assignment;

d. Case study;

e. Tutorial;

f. Self-paced CBT (asynchronous); and

g. Video teleconference.

5. **Training Aids**:

a. MLPs;

b. Universal Classroom or GP Net Capable workspace;

c. IP Addresses for HALFAX Class Ship's DWAN Riverbed Steelheads;

d. Monitoring Passwords for HALFAX Class Ship's Riverbed Steelheads;

e. Monitoring Passwords for MARLANT and MARPAC NOC Cascade;

f. Riverbed Steelhead Lab #1 and Lab #2; and

g. Tour of NOC.

6. **Test Details**: EO 009.02 will be tested in EC 009.02 and PC 009.

7. **Remarks**:

a. Emphasis on ability to navigate the Riverbed Steelhead appliances to determine QoS Settings, HSDC Router Settings, and Bandwidth Management metrics shall be made.

8. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Interpret IS Security Orders IAW ref.						
1) Interpret the IS security orders, by reviewing: a. Unclassified networks; b. Classified networks; and c. Allied networks.	**120**				S1: Reading Assignment S2: DL S3: Nil.	Self-Study followed by Interactive Lecture

Describe the function of network management authorities IAW ref.						
2) To include: a. Defining the levels of responsibilities b. CFNOC c. NOC, d. BIS, e. F3 Ops f. Level 0 through L3 Troubleshooting Ticket elevation process.	**30**				S1: Interactive lecture S2: Nil. S3: Nil.	Instructor to provide in-depth coverage of TS escalation process.
Describe IS Administrative Requirements IAW ref						
3) Identify Shipboard networks, role, sponsoring and supporting agency: a. DWAN / ShipLAN b. GP Net c. Tactical Common Datalink (TCDL) d. CSNI / SECLAN e. NSWAN f. CENTRIXS enclaves g. CMFC h. CMFP i. MLEC j. GCTF k. CDMN l. ICI / Pegasus m. Stone Ghost n. PSSS	**60**				S1: Interactive Lecture S2: ADM(IM) reference on CAF networks S3: Diagram	Instructor should be able to explain the relationship between each network, what level of processing, and who the OA and TA are.
Identify CIS Application requirements by department and their shipboard uses IAW ref:						
4) Baseline software to include: a. Monitor Mass b. DRMIS c. FMAS d. CFHIS e. PeopleSoft f. Other software as appropriate	**30**	**60**			S1:Lecture Overview of requirements for each software S2: QL5B Content S3: Synergy between QL5B, CISO, and ADM(IM) information	Instructor to draw from ADM(IM) approved software list
Configuring Bandwidth Management Hardware IAW ref						

5) Replication: a. Overview b. Domino c. Web enabled database replication d. Replication architecture: i. Hub-spoke ii. Meshed iii. Federated hub-spoke iv. Scalability v. Survivability e. Replication process: i. What is replicated ii. Transaction logging iii. Replication cycle/time iv. Force replication v. Streaming replication vi. Network compression vii. Concept for employment	**120**				S1:Lecture S2: DL S3: Computer Lab with virtual tools, or physical hardware	
Configuring Bandwidth Management Software IAW ref.						
6) Network Bandwidth Quality of Service (QoS): a. Description of QoS: i. Predictability and consistency ii. Guarantee iii. Management iv. End-to-end v. Link throughput/bottleneck vi. Maximizing link throughput: vii. Latency viii. TCP congestion control.	**60**		**240**		S1:Lecture S2: Nil. S3: Computer Lab with virtual tools, or physical hardware	4 hours dedicated for Labs

ix. Controlling less urgent traffic x. Objective: xi. Visibility xii. Control xiii. Compression xiv. QoS solutions						
7) Network Bandwidth Requirements: a. Required Bandwidth for: i. DIN/Unclas services; ii. MTWANs; iii. VOIP; iv. VTC; and v. Other network services. b. Bandwidth management tools as installed per ship c. Bandwidth monitoring tools as installed per ship	**60**		**240**		S1:Lecture S2: DL S3: Computer Lab with virtual tools, or physical hardware	4 hours dedicated for Labs
8) Network Operations Center (NOC) Tour a. Servers Houses at NOC (what networks, why) b. Services Offered to HMC Ships c. How CIS Service Request messages are processed/actioned d. How the HSDC Router is set to the proper bandwidth allocation based on SATCOM allocation e. How the NOC managed QoS settings for ships.	**120**				S1: Tour S2: Tour S3: Tour	

Enabling Check – EO 009.02 Manage Network Performance and Administration

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 009.02 will be a practical assessment based on content from EO 009.02.
3. To pass this EC, the trainee must achieve 13/18 SATs.
4. Time required: 3 x 60 min period(s).
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 009.02 will be a practical assessment based on content from EO 009.02.
2. During this practical assessment, you will be required to identify CAF networks, provide the Operational Authority, and Technical Authority for each, and explain the troubleshooting process and what each level can TS. You must pass this EC to pass EO 009.02.
3. References will not be provided.
4. Pass mark is 13/18 SATs IAW ECL 009.02.
5. You will have 3 x 60 min period(s) to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Part 1 is to identify and explain Networks and their purposes. Part 2 is to identify the OA and TA of the networks Part 3 is to outline and explain the network troubleshooting process for DWAN. Part 4 is to provide directions on how to assess the QoS, and Bandwidth management settings in the Riverbed Steelhead appliance.

PART IV – ENABLING CHECK LIST

Item	EC 009.02 Manage Network Performance and Administration	SAT	UNSAT	Remarks
1. Identify and explain purpose of CAF Networks				
DWAN Explanation		☐	☐	
CSNI Explanation		☐	☐	
CDMN Explanation		☐	☐	
PSSS or Stoneghost (depending on exam)		☐	☐	
CMFC/CMFP (depending on exam)		☐	☐	
2. Operational and Technical Authorities for the following Networks:				
DWAN (OA and TA)		☐	☐	
CSNI (OA and TA)		☐	☐	
NSWAN (OA and TA)		☐	☐	
MLEC (OA and TA)		☐	☐	
3. Troubleshooting Process				
Level 0 – ISA, Capabilities, Escalation		☐	☐	
Level 1 – NOC, Capabilities, Escalation		☐	☐	
Level 2 – CFNOC/NETE, Caps, Escalation		☐	☐	**CFNOC, QETE or NETE depending on Network**
Level 2 – CFNOC/NETE, Caps, Escalation		☐	☐	**CFNOC, QETE or NETE depending on Network**
4. Bandwidth Management and QoS Assessment				
Identified QoS Settings		☐	☐	(screenshot)
Confirmed QoS Turned ON		☐	☐	(screenshot)
Determined top Three protocols in use		☐	☐	(screenshot)
Determined what percentage of traffic was Unclassified vs Classified		☐	☐	(screenshot)
Determined if Classified traffic was optimised or not		☐	☐	(screenshot)

Score: ***/***

Trainee must achieve 13/18 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 010.02 – Overall Assessment – Circle One

Item	EC 009.02 Manage Network Performance and Administration	SAT	UNSAT	Remarks

Met Standard		Did Not Meet Standard	
Assessor I have discussed the results of EC 010.02 with the candidate.		**Candidate** I have been debriefed on the result of EC 010.02.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 009 Manage Network Operations

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. PC 009 will be a practical assessment based on content from PO 009.

3. To pass this PC, the trainee must achieve 7/9 SATs.

4. Time required: 3 x 60 min period(s).

5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 009 will be a practical assessment based on content from PO 009.

2. During this practical assessment, you will be required to Manage Network Operations. You must pass this PC to pass PO 009.

3. References will not be provided.

4. Pass mark is 7/9 SATs IAW PCL 009.

5. You will have 3 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The CISO student will coordinate a plan for the installation of MTWAN pertinent to the scenario. This will include accounts, TACLANE, security considerations, bandwidth assessment for the network and a request to the NOC to reconfigure the HSDC router and Steelhead appliance. The student will brief the details of their plan.
IAW threaded scenario the CISO student is posted to a ship sailing in an exercise.

To be given at start of the PO after Warning Order brief. Ship has received warning order removing them from JW and tasking them to proceed to a deployed operation scenario (i.e. South China Sea, Korean Peninsula, Black Sea, Baltics, etc.) The ship will need to be fitted with new network equipment that was not installed for the exercise, but is required for deployment.

PART IV – PERFORMANCE CHECK LIST

Item	PC 009 Manage Network Operations	MET	NOT MET	Remarks
1. Assessment of Mission Networks				
Identify what Network is required for the mission		☐	☐	
Layout of the process to achieve installation of network on the ship		☐	☐	
Requirements for accounts and POC		☐	☐	
2. Paperwork to get Mission Network on-board				
PSS Requirements		☐	☐	
CA&A (standing?)		☐	☐	
TCI for network		☐	☐	
3. Proposed plan IAW Frag-Ord and CJOC Mission Orders				
Identification of Port for install		☐	☐	
Layout required work for Section to prepare for new network		☐	☐	
Considerations for un-installing other Mission Networks.		☐	☐	

Score: **/**

The trainee must achieve 7/9 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 009 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 009 with the candidate.	**Candidate** I have been debriefed on the result of PC 009.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 009.02 Manage Network Performance and Administration

1. **Performance**: EO 009.02 shall be taught and administered IAW the TDL 100 and TDL 200 Qualification document.

PO 010 Coordinate Defensive Cyber Operations – Internal Defensive Measures

EO 10.01 Brief Command on Cyber Impact to Operations

1. **Performance**: Brief Command on cyber impact to operations (N0035).

2. **Conditions**:

   g. Given:

      (1) DWAN Access;

      (2) Classified Network Access;

      (3) Cyber Threat Assessment;

      (4) References.

   h. Denied: Nil.

   i. Environment: Classroom.

3. **Standard**: The trainee shall brief Command on cyber impact to operations (N0035) IAW reference by:

   a. Identifying a cyber-attack IAW NATO Cyber DL to include:

      (1) Cyberspace definition

      (2) OCO, DCO, and Network Operations

      (3) areas of cyberspace

      (4) Attack types and characteristics

      (5) Vector types

      (6) Access types

   b. Interpreting a cyber threat assessment IAW references by:

      (1) Comparing the threat to equipment and networks

      (2) Comparing the threat to IS Security, NETSEC, TRANSEC, and EMSEC orders

(3) Comparing the threat to physical and personnel security

c. Developing a cyber security plan IAW references by:

(1) Identifying most likely and most dangerous attack vectors

(2) Determining the required precautions, settings, or increases to security

(3) Liaising with authorities on the solution

(4) Briefing Command on the cyber security plan

d. Implementing a cyber security plan IAW references by:

(1) Facilitating changes to equipment settings

(2) Briefing ship's personnel on cyber threat

4. **Time Required**:

a. EO – 12 x 60 min periods;

b. EC – 1 x 60 min periods; and

c. PC – 2 x 60 min periods.

5. **Instructional Methods**:

a. Interactive lecture;

b. Study assignment;

c. Case study; and

d. Video teleconference.

6. **Training Aids**:

a. MLPs;

b. Cyber Threat Brief based on training scenario;

c. NavIS Architecture diagrams;

d. Ship's Security Orders; and

e. List of port visits based on training scenario.

7. **Test Details**: EO 010.01 will be tested in EC 010.01 and PC 010.

8. **Remarks**:

a. EC 010.01 is a 5 minute presentation on a case study of a recent Cyberspace activity, attack/hack, or malware. The students should be assigned the topic for research at the start of this EO so that they may compare and apply the readings and discussions they will accomplish throughout the EO to their presentation for the EC.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Identifying a cyber-attack IAW NATO Cyber DL:						
1) Cyberspace definition a. OCO, DCO, and Network Operations b. areas of cyberspace c. Attack types and characteristics d. Vector types e. Access types	**180**				S1: NATO DL S2: CAF DL S3: CAF DL mixed with Computer Lab where software can be exploited by controlled hacks.	NATO DL Account Required to access Course
2) Cyber Attack Case Study a. Open source Case Study based on NATO DL content			**120**		S1: In-Class Case Study and Presentation S2: DL Discussion Group S3: Computer Lab with exploitable software to demonstrate hacks	Assign Students Case Study from recent news source where Cyber "attack", exploit, or organization used cyberspace to cause an effect
Interpreting a cyber threat assessment IAW ref:						
3) Interpreting cyber threat assessments a. Comparing the threat to equipment and networks b. Comparing the threat to IS	**30**		**120**		Students assigned recent Cyber Threat assessment from TRINITY	TRINITY Threat Assessment, Ship's Configuration, and NDSODs

Security, NETSEC, TRANSEC, and EMSEC orders c. Comparing the threat to physical and personnel security					and compare it as required.	required to accomplish.
Developing a cyber security plan IAW ref:						
4) Identifying most likely and most dangerous attack vectors a. Determining the required precautions, settings, or increases to security b. Liaising with authorities on the solution c. Briefing Command on the cyber security plan	**30**		**60**			
5) CFIOG/CFNOC SME a. CFNOC Responsibilities b. Incident Detection c. Incident Response d. Case Study	**120**				S1: Secure VTC S2: Local NOC SME S3: Cyber Operator SME	VTC with CFIOG, CFNOC SME on Cyber Defenses
Implementing a cyber security plan						
6) Facilitating changes to equipment settings a. Briefing ship's personnel on cyber threat	**30**		**30**			

Enabling Check – EO 010.01 Brief Command on Cyber Impact to Operations

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom/Ship.

Arrangements and Procedures

2. EC 010.01 will be a practical assessment based on content from EO 010.01.
3. To pass this EC, the trainee must achieve 4/6 SATs or %.
4. Time required: 5 minutes per trainee, 1 x 60 min period total.
5. References will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 010.01 will be a practical assessment based on content from EO 010.01.
2. During this practical assessment, you will be required to brief command on cyber impact to operations. You must pass this EC to pass EO 010.01.
3. References will not be provided.
4. Pass mark is 4/6 SATs IAW ECL 010.01.
5. You will have 5 minutes to complete this assessment.
6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The Trainee shall prepare a cyber security brief on the Cyber TRA for the port, as assigned in class and present it to the course in a 5 minute quick rundown of the threat to the ship and to the ship's company.

PART IV – ENABLING CHECK LIST

Item	EC 010.01 Brief Command on Cyber Impact to Operations	SAT	UNSAT	Remarks
1. TRINITY Threat Assessment				
Recognized Threat to ship		☐	☐	
Briefed Threat to ship		☐	☐	
Ordered precautionary measures to Ship's Company (DWAN, CSNI, etc. protections)		☐	☐	
2. Briefing to Ship's Company				
Recognized Threat to Personnel		☐	☐	
Briefed Threat to Personnel		☐	☐	
Suggested precautionary measures		☐	☐	

Score: **/**

Trainee must achieve 4/6 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 010.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 010.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 010.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 010 Coordinate Defensive Cyber Operations – Internal Defensive Measures

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom

Arrangements and Procedures

2. PC 010 will be a practical assessment based on content from PO 010.
3. To pass this PC, the trainee must achieve 6/8 SATs.
4. Time required: 2 x 60 min period(s).
5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

6. PC 010 will be a practical assessment based on content from PO 010.
7. During this practical assessment, you will be required to coordinate defensive cyber operations – internal defensive measures. You must pass this PC to pass PO 010.
8. References will be provided.
9. Pass mark is 6/8 SATs IAW PCL 010.
10. You will have 2 x 60 min period(s) to complete this assessment.
11. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Spiral 1: Continuation of the threaded scenario that includes a shift from an Exercise sail to a named CJOC mission where there exists a cyber threat. The CISO trainee must then use the TRA, Network Configurations, and threats to determine the probable attacks, recommended security increases, immediate actions, and required information that must be conveyed to the ship's Company. Additionally, there will be a cyber-attack on the ship while in transit and they must explain the response that they would take (or have the ISA/NOC/CFNOC) take to defend the ship's Networks.

Spiral 3: Mock ship's network where the CISO student configures the IT/IS to counter the cyber threat, briefs the class on the measures they have taken, and then the instructor runs a cyber-attack on the training network where the CISO is required to react to the attack IAW doctrine.

PART IV – PERFORMANCE CHECK LIST

Item	PC 010 Coordinate Defensive Cyber Operations – Internal Defensive Measures	MET	NOT MET	Remarks
1. Cyber Attack Description				
Summary of Threat Brief		☐	☐	
How the attack was recognized		☐	☐	
Description of attack type		☐	☐	
2. Immediate Actions Conducted				
Description of immediate response		☐	☐	
Summary of contact with response authority		☐	☐	
3. Cyber Security Plan				
Recommendations for alteration to ship's network configuration		☐	☐	
Potential impact of plan on operations		☐	☐	
Plan to brief ships personnel		☐	☐	

Score: **/**

The trainee must achieve 6/8 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 010 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 010 with the candidate.	**Candidate** I have been debriefed on the result of PC 010.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 011 Coordinate a Communications and Information System Plan

EO 011.01 Coordinate the installation of CIS resources

1. **Performance**: Coordinate the installation of CIS resources (N0037).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation; and

      (2) References.

   b. Denied: Nil.

   c. Environment: Classroom.

3. **Standard**: The trainee shall coordinate the installation of CIS resources (N0037), IAW references by:

   a. Arranging Formation CIS support, IAW references;

   b. Arranging Fleet CIS support, IAW references; and

   c. Arranging external agency support, IAW references.

4. **Time Required**:

   a. EO – 5.5 x 60 min periods;

   b. EC – Nil; and

   c. PC – 3 x 60 min periods.

5. **Instructional Methods**:

   a. Interactive lecture; and

   b. Self-study.

6. **Training Aids**:

   a. MLPs;

   b. BIS ship support project checklist;

   c. Fleet Communications Planning MIP; and

   d. CIS installation flow chart.

7. **Test Details**: EO 011.01 will be tested in EC 011.02 and PC 011.

8. **Remarks**:

   a. Trainees will be scheduled to work alongside the CISN Manager QL6B course for this part of their course with an n emphasis on teamwork, team goals, and planning requirements for the CJOC mission.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Arranging Formation/Fleet CIS support						
1) CIS installation process a. Review of Assyst process, IMR generation b. Liaison between CISO and CSEO c. Work period progress meetings		**30**			S1: Instructor Demonstration	
2) N6 and BIS a. Org chart b. Services provided	**120**				S1: Review BIS and N6 Web spaces then Guest Lecture S2: Nil. S3: Nil.	Guest lecturers
Arranging Fleet CIS support						
3) Fleet NAV COMM a. Org chart b. Services provided	**60**				S1: Guest Lecture S2: Nil. S3: Nil.	Guest lecturer
Arranging external agency support						
4) 76 Comms Regiment a. Org chart b. Services provided	**60**				S1: Halifax, or Esquimalt Det SME S2: DL S3: DL in PIP	Possible requirement for VTC depending on SME availability
5) CFCSS det West/East c. Org chart	**60**					Guest lecturer

d. Services provided						

EO 011.02 Prepare a Communications Plan

1. **Performance**: Prepare a communications plan (N0038).
2. **Conditions**:
   a. Given:
      (3) DWAN enabled workstation; and
      (4) References.
   b. Denied: Nil.
   c. Environment: Classroom, Simulator.
3. **Standard**: The trainee shall prepare a communications plan (N0038) by:
   a. Identifying shipwide C4 resource requirements (T0003), IAW references by;
      (1) Contributing CIS tasks to an HMC Ship Master Implementation Plan (MIP);
      (2) Generating an Information Exchange Requirements (IER) matrix; and
      (3) Developing a Primary, Alternate, Contingency, Emergency (PACE);
   b. Extracting operational CIS information, IAW references;
   c. Managing CIS operational correspondence, IAW references;
   d. Executing Operational Planning Process (OPP) board game; and
   e. Conducting a maritime tactical estimate.
4. **Time Required**:
   a. EO – 20.5 x 60 min periods; and
   b. EC – 1 x 60 min periods.
5. **Instructional Methods**:
   a. Interactive lecture;
   b. Demonstration - performance;
   c. Study assignment; and
   d. Self-study.
6. **Training Aids**:

a. Mission Implementation Plan;

b. Maritime Tactical Planning Tool;

c. PACE matrix;

d. Ship's Equipment Status slide;

e. Planning Template, including Quad Slide;

f. Checklists;

g. Ship's IT configuration (for OP STAT UNIT CIS Supp); and

h. MLP.

10. **Test Details**: EO 011.02 will be tested in EC 011.02 and PC 011.

11. **Remarks**:

    a. Trainees will be scheduled to work alongside the CISN Manager QL6B course for this part of their course with an n emphasis on teamwork, team goals, and planning requirements for the CJOC mission.

12. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Identifying shipwide C4 resource requirements IAW ref.						
1) Contributing CIS tasks to an HMC Ship Master Implementation Plan (MIP)	**15**		**15**	CCFP/CCFL Website	S1: Lecture, Practical In-class planning No S2 or S3.	Instructor to introduce MIP and require it be completed as a part of this PO.
2) Generating an Information Exchange Requirements (IER) matrix	**60**		**120**	CFSCE SharePoint Site	S1: Reading, Lecture, and Fill out S2: Nil. S3: Continuous Improvement	Instructor to introduce IERs and require that they be completed for the CJOC Mission. MARLANT and MARPAC may vary

3) Developing a Primary, Alternate, Contingency, Emergency (PACE)	**15**		**45**			PACE required to be completed as a part of the Scenario
Extracting operational CIS information IAW ref						
4) Extract from planning documents Standing OPGEN, OPORD and EXDIR/FRAGORD information relating to CIS a. OPGEN b. OPORD (and Annexes) c. EXDIR d. EXINST e. FRAGORD f. CONPLANs.	**30**		**30**	APP 11 CONPLAN JUIPITER, RENNAISANCE, ANGLE,		60 min reading time
5) Extract from existing COMPLAN information relating to CIS a. MARLANT standing COMPLAN b. MARPAC standing COMPLAN c. MARCOM COMPLAN d. COMBEXAG V	**15**	**15**	**30**		Students to take COMPLAN and generate CCR/Ops Room quick reference guide, including line numbers, radio circuits and frequencies	Assigned Readings, followed by in-class discussion and time to complete own messages/ ingestion
6) Extract cryptologic information from a. OPTASK COMMS b. OPTASK LINK c. CRYPTO CALLOUT message	**30**				S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
Generate CIS operational correspondence IAW ref						
7) Generate required OPTASKs and OPSTATs a. OPTASK COMMS b. OPTASK IM c. OPTASK EW d. OPSTAT UNIT	**60**		**120**	APP 11	S1: Reading Assignment, in-class discussion, assignment S2: Nil. S3: Nil.	60 min reading time Instructor to assign OPSTAT UNIT CIS

e. OPSTAT UNIT CIS SUPP						SUPP to trainees
8) Generate a C4ISR requirements message	**15**		**15**		S1: Reading Assignment, in-class discussion, assignment S2: Nil. S3: Nil.	Instructor to assign C4ISR Req. Msg to trainees
9) Generate SATCOM requests IAW current process	**60**		**30**		S1: Reading Assignment, in-class discussion, assignment S2: Nil. S3: Nil.	
10) Generate a Non-tactical Frequency Clearance Request Message: a. Requirements b. References c. Action / Information Addressees d. Format e. Request Timelines	**15**		**30**		S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	Instructor to assign Non-Tactical Freq-Clearance message to trainees
11) Generate a Tactical Frequency Clearance Request Message: a. Requirements b. References c. Action / Information Addressees d. Format e. Request Timelines	**15**		**30**		S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
12) Direct cryptographic requirements: a. Increase in entitlements b. Cryptographic material request (CMR) c. Replace equipment d. Rekeying considerations i. STE ii. VIPER	**45**	**15**			S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	Not to be confused with Crypto Content. Specific scheduling requirements for COMSEC planning

iii. KSV-22						
Executing Operational Planning Process (OPP) board game IAW ref:						
13) Planning conference process a. IPC (initial planning conference) b. MPC (main planning conference) c. FPC (final planning conference)	**60**				S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
14) Operational Planning Process Board Game			**240**		S1: In-class S2: Game Pieces other than paper S3: Application	
Conducting a maritime tactical estimate IAW ref:						
15) Maritime Tactical Estimate process IAW CFMWC Doctrine a. Introduction to tool b. When it applies c. Example of use	**60**				S1: Guest Lecture (VTC) S2: Experienced SME S3: Doctrine	New tool that not many personnel know about that CFMWC is trying to get used more. Suggest VTC for NFS (P) to interact with SME from CFMWC.

Enabling Check – EO 011.02 Prepare a Communications Plan

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

13. Classroom.

Arrangements and Procedures

14. EC 011.02 will be a practical assessment based on content from EO 011.02.

15. To pass this EC, the trainee must achieve 6/8 SATs.

16. Time required: 1 x 60 min period.

17. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 011.02 will be a practical assessment based on content from EO 011.02.

2. During this practical assessment, you will be required to generate a unit-specific Communications Plan based on a given OPTASK COMMS message. You must pass this EC to pass EO 011.02.

3. References will be provided.

4. Pass mark is 6/8 SATs IAW ECL 011.02.

5. You will have 1 x 60 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Provide students with the OPTASK COMMS being used for the mission planning scenario and have them extract information based on an HCM Frigate to generate a Power Point slide which identifies line numbers, radio channels and frequencies which would be made available to the Ops Room and CCR.

PART IV – ENABLING CHECK LIST

Item	EC 011.02 Prepare a Communications Plan	SAT	UNSAT	Remarks
1. OPTASK COMMS				
Identified COMPLAN		☐	☐	
Identified Duties		☐	☐	
Identified TG Comd COMMO		☐	☐	
Identified Restrictions		☐	☐	
Identified Safety Information		☐	☐	**CRITICAL**
2. IERs				
Identified TG Comd IERs		☐	☐	
Identified IERs ship would not meet		☐	☐	
Made recommendation to achieve IERs		☐	☐	

Score: ***/***

Trainee must achieve 6/8 SATs on non-critical factors **AND** all critical factors to pass this EC.

Positive Feedback	Improvement Feedback

EC 011.02 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 011.02 with the candidate.	**Candidate** I have been debriefed on the result of EC 011.02.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 011 Prepare a Communications Plan

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

18. Classroom.

Arrangements and Procedures

19. PC 011 will be a practical assessment based on content from PO 011.

20. To pass this PC, the trainee must achieve 11/13 SATs.

21. Time required: 3 x 60 min period(s) (15 minutes per trainee)

22. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 011 will be a practical assessment based on content from PO 011.

2. During this practical assessment, you will be required to prepare a communications plan. You must pass this PC to pass PO 011.

3. References will be provided.

4. Pass mark is 11/13 SATs IAW PCL 011.

5. You will have the provided in-class time and 15 minutes to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

The CISO student will coordinate a communications and information systems plan. They will need to draft, and create all proper documentation for the Operation that would be expected of a ship. They will need to feed into the Combat Officer's MIP, coordinate IERs, satellite access, foreign port frequency clearance requests (CBTO, NAVO, and LogO), etc.; complete a full Operation communications plan, and prepare a brief (quad slide format) with all supporting documentation to be handed into the instructor; and be able to brief on communications at the "mission brief" for the scenario. Two slides maximum which will be included in the ORO Course's Mission Brief on communications. Brevity is essential.

PART IV – PERFORMANCE CHECK LIST

Item	PC 011 Coordinate a Communications and Information System Plan	MET	NOT MET	Remarks
1. Deliverables				
IER Spreadhseet		☐	☐	
C4ISR Requirements Message		☐	☐	
Satellite Access Request Message		☐	☐	
Non-Tactical Frequency Clearance Request Message		☐	☐	
OPSTAT UNIT CIS SUPP		☐	☐	
2. Mission Brief				
Quad Slide Format		☐	☐	
Identified MIP Requirements		☐	☐	
Identified Completed Messages		☐	☐	
Ship's Equipment States		☐	☐	
Special Considerations (TAVs)		☐	☐	
Information System Install schedule/plan		☐	☐	
3. Operational Planning Process				
Used OPP in approach to planning process		☐	☐	
Mission Brief met requirements of OPP Format		☐	☐	

Score: /

The trainee must achieve 11/13 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 011 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of PC 011 with the candidate.		**Candidate** I have been debriefed on the result of PC 011.	
PC Administrator	Name:	Date:	Signature:

180/-

Item	PC 011 Coordinate a Communications and Information System Plan	MET	NOT MET	Remarks
Candidate	Name:	Date:		Signature:

EO 011.02 Prepare a Communications Plan

1. **Performance**: Prepare a communications plan (N0038).

2. **Conditions**:

   d. Given:

      (5) DWAN enabled workstation; and

      (6) References.

   e. Denied: Nil.

   f. Environment: Classroom, Simuolator.

3. **Standard**: The trainee shall prepare a communications plan (N0038) by:

   f. Identifying shipwide C4 resource requirements (T0003), IAW references by;

      (1) Contributing CIS tasks to an HMC Ship Master Implementaion Plan (MIP);

      (2) Generating an Information Exchange Requirements (IER) matrix; and

      (3) Developing a Primary, Alternate, Contingency, Emergency (PACE);

   g. E ;

   h. Executing Operational Planning Process (OPP) board game; and

   i. Conducting a maritime tactical estimate.

4. **Time Required**:

   c. EO –19.5 x 60 min periods;

   d. EC – Number x 60 min periods; and

   e. PC – Number x 60 min periods.

5. **Instructional Methods**:

   e. Interactive lecture;

   f. Demonstration - performance;

g. Study assignment; and

h. Self-study.

6. **Training Aids**:

i. Mission Implementation Plan;

j. Maritime Tactical Planning Tool;

k. PACE matrix;

l. Ship's Equipment Status slide;

m. Planning Template, including Quad Slide;

n. Checklists;

o. Ship's IT configuration (for OP STAT UNIT CIS Supp); and

p. MLP.

7. **Test Details**: EO 011.02 will be tested in EC 011.02 and PC 011.

8. **Remarks**:

b. Trainees will be scheduled to work alongside the CISN Manager QL6B course for this part of their course with a n emphasis on teamwork, team goals, and planning requirements for the CJOC mission.

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Identifying shipwide C4 resource requirements IAW ref.						
16) Contributing CIS tasks to an HMC Ship Master Implementaion Plan (MIP)	**15**		**15**	CCFP/CCFL Website	S1: Lecture, Practical In-class planning No S2 or S3.	Instructor to introduce MIP and require it be completed as a part of thei PO.
17) Generating an Information Exchange Requirements (IER) matrix	**60**		**60**	CFSCE SharePoint Site	S1: Reading, Lecture, and Fill out S2: Nil.	Instructor to introduce IERs and require that they be

					S3: Continuoius Improvement	completed for the CJOC Mission. MARLANT and MARPAC may vary
18) Developing a Primary, Alternate, Contingency, Emergency (PACE)	**15**		**45**			PACE required to be completed as a part of the Scenario
E IAW ref						
19) Extract from planning documents Standing OPGEN, OPORD and EXDIR/FRAGORD information relating to CIS g. OPGEN h. OPORD (and Annexes) i. EXDIR j. EXINST k. FRAGORD l. CONPLANs.	**30**		**30**	APP 11 CONPLAN JUIPITER, RENNAISANCE, ANGLE,		60 min reading time
20) Extract from existing COMPLAN information relating to CIS e. MARLANT standing COMPLAN f. MARPAC standing COMPLAN g. MARCOM COMPLAN h. COMBEXAG V	**15**	**15**	**30**		Students to take COMPLAN and generate CCR/Ops Room quick reference guide, including line numbers, radio circuits and frequencies	Assigned Readings, followed by in-class discussion and time to complete own messages/ ingestion
21) Extract cryptologic information from d. OPTASK COMMS e. OPTASK LINK f. CRYPTO CALLOUT message	**30**				S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
Generate IAW ref						

22) Generate required OPTASKs and OPSTATs f. OPTASK COMMS g. OPTASK IM h. OPTASK EW i. OPSTAT UNIT j. OPSTAT UNIT CIS SUPP	**60**		**120**	APP 11	S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	60 min reading time
23) Generate a C4ISR requirements message	**15**		**15**		S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
24) Generate SATCOM requests IAW current process	**60**		**30**		S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
25) Generate a Non-tactical Frequency Clearance Request Message: f. Requirements g. References h. Action / Information Addressees i. Format j. Request Timelines	**15**		**30**		S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
26) Generate a Tactical Frequency Clearance Request Message: f. Requirements g. References h. Action / Information Addressees i. Format j. Request Timelines	**15**		**30**		S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
27) Direct cryptographic requirements: e. Increase in entitlements f. Cryptographic material request (CMR) g. Replace equipment	**45**	**15**			S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	Not to be confused with Crypto Conent. Specific scheduling requirements

h. Rekeying considerations i. STE ii. VIPER iii. KSV-22						for COMSEC planning
Executing Operational Planning Process (OPP) board game IAW ref:						
28) Planning conference process d. IPC (initial planning conference) e. MPC (main planning conference) f. FPC (final planning conference)	**60**				S1: Reading Assignment and in-class Discussion S2: Nil. S3: Nil.	
29) Operational Planning Process Board Game			**240**		S1: In-class S2: Game Pieces other than paper S3: Application	
Conducting a maritime tactical estimate IAW ref:						
30) Maritime Tactical Estimate process IAW CFMWC Doctrine d. Introduction to tool e. When it applies f. Example of use	**60**				S1: Guest Lecture (VTC) S2: Experienced SME S3: Docrine	New tool that not many personnel know about that CFMWC is trying to get used more. Suggest VTC for NFS(P) to interact with SME from CFMWC.

Enabling Check – EO 011.02 Title

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Classroom.

Arrangements and Procedures

EC 011.02 will be a practical assessment based on content from EO 011.02.

To pass this EC, the trainee must achieve x/y SATs or %.

Time required: 1 x 60 min period.

References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

7. EC 011.02 will be a practical assessment based on content from EO 011.02.

During this practical assessment, you will be required to generate a unit-specific Communications Plan based on a given OPTASK COMMS message. You must pass this EC to pass EO 011.02.

References will be provided.

Pass mark is x/y SATs IAW ECL 011.02.

You will have 1 x 60 min period to complete this assessment.

Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Provide students with the OPTASK COMMS being used for the mission planning scenario and have them extract information based on an HCM frigate to generate a PointPoint slide which identifies line numbers, radio channels and frequencies which would be made available to the Ops Room and CCR.

PART IV – ENABLING CHECK LIST

Item	EC 011.02 Performance Statement	SAT	UNSAT	Remarks
3.				
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
4.				
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	

Score: /

Trainee must achieve x/y% SATs on non-critical factors **AND** all critical factors to pass this EC.
The trainee must achieve x/y% SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 011.02 – Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of EC 011.02 with the candidate.		**Candidate** I have been debriefed on the result of EC 011.02.	
EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 011 Title

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom/Ship/Submarine.

Arrangements and Procedures

8. PC 011 will be a practical/written assessment based on content from PO 011.

To pass this PC, the trainee must achieve x/y SATs or %.

Time required: # x 60 min period(s) per trainee (# periods total (time per trainee x number of trainees)).

References will/will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

9. PC 011 will be a practical assessment based on content from PO 011.

During this practical assessment, you will be required to performance statement. You must pass this PC to pass PO 011.

References will/will not be provided.

Pass mark is x/y SATs IAW PCL 011.

You will have # x 60 min period(s) to complete this assessment.

Do you have any questions before you start?

PART II – WRITTEN ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

10. PC 011 will be a written assessment based on content from PO 011.

References will/will not be provided.

Pass mark is %.

You will have # x 60 min period(s) to complete this assessment.

Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Give guidance for the development of the assessment.

PART IV – PERFORMANCE CHECK LIST

188/-

Item	PC 011 Coordinate a Communications and Informations System Plan	MET	NOT MET	Remarks
4.				
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
5.				
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	
		☐	☐	

Score: **/**

Trainee must achieve x/y SATs on non-critical factors **AND** all critical factors to pass this PC. The trainee must achieve x/y SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 011 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 011 with the candidate.	**Candidate** I have been debriefed on the result of PC 011.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 012 Direct Shipboard Communications Tactics

EO 012.01 Develop Communications Tactics

9. **Performance**: Develop communications tactics (N0040).

10. **Conditions**:

    a. Given:

        (1) DWAN enabled workstation;

        (2) Classified Network Access;

        (3) Intelligence Brief;

        (4) Threat briefs; and

        (5) References.

    a. Denied: Nil.

    d. Environment: Classroom, trainer.

11. **Standard**: The trainee shall develop communications tactics (N0040) by:

    a. Assessing communications threat information, IAW references to include:

        (1) Communications tactics;

        (2) Tactics development; and

        (3) Threats to communications.

    b. Interpreting threat analysis with regards to communications, information systems, IAW references;

    c. Interpreting ELINT analysis to include;

        (1) Communications intercept data and information; and

    d. Respond to communications intercepts (DT0200), IAW references.

12. **Time Required**:

    a. EO – 18 x 60 min periods;

    b. EC – 1 x 60 min periods; and

    c. PC – 2 x 60 min periods.

13. **Instructional Methods**:

    a. Interactive lecture;

    b. Demonstration - performance;

    c. Study assignment; and

    d. Self-study.

14. **Training Aids**:

    a. Threat brief;

    b. Mission planning scenario;

    c. Platform capability references (Jane's);

    d. MLP; and

    e. Battle lab.

15. **Test Details**: EO 012.01 will be tested in EC 012.01 and PC 012.

16. **Remarks**:

    a. Trainees will be scheduled to work alongside the CISN Manager QL6B course for this part of their course with an emphasis on teamwork, team goals, and planning requirements for the CJOC mission.

17. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Develop communications tactics, IAW refs						
1. Communications Tactics, to include: a. Lowest probability of intercept (LPI); b. Direction finding (DF); c. Spread spectrum; d. Frequency hopping; e. Countermeasures, f. Standard operating procedures (SOP); g. Doctrine, to include	**240**		**180**	A1 Ch. 6 A30 B8 A14	S1: Self Study and instructor led discussion S2: continuous improvement S3: Integration of Encounter EX scenario in Battle Lab	

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
1. Allied; 2. CAN/US; and 3. NATO.					with LPI and DF focus	
2. Tactics development , to include: a. MARTIs; b. TACNOTE; and c. In-theatre.	**60**		**60**	A35	S1: Self Study and instructor led discussion S2: Continuous improvement S3: Critical analysis of doctrine in Battle Lab	CFMWC process to be explained and then examples provided by instructor to students.
Assess communications threat information, IAW refs						
3. Threats to communications, to include: a. Adversary capabilities; b. “near peer” vs. asymmetric threats; and c. Open source threat capability.	**120**			Trinity Brief	S1: Table-top between students and instructor S2: simulation or video demonstration of capabilities S3: Integration of tactics into Battle Lab simulation	Guest lecture (1 hour) followed by table top TRINITY Comms Int trade SDA required
4. Scenario threat brief, to include: a. Known capabilities; b. Possible capabilities; c. Probability of use, to include examples; d. Frequencies; and e. EM spectrum pattern of life.	**60**				Instructor or SME to provide SECRET threat brief based on course scenario to students.	Comms information systems and cyber threat must be covered in this brief.
Interpret threat analysis for communications, information systems, IAW refs						
5. Comparison of threat and own ship capabilities, to include: a. CONPLAN; b. ORBAT; c. Battle Orders;	**120**			A32 A36 A37 A14	S1: Table-top between students and instructor based on scenario,	Trainees are expected to ingest the threat brief and compare the scenario

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
d. Doctrine; e. MARTI; f. TACNOTE; g. Configurations; h. EMCON; i. SOPs; and j. Aim of countering threats.					class of own ship, threat brief. S2: Software to predict EM signature based on configuration. S3: Integration of tactics into Battle Lab simulation	threats to assigned ship's capabilities with tactical consideration

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Interpret ELINT analysis, IAW refs						
6. Communications intercept data and information, to include: a. Role within Combat Information Organization (CIO); b. Responsibility to feed into CIO for EM spectrum; and c. Recognition of EMI based on EWS reports or ELINT information.	**60**		**60**	A42	S1: Self Study via reading assignment and instructor led discussion S2: Secret level exercise in frequency identification of own force S3: Integration into Battle Lab simulation	Instructor is to emphasize the WSUP and SNC feed into CIO based on known comms frequencies of own force vs VOIs or adversary.
Responding to communications intercepts, IAW refs						
7. Recommendations for action when comms are known to be intercepted, to include: a. Doctrine; b. Pre-plans; c. Considerations; d. Risk; and e. Authority of CO vs circuit or system.	**60**		**60**	A37 A14 A32 A38 A16 A32 A36	S1: Reading Assignment followed by instructor led discussion S2: Case study with analysis of actions in DMOC S3: Integration of comms intercept into Battle Lab scenarios	Instructor to facilitate discussion, presenting scenarios to students and then they determine the best way forward using SARR format

Enabling Check – EO 012.01 Develop Communications Tactics

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 012.01 will be a practical scenario based assessment based on content from EO 012.01.

3. To pass this EC, the trainee must achieve 6 of 8 SATs.

4. Time required: 1 x 60 min period.

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 012.01 will be a practical assessment based on content from EO 012.01.

2. During this written assessment, you will develop and modify communications tactics. You must pass this EC to pass EO 012.01.

3. References will be provided.

4. Pass mark is 6 of 8 SATs IAW ECL 012.01.

5. You will have 1 x 60 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Spiral 1: Shall have four scenarios which cover the following topics:

    a. Brief to Comd based on scenario for Comms Intercept using SARR (Situation, Assessment, Risk, Recommendation)

    b. Provide recommendation to ORO as a part of CIO for tactical situation

2. Spiral 3: Scenarios will be incorporated into Battle Lab and will be made more practical vice written scenarios.

PART IV – ENABLING CHECK LIST

Item	EC 012.01 Develop Communications Tactics	SAT	UNSAT	Remarks
1. Application of MARTI or TACNOTE				
Application of doctrine		☐	☐	
Recommendation IAW doctrine		☐	☐	
2. Command brief for communications intercept				
Application of SARR		☐	☐	
Risk assessment		☐	☐	
3. CIO tactical situation and recommendation				
Situation assessment		☐	☐	
Recommendation		☐	☐	
4. Battle Orders draft change				
Recognized issue in orders		☐	☐	
Proposed changes IAW OPORD		☐	☐	

Score: / 8

The trainee must achieve 6 of 8 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 012.01 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 003.01 with the candidate.	**Candidate** I have been debriefed on the result of EC 003.01.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 012.02 Manage the Configuration of CIS Resources

1. **Performance**: Managing the configuration of CIS resources (N0039).

2. **Conditions**:

   a. Given:

      (1) DWAN enabled workstation;

      (2) Classified Network Access;

      (3) Intelligence Brief;

      (4) Threat briefs; and

      (5) References.

   b. Denied: Nil.

   c. Environment: Classroom, trainer.

3. **Standard**: The trainee shall manage the configuration of CIS resources (N0039), IAW references by:

   a. Advising Command on tactical employment of communications equipment, IAW references;

   b. Drafting communications battle orders, IAW references;

   c. Managing the implementation of EMCON states, IAW references

4. **Time Required**:

   a. EO – 21 x 60 min periods;

   b. EC – 1 x 60 min periods.

5. **Instructional Methods**:

   a. Interactive lecture;

   b. Study assignment;

   c. Self-study;

   d. Simulation; and

   e. Case study.

6. **Training Aids**:

   a. Indoctrination brief on tactical situation;

   b. Scenario brief;

   c. Intelligence brief;

   d. MLP; and

   e. Threat brief.

7. **Test Details**: EO 012.02 will be tested in EC 012.02 and PC 012.

8. **Remarks**:

9. **Teaching Points**:

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Advise Command on the tactical employment of communications, IAW refs						
1. Communications configuration and employment to include: a. Degrees of readiness; b. States of preparedness; c. Tactical picture;	**120**		**120**	A16 A36 Trinity Threat Assessment	S1: Self-study with instructor led discussion S2:Nil S3: Ability to reconfigure Comms in Battle Lab	Trainees to read reference material and be prepared to discuss in class with instructor.
2. Tactical communications data, to include: a. Electromagnetic (EM) signature of own ship; b. Tactical EM considerations; c. Direction finding; and d. Power outputs and predicted ranges.	**120**		**120**	A44 A43 B8 A14 D3		
Draft communications battle orders, IAW refs						
3. Draft communications battle orders, to include: a. Considerations;	**60**		**120**	Direction from Captain, Higher	S1: Interactive lecture.	Trainees to read reference material and

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
b. Command Direction; and c. Battle rhythm.				Commander's Intent, 1up, 2up A36	S2: Integrate into Scenarios with ORO course drafting orders S3: trialed orders in IW Battle Lab	be prepared to discuss in class with instructor.

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Compile communications orders of battle (COB) IAW refs						
4. Communications Orders of Battle, to include: a. ORBAT messages; b. Communications Intercept for OPFOR	**60**		**180**	B1 Annex A Appendix 1	S1: trainees to apply messages to scenario. Instructor to provide feedback to in-class work time S2: Continuous improvement S3: Integrate into Battle Lab simulations.	Reading time 3 hours 1 hour to apply to scenario in-class
Advise Command on implementation of EMCON states, IAW refs						
5. EMCON policy and procedures, to include: a. EMCON restrictions b. EMCON plan c. Warfare Threat; d. Collection; e. Proximity to nation states; and f. Assigned duties g. Other considerations.	**120**		**120**	B8 A44 A14, Tactical and non-tactical frequency clearance requests B6, assigned duties	S1: Self-study with Instructor led discussion to generate checklist S2: Tables of own platform information integrated into lesson (continuous improvement) S3: Battle lab simulation of ranges to demonstrate EMCON vs adversaries	Instructor to facilitate student's creation of an EMCON consideratio n checklist based on references S2: Once available, the EM characteristic s of platforms need to be integrated into this lesson.

EC 012.02 Manage the Configuration of CIS Resources

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 012.02 will be a practical assessment based on content from EO 012.02.

3. To pass this EC, the trainee must achieve 2/3 SATs.

4. Time required: 1 x 60 min period.

5. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 012.02 will be a practical assessment based on content from EO 012.02.

2. During this practical assessment, you will manage the configuration of CIS resources. You must pass this EC to pass EO 012.02.

3. References will be provided.

4. Pass mark is 2/3 SATs IAW ECL 012.02.

5. You will have 1 x 60 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Spiral 1: will consist of four (4) tactical situations where communications may need to be altered in order to meet a threat or prevent collection. The scenarios shall include the following; geo-political situation where power or circuits must be reduced due to TTW or EMI, EMCON configuration must be changed based on tactical situation, ELINT collection threat exists and a recommendation must be made to Command.

2. Spiral 3: Practical EC in Battle Lab to be developed and include, geo-political situation where power or circuits must be reduced due to TTW or EMI, EMCON configuration must be changed based on tactical situation, ELINT collection threat exists and a recommendation must be made to Command. The Battle Lab should demonstrate the change and create a visual representation via chart and ranges.

PART IV – ENABLING CHECK LIST

Item	EC 012.02 Manage Configuration of CIS Resources	SAT	UNSAT	Remarks
1.	Geo-political situation (Sovereignty)	☐	☐	
2.	EMCON configuration	☐	☐	
3.	ELINT collection threat	☐	☐	

Score: / 4

The trainee must achieve 2 of 3 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 012.02 – Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of EC 012.02 with the candidate.	**Candidate** I have been debriefed on the result of EC 012.02.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

EO 012.03 Advise Command on Command and Control in a Denied or Degraded Environment (C2D2E)

1. **Performance**: Advise Command on C2D2E (N0041)).

2. **Conditions:**

   a. Given:

      (1) DWAN access;

      (2) Secret network access;

      (3) Intelligence brief;

      (4) Threat assessment;

      (5) Scenario; and

      (6) References.

   b. Denied: Nil.

   c. Environment: All naval environmental conditions at sea, alongside or ashore.

3. **Standard:** The trainee shall advise Command on C2D2E (N0041), IAW A19, A37, A39, and B7, by:

   a. Assessing changes in the electromagnetic environment ();

   b. Assessing communications jamming (to include;

      (1) Harmful Interference and infringement procedures ();

      (2) Electromagnetic interference (EMI) (; and

      (3) Interference to SATCOM (); and

   c. Coordinating anti-jam procedures ().

4. **Time Required:**

   a. EO – 14 x 60 min periods;

   b. EC – 1 x 60 min periods; and

5. **Instructional Methods:**

   a. Interactive lecture;

b. Self-study;

c. Simulation; and

d. Case study.

6. **Training Aids:** MLP.

7. **Test Details:** EO 012.03 will be tested in EC 012.03 and PC 012.

8. **Remarks:** Nil.

9. **Teaching Points:**

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Advise Command on C2D2E, IAW refs						
1. C2D2E a. Definition b. Impact to operations c. Real world examples d. exercises	**120**		**120**	B7 -797 A37 Chapter 6 (TACNOTES) A39 6650 on maritime IW (note secret CEO addition)	S1: Self-study and Guest Lecture S2: tie into NATO resources (continuous improvement) S3: Integration into Battle Lab scenarios	Guest Lecture from CFMWC
Assessing changes in the electromagnetic environment, IAW refs						
2. Understanding pattern of life, to include: a. Recognizing EM changes through trouble shooting; and b. Reporting.	**60**		**120**	A19 Post Deployment Reports	S1: Guided discussion S2: Video or DMOC demonstration S3: Integration into Battle Lab scenario	Instructor should use real exercise and OP examples from PDRs

Main Teaching Point	Instructional Technique (Time in min)			References	Delivery Approach/ Spirals	Comments
	T	D	P			
Assessing communications jamming (), IAW						
3. Electromagnetic interference (EMI) vs. harmful interference, to include: a. Infringement procedures; and b. Blue force EMI.	**30**	**30**	**120**	CF spectrum manual	S1: Case Study S2: continuous improvement S3: Integration into Battle Lab	Instructor to assign cases 5 days prior
4. Interference to SATCOM, to include: a. C2D2E and SATCOM; b. C2 systems; c. Global positioning system (GPS); and d. Impact to Ops.	**60**			B7 -797 A37 Chapter 6 (TACNOTES) A39 6650 on maritime IW (note secret CEO addition)	S1: Guided discussion. S2: video S3: Integration into Battel Lab	
Coordinating anti-jam procedures (DT0210), IAW refs						
5. Communications jamming a. Detection methods b. Mitigation measures c. Pre-planned responses	**60**		**120**	A14 B8 Vol 1	S1: Guided discussion. S2: video S3: Integration into Battel Lab	

EC 012.03 Advise Command on Command and Control in a Denied or Degraded Environment (C2D2E)

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

1. Trainer/Classroom.

Arrangements and Procedures

2. EC 012.03 will be a practical/written assessment based on content from EO 012.03.

3. To pass this EC, the trainee must achieve 7 of 9 SATs.

4. Time required: 1 x 60 min period.

5. References will/will not be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. EC 012.03 will be a practical assessment based on content from EO 012.03.

2. During this practical assessment, you will Advise Command on C2D2E. You must pass this EC to pass EO 012.03.

3. References will not be provided.

4. Pass mark is 7 of 9 SATs IAW ECL 012.03.

5. You will have 1 x 60 min period to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

1. Spiral 1: Written scenario based EC where the trainees are provided three scenarios; Blue Force EMI, purposeful jamming from an adversary, and maintaining C2 in a denied or degraded environment (SATCOM loss/degraded). They will provide their answers in SARR format.

2. Spiral 3: Practical scenario based EC where the Battle Lab is used to carry out the scenarios with trainees reacting in near real-time.

PART IV – ENABLING CHECK LIST

Item	EC 012.03 Advise Command on C2D2E	SAT	UNSAT	Remarks
1. Blue Force EMI				
Recognized issue		☐	☐	
Used SARR format		☐	☐	
Recommendation IAW doctrine		☐	☐	
2. Purposeful Jamming				
Recognized issue		☐	☐	
Used SARR format		☐	☐	
Recommendation IAW doctrine		☐	☐	
3. C2D2E Situation				
Recognized issue		☐	☐	
Used SARR format		☐	☐	
Recommendation IAW doctrine		☐	☐	

Score: / 9

The trainee must achieve 7 of 9 SATs to pass this EC.

Positive Feedback	Improvement Feedback

EC 012.03 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of EC 012.03 with the candidate.	**Candidate** I have been debriefed on the result of EC 012.03.

EC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

Performance Check – PO 012 Direct Shipboard Communications Tactics

PART I – ADMINISTRATIVE INSTRUCTIONS

Material and Facilities Instructions

10. Trainer/Classroom/Ship/Submarine.

Arrangements and Procedures

11. PC 012 will be a practical assessment based on content from PO 012.

12. To pass this PC, the trainee must achieve 5/6 SATs.

13. Time required: 2 x 60 min period(s).

14. References will be required.

PART II – PRACTICAL ASSESSMENT INSTRUCTIONS GIVEN TO TRAINEE

1. PC 012 will be a practical assessment based on content from PO 012.

2. During this practical assessment, you will be required to performance statement. You must pass this PC to pass PO 012.

3. References will/will not be provided.

4. Pass mark is 5/6 SATs IAW PCL 012.

5. You will have 2 x 60 min period(s) to complete this assessment.

6. Do you have any questions before you start?

PART III – ENABLING CHECK NOTES

Example Scenario:
During Operation _____ your ship is operating in a C2D2 environment. There have been reports from Ops that the ship is being spoofed and jammed. Assess the situation according to the threat brief and reports provided. Provide Command with recommendations for how to direct shipboard communication tactics to respond to the threat including:

Explaining the current C2D2E environment
Summarizing the communications portion of the ELINT analysis
Amending the EMCON plan
Coordinating the equipment reconfiguration based on the amended EMCON plan

PART IV – PERFORMANCE CHECK LIST

Item	PC 012 Direct Shipboard Communications Tactics	MET	NOT MET	Remarks
1. C2D2E Environmental Assessment				
Identification of threat		☐	☐	
Frequencies affected		☐	☐	
Operational Impact assessment		☐	☐	
Recommendation for action		☐	☐	
2. EMCON Plan				
Recommendation IAW doctrine		☐	☐	
Provided time assessment of changes		☐	☐	

Score: **/**

The trainee must achieve 5/6 SATs to pass this PC.

Positive Feedback	Improvement Feedback

PC 012 – Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of PC 012 with the candidate.	**Candidate** I have been debriefed on the result of PC 012.

PC Administrator	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

PO 013 Coordinate All Aspects of Ship's COMSEC

EO 013.01 Coordinate All Aspects of Ship's COMSEC

15. **Performance**: Coordinate all aspects of ship's COMSEC (B0480).

16. **Conditions**:

    a. Given: IAW COMSEC Custodian QSP.

    b. Denied: IAW COMSEC Custodian QSP.

    c. Environment: IAW COMSEC Custodian QSP.

3. **Standard**: The trainee shall complete the COMSEC Custodian course IAW the COMSEC Custodian QSP.

4. **Time Required**:

    a. EO – 84 x 60 min periods; and

    b. PC – IAW COMSEC Custodian PCL.

5. **Instructional Methods**: IAW COMSEC Custodian QSP

6. **Training Aids**:

    c. IAW COMSEC Custodian QSP

7. **Test Details**: IAW COMSEC Custodian QSP

8. **Remarks**: Nil.

9. **Teaching Points**: IAW COMSEC Custodian QSP

Annex A – Main References

A CF or other official Canadian government publication
B Official Allied Force or Allied government publication
C Commercial publication
D Other

Ref Code	Publication
A1	02 - ARMY_CFSCE_ACISS/SSCIA_DP2.0/PP2.0_Reserve Mod1 (ARMY_CFSCE_AKOX_MOD1_RES 2016, 2.2)
A2	ACP-125 CANSUPP(C) Radio Telephone Procedures (CFSCE_ACP125_RTP_WBT_V1_EN, 1.0) [Voice Procedures]
A3	A-IM-100-00/AG/001 Guideline for Certification and Accreditation Of Information Systems
A4	Analog to Digital Conversion (CFSCE_04_ADC_CEBS_111116_EN, 2.0)
A5	B-GG-005-004/AF-000 CF Operations
A6	B-GG-005-004/AF-010 CF Information Operations
A7	B-GJ-005-500/FP-00 CF Operational Planning Process
A8	C&E Signals Basics (CFSCE_CESB_CBT_111116_EN)
A9	C-03-005-012/AM-001 (Engineering Change Proposal)
A10	C4ISR (CFSCE_18_C4ISR_CEBS_111116_EN, 2.0)
A11	Canadian Navy's Communications Blueprints 2010
A12	CF Address Book (website)
A13	CFCD 101 Operational Analysis
A14	CFCD 102 Combat Readiness Requirements
A15	CFCD 106 Maritime Tactical Instructions
A16	CFCD 107 Maritime Command Communications Orders
A17	CFCD 114 EW Instruction
A18	CFCD 117 Guidance to Commanders
A19	CFCD 120 Tactical Aids
A20	CFCD 124 Maritime Operational Test and Evaluation Guide
A22	(E)
A23	Characteristics of Waves and Waveform Displays (CFSCE_02_CWWD_CEBS_111116_EN, 2.0)
A24	CIS/01/203 Status Manual for COMSEC
A25	CIS/01/509 Common Fill Devices Maintenance Manual
A28	
A29	Consolidated Secret Network Infrastructure (CSNI) Security Orders
A30	Cryptography (CFSCE_06_CRYPTO_CEBS_111116_EN, 2.0)
A31	DAOD 6000 Information Management & information technology
A32	DAOD 6001 Information Management
A42	
A46	
A47	Defence Wide Area Network (DWAN) Security Orders
A48	DNDP-35 Management of the Radio Spectrum (B-GT-035-001/AG-000)
A49	Electromagnetic (EM) Spectrum (CFSCE_09_EMS_CEBS_111116_EN, 2.0)
A50	EMSEC Policy Rev 2102-136 (DIR-IM-SECUR 5-2-3)
A52	
A53	GoC EKMS LMD/KP Operator's Aide Memoire
A54	Government Security Policy (GSP)
A55	HFX Class SATCOM VINSON (UNS-8)

A56	IMD 100 (Information Management Accountability Frame Work)
A57	INFOSEC 2 COMSEC Instructions
A58	INFOSEC 601
A59	Interference, Compatibility and Nuclear Radiation Effects (CFSCE_13_ICNRE_CEBS_111116_EN, 2.0)
A60	Introduction to Computers (CFSCE_14_IC_CEBS_111116_EN, 2.0)
A61	IS Security Response Manual
A63	
A64	Methods of Modulation (CFSCE_07_MM_CEBS_111116_EN, 2.0)
A65	Miscellaneous Communications Systems (CFSCE_12_MCS_CEBS_111116_EN, 2.0)
A66	Multiplexing and Multiple Access Systems (CFSCE_08_MMAS_CEBS_111116_EN, 2.0)
A67	
A69	
A70	Naval Multiband Terminal (NMT) SOP
A71	NAVORD 11900-2 Naval Orders – Maritime Evaluation
A72	
A74	
A77	Networking (Part 1 of 2) (CFSCE_15A_NET_CEBS_111116_EN, 2.0)
A78	Networking (Part 2 of 2) (CFSCE_15B_NET_CEBS_111116_EN, 2.0)
A79	Networking Models and Topologies (CFSCE_NETWORKING_01_E_2016, 1.0)
A80	Operations and Maintenance Manual and ICD (RT-1828) Rev 4, 27 Apr 2007
A81	Origins of the Internet (CFSCE_NETWORKING_00_E_2016, 1.0)
A82	Radar (CFSCE_17_RDR_CEBS_111116_EN, 2.0)
A83	
A84	RDIMS 370925 RCN SEMS Part II Directive S9-RCN Radio Frequency Safety Program
A86	Satellite Access Authorisation (SAA) Messages
A87	Satellites (Part 1 of 2) (CFSCE_11A_SAT1_CEBS_111116A_EN, 2.0)
A88	Satellites (Part 2 of 2) (CFSCE_11B_SAT2_CEBS_111116B_EN, 2.0)
A89	Ship's Standing Orders (SSOs)
A90	
A91	SHIPLAN Training Manual
A93	Signal Sources (CFSCE_03_SS_CEBS_111116_EN, 2.0)
A94	Telephony (CFSCE_05_TEL_CEBS_111116_EN, 2.0)
A95	TP 001.03b TCPIP Internet Protocols (CFSCE_NDA_TP_001.03B_00022016_EN, 1.0)
A96	TP 001.03d Sending and Receiving Packets (CFSCE_NDA_TP001.03D_00032017_EN, 1.0)
A97	ViaSat DAMA Training 05 (PowerPoint)
A98	Virtualization (CFSCE_16_VR_CEBS_111116_EN, 2.0)
B1	AComP-1 NATO Communications Glossary
B2	ACP 100 Call Signs and Address Groups-Instructions and assignments
B3	ACP 110 Tactical Call Signs
B4	ACP 113 Tactical Call Signs for Ships
B5	ACP 117 Allied Routing Indicators
B6	ACP 117 CAN-Supp Allied Routing Indicator Book Canadian Supplement
B7	ACP 121 Communication Instructions-General
B8	ACP 122 Communication Instructions-Security
B9	ACP 123 Common Message Strategy and Procedures

B10	ACP 124 Radio Telegraph
B11	ACP 125 Radio Telephone
B12	ACP 126 Teletypewriter Procedures
B13	ACP 127 Tape Relay
B14	ACP 128 CANSUPP - 1 ADDN Operating Procedures
B15	ACP 128 Telecomm Record System (ALTERS)
B16	ACP 130 Signalling Procedures in the Visual Medium
B17	ACP 131 Operating Signals
B18	ACP 132 Field Generation and OTAD of COMSEC Key
B19	ACP 133 Directing Services
B20	ACP 134 Telephone Switchboard Procedures
B21	ACP 135 Communication Instructions-Distress and Rescue Procedures
B22	ACP 150 Recognition and ID Instructions
B23	ACP 165 Operational Brevity Codewords
B24	ACP 167 Glossary of Communications-Electronics Terms
B25	ACP 176 Allied Naval and Maritime Air Communication Instructions
B26	ACP 176 NATO Supp 1 Allied Communication Publication NATO
B27	ACP 180 EW Instructions (Replaces ACP 177/178/179)
B28	ACP 190 Guide to Frequency Planning
B29	ACP 191 Ionospheric Sounding
B30	ACP 200 Maritime Tactical WANs
B31	ACP 220 Video Tele Conferencing (VTC)
B32	ACP-145 Gateway to Gateway Implementation Guide for ACP-123/SANAG 4406 Messaging Services
B33	ADatP-2 NATO Information Technology Glossary
B34	AJP 6 Allied Joint Doctrine for Communication and Information Systems
B35	AN/CYZ-10 Students Handbook
B36	APP-1 Allied Maritime Voice Procedures
B37	APP-11 NATO Message Catalogue
B38	ATP 1 Vol I Allied Maritime Tactical Instructions and Procedures
B39	ATP 1 Vol II Allied Maritime Tactical Signal and Manoeuvering Book
B40	ATP 17 Arctic Instructions
B41	ATP 18 Submarine Instructions
B42	AXP 3 Allied Naval Communications Exercises
B43	B-GT-516-001/KG-K01 CF High Frequency Skywave Prediction
B44	CCD-08 Canadian Cryptographic Doctrine for the KG-175 TACLANE
B45	CFJP 3-13.1 Joint Doctrine for C2W
B46	CFJP 6-0 Doctrine for Command Control Communications and Computer
B47	CIS/01/3 Communications-Electronics Security Operations and Training
B48	CIS/01/306 Field Generated and OTAD of Tactical Keying Material
B49	JP 6-0 Joint Communication and Information Systems
B50	KAO 207(B) Guidelines for the Use and Operation of KW46
B51	KAO 210A Guidelines for the Use and Operation of the KG84C
B52	KIV 7 Integration and Operation Manual
B53	MARLANT ORDS
B54	MARPAC ORDS

B55	PCP-1 PfP Naval and Maritime Air Communication Instructions
B56	RCN Orders (NAVORDS)
B57	TL-063-10 TACLANE Operator Manual
B58	USN ANDVT Manual
B59	B-GA-209-001/FP-001 National SAR Manual
B60	C-02-015-001/AG-000
B61	C-51-552-000/MS-001 (1280R)
B62	C-51-575-000/MS-001 CCMS
B63	C-51-579-000/MS-001 LFSR
B64	C-51-586-000/MS-001 SATCOM System
B65	C-51-658-000/MS-001 DAMA
B66	C-51-660-000/MS-001 WSC-3 (UHF Transceiver)
B67	C-51-890-000/MB-001 Saturn BM Marine MX2 Operator Manual
B68	C-52-289-001/MS-001 SHINCOM Level II Terminal
B69	CCD-55 KG-175D Doctrine and KG-175D Operating Manual
B70	CGS-200 Operator Manual (MHDS)
B71	CIS/01/203 Status Manual for COMSEC Material
B72	CompTIA A+ Certification Exam Guide
B73	CompTIA Network+ Study Guide
B74	How Computers Work
B75	HSDC Troubleshooting Guide
B76	ICOM VHF Marine Transceiver User Manual
B77	INNWOAT MX 2400 Operators Manual
C1	ITPRO.TV website (IT Fundamentals) https://itpro.tv/course-library/fundamentals/fundamentals-overview/
C2	ITSG-10 IT Security Guidance for COMSEC Material and Control Manual
C3	MHS/MPS/MHDS Operators Manual
C4	Motorola STU III Operating Manual
C5	MS Windows 2000 Professional
C6	Navy UHF Satellite Communication System Description
C7	Network Plus Training Kit
C8	OSSIS (Operational Security Standard for Information Systems)
C9	Radio Communications in the Digital Age vol. 1 HF Technology
C10	Radio Communications in the Digital Age vol. 2 VHF/UHF Technology
C11	RD-1310-13 Exciter Instruction Manual
C12	Relevant Current IS/IT Intrusion/Detection Material
C13	RF-5710 HF Modem (PSK) Instruction Manual
C14	RF-590 Receiver Instruction Manual
C15	Sailor 250/500 FBB User Guide
C16	VSD-612339-02-169 Operations and Maintenance Manual and ICD for RT-1829
D1	XTS 5000 Operator Manual

Annex B – Task List

Task #	Task Statement	T/NT	PO
	NWO MTL - CISO		
B0130	Prepare operational messages	T	004
B0140	Prepare evaluation operational reports	T	002
B0350	Advise command on communications	T	014
B0470	Manage ~~IS systems~~ Ship's IS/IT Configuration	T	009
B0480	Coordinate all aspects Ship's COMSEC	T	015
B0490	Organize intelligence	NT	
B0540	Identify systems performance deficiencies, malfunctions or failures	T	002
	CISO OSS		
T0001	Provide advice to command on Command, Control, Communications, Computers		
T0002	Oversee Ship's communications and IS/IT	T	006
T0003	Identify ship-wide C4 resource requirements	T	013
T0004	Direct enforcement of Communications and Information Security	T	003
	Draft QSP		
N0005	Lead NAVCOMM section	T	001
N0006	Manage NAVCOMM personnel	T	001
N0007	Coordinate ship's CIS training	T	001
N0008	Draft communications improvements documents	T	002
N0009	Recommend CIS improvements	T	002
N0010	Advise Command on CIS security	T	003
N0011	Direct information security procedures	T	003
N0012	Direct operational messaging	T	004
N0013	Direct operational message traffic	T	004
N0014	Advise command on electromagnetic spectrum warfare	T	005
N0015	Manipulate RF prediction software	T	005
N0016	Correlate RF information from operational sources	T	005
N0017	Explain RF theory	T	005
N0018	Identify communications monitoring faults	T	006
N0019	Configure ship's personal radio communicators	T	006
N0020	Maintain ship's PRCs	T	006
N0021	Maintain SubNet Relay	T	006
N0022	Manage PRC 117 equipment	T	006
N0023	Manage satellite communications equipment	T	007
N0024	Maintain satellite communications	T	007
N0025	Manage cryptographic equipment	T	008
N0026	Maintain cryptographic equipment	T	008
N0030	Manage network operations	T	009
N0032	Manage network performance and administration	T	009
N0034	Coordinate defensive cyber operations – internal defensive measures	T	011
N0035	Brief Command on cyber impact to operations	T	011
N0037	Coordinate the installation of CIS resources	T	012
N0038	Prepare a communications plan	T	012

N0039	Manage the configuration of CIS resources	T	013
N0040	Develop communications tactics	T	013
N0041	Advise Command on Command and Control in a Denied or Degraded Environment	T	013
N0042	Manage the CIA of IS and networks	T	013
N0043	Coordinate Naval SATCOM	T	007

Annex C – QSP Working Group Records of Proceedings
Dates: 18 April – 26 May, 2017

Location: Naval Training Development Center Atlantic, Building S-15, Room 326

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Hutton	HJ	LT (N)	00207	NTDC (A)
Drover	A	LT (N)	00207	NTDC (A)
Whelan	SMP	LCDR	00207	MARLANT HQ – N61
Dickinson	JR	LT (N)	00207	NTDC(A)
Grinton	A	LT (N)	00207	NFS (A)
Chaisson	MM	LT (N)	00207	HMCS Toronto
Sipos	SB	LT (N)	00207	DGNSR DNIW
Richardson	IL	LT (N)	00207	NPTG HQ
Heckman	CP	LCDR	00207	NTDC (A)
Watt	BJ	SLT	00211	NTDC (A)

Working Group Objective:

1. Review, update and re-write the following course content:

    a. QSP IAW updated master task list; and
    b. Technology enabled learning IAW Future Naval Training Strategy.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
B0490	Organize intelligence	This task is performed by the IMO
T0001	Provide advice to command on command, control, communications and computers	Task was too vague. New tasks were generated based on the requirements of the job.

Discussions / Decisions:

1. 18 April - The board commenced at 0900. The TDO delivered a CFITES brief outlining the process required to complete Chapter 2 of the QSP. The Board Chair delivered a brief summarizing the state of the current CISO QSP and the direction given by Command regarding the future of the qualification. The board briefly adjourned so that the Board Chair and TDO could attend a meeting with the Commandant. Upon recommencing the board, the members conducted a document review. The board then reviewed a task list composed of tasks from the NWO specification, the Occupation Specialty Specification and tasks from the CISO QSP v2.2. Train / No Train decisions were made and are recorded below.

The board began grouping tasks and decided that the best way to proceed was to develop tentative modules. Five modules were decided upon, including:

a. Module 1: Common Phase;
b. Module 2: Equipment Phase;

c. Module 3: Cyber Phase;
d. Module 4: Communication Operations; and
e. Module 5: COMSEC Course

LCdr Whelan joined the board and provided his recommendations regarding:

a. SNR – NETE;
b. Fly away kits;
c. PRC 117 JDCS; and
d. Focus on practical application rather than knowledge on course.

The board finished assigning tasks to modules and adjourned for the day.

Attendance: LCdr Whelan, Lt(N) Hutton, Lt(N) Grinton, SLt Watt,

2. 19 April - The board commenced at 1030 and began to scalar module 4. Module 4 was divided into a Planning section and a Tactics section. The board discussed the possibility that the Planning section could be shared with the CISN Manager QSP.

Attendance: Lt(N) Hutton, Lt(N) Grinton, SLt Watt

3. 20 April - The board commenced at 0800 and began to scalar module 2. There were discussions as to whether the PRC 117 was significantly different from other RF communications equipment. It was decided that due to its power and capabilities it was significantly different and should have its own PO. The board conducted a focus group with current CISO students regarding what they thought was most useful on the current course as well as topics they felt offered insufficient training. The board began to scalar Module 3 and spoke with Lt(N) Sipos of DNIW to discuss the role that the CISO will fill in Cyber warfare. Important notes from the conversation include:

a. The CISO occupation should have a focus on defensive cyber;
b. The CISO should be able to identify a threat and carry out defensive measures;
c. A Cyber Awareness Course is available through MilPersGen; and
d. DNIW does not have a policy framework but it is being built.

Attendance: Lt(N) Hutton, Lt(N) Grinton, SLt Watt,

4. 21 April - The board commenced at 0830 and began to scalar Module 1. The board discussed why there is no task for responsibilities of a Divisional Officer in the specification and decided that NOPQ NWOs do not have enough experience to fulfill the duties without further training. The board decided that Reserve NAVCOMM career progression should be taught on course in PO001 IAW the One Navy Policy. The board discussed expanding the RF theory portion in relation to tactics and employment of RF equipment and the possible overlap with CSEO responsibilities. The board contacted Lt(N) Sipos of DNIW to discuss whether the management of EW resides with the CISO or the AWWD.

The board completed the scalar for Module 1 and developed a rough course timeline based on the current scalar.

Present: Lt(N) Hutton, Lt(N) Grinton, SLt Watt,

5. 24 April - The board commenced at 0830. The TDO briefed the board explaining how to complete Chapter 4 of the QSP. The board then reviewed the scalars created last week to ensure that there were no duplicated tasks, no redundant tasks, no missing tasks and to ensure that the scalar represented the correct job performance sequence. Two changes were made to the scalar including:

a. Moving the task “Coordinate Watch and Station Bill requirements” to PO001 because it was decided that this task was performed as part of section administration.
b. Changed the task “Conduct SAR/Distress/SUBSMASH communication procedures” to “Direct emergency communication procedures” as the CISO does not conduct these procedures but directs them.

The board began working on completing chapter 2. The board adjourned at 1500.

Attendance: Lt(N) Hutton, Lt(N) Grinton, Lt(N) LaMarche, SLt Watt

6. 8 May – The board discussed Subnet Relay. This system is not currently instructed as it is a newer capability and is typically only fitted on deployed HMC ships. The board decided that the CISO needs to be able to troubleshoot the Subnet Relay system as this responsibility is falling to the CISO onboard HMC ships. The board discussed the limited ability of the RCN to train this capability as no trainer currently exist. A SNR trainer is planned and this capability is expected to expand in the near future.

Attendance: Lt(N) Hutton, Lt(N) Grinton, SLt Watt

7. 9 May – The board discussed the PRC 117. This is a capability that HMC ships currently have, however no training has been offered in the past. As well, the newly stood up MTOG unit is also using this equipment and CISOs in the fleet have been required to assist in the set up and maintenance of these connections. Training for CISOs on this equipment is required in order to ensure the proper operation and maintenance of this capability. A course is currently being developed which will provide CISOs with skills needed to manage this equipment.

Attendance: Lt(N) Hutton, SLt Watt

8. 10 May – The board decided that the QSPWG will develop a checklist for the PRC 117 in case the PRC 117 course is not developed by the time the CISO pilot course runs. As well, it was discussed that as a back-up plan, the students could practice set up of the PRC 117 onboard an HMC ship under the direction of a senior NAV COMM.

The board discussed the depth of knowledge needed for CISOs in regards to multi-link. After reviewing the QSPs, it was decided that completion of TDL100 and TDL200 will provide students with an adequate base of knowledge. The addition of 1 training day to allow for instruction on multi-link tasks specific to the CISO occupation.

Attendance: Lt(N) Hutton, SLt Watt

10. 16 May – The board discussed whether the stand-alone courses (TDL 100, TDL 200, PRC 117, TACLANE, KIV 7) should be embedded in the training time for the course as opposed to requiring them as pre-requisites. It was decided that it is better for the students to be given time on the course to complete these courses as it is likely that if they were pre-requisites,

students would have a difficult time completing them at their unit. This is because NWOs are often very busy alongside with their primary and secondary duties.

Attendance: Lt(N) Hutton, SLt Watt

11. 19 May – The board received feedback through email from Cdr Sohn from DNIW regarding changes to the current version of the QSP. His recommendations were as follows:

a. PO 010 Manage PRC 117 Equipment should be included in PO 006 Manage LF/MF/HF/VHF/UHF Equipment.
b. PO 003 title should read Security and then be broken out into EO's covering ITSEC, PHYSEC, CYBER SEC, OPSEC, TRANSEC, etc.
c. PO 005 should read Advise Command on Advise Command on Electromagnetic Spectrum Warfare vice Radio Frequency Spectrum Warfare.

These suggestions were adopted and are reflected in the QSP.

The board decided to stop meeting on a regular basis due to low attendance from members and the increasing demands of secondary duties of the regularly attending members. It was decided that individual work on the QSP will continue using SharePoint for version control and email and telephone for correspondence.

12. 31 May – LCdr Heckman is attached posted to NTDC (A) and has joined the board until 23 June. While the schedule of board members will not allow for regular meetings to work on the QSP as a group, it was decided that LCdr Heckman will be able to provide advice using the comments function in MS word and SharePoint. Specifically, LCdr Heckman will provide guidance on references and EO content.

13. 11 July – Multi-Link was removed from QSP after a discussion between Lt(N) Hutton and Lt(N) Dickinson regarding its relevance to the CISO occupation. It was decided that what the CISO needs to be able to do regarding Multi-link deals with the cryptographic requirements which will be covered in PO 008.

Guidance to Development and/or Instructional Staff:

14. TTDraft Performance Check assessment plans were developed by the board and can be found on NTDC (A) Sharepoint. These documents contain amplifying information regarding the conduct of assessments including activity information, scenario information and references.

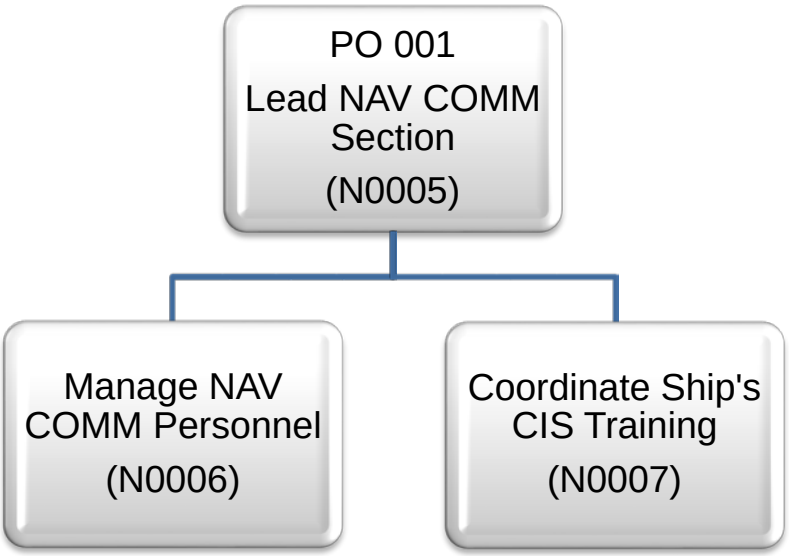
Follow Up Action:

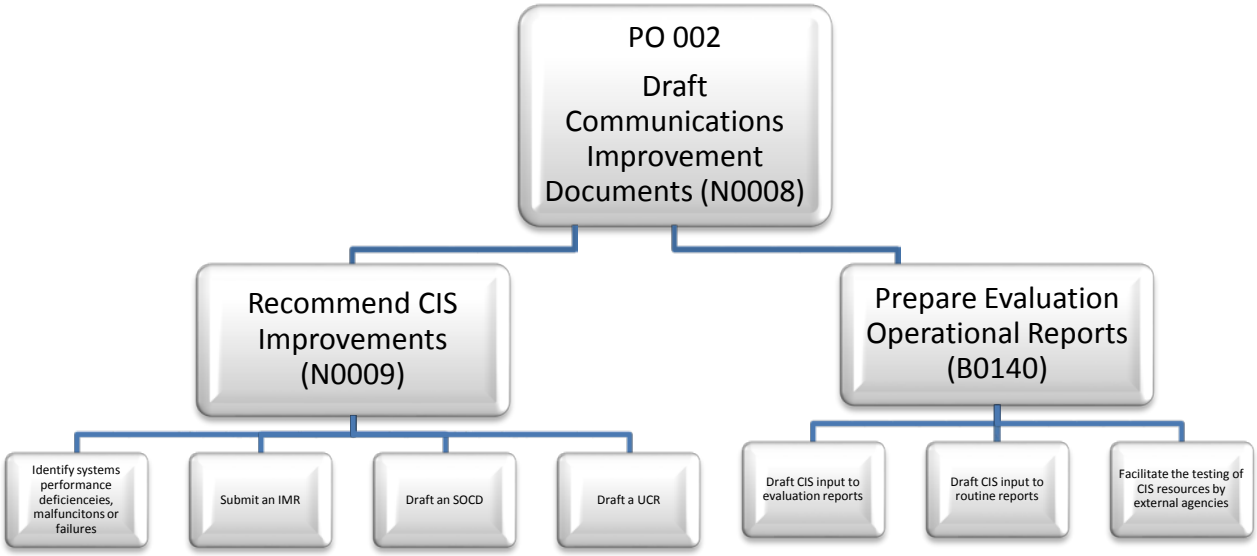
Item	Action	OPI	Expected Date of Completion
1	**SPQR forms for tasks N0005-N0043**	**Lt(N) Dickinson, and SLt Watt**	**Oct 2017**
2	**Population of References**	**Lt(N) Dickinson**	**Aug 2017**
3	**Additional Resources Annex**	**Lt(N) Dickinson**	**Aug 2017**
4	**Statements of Requirements**	**Lt(N) Dickinson**	**Oct 2017**

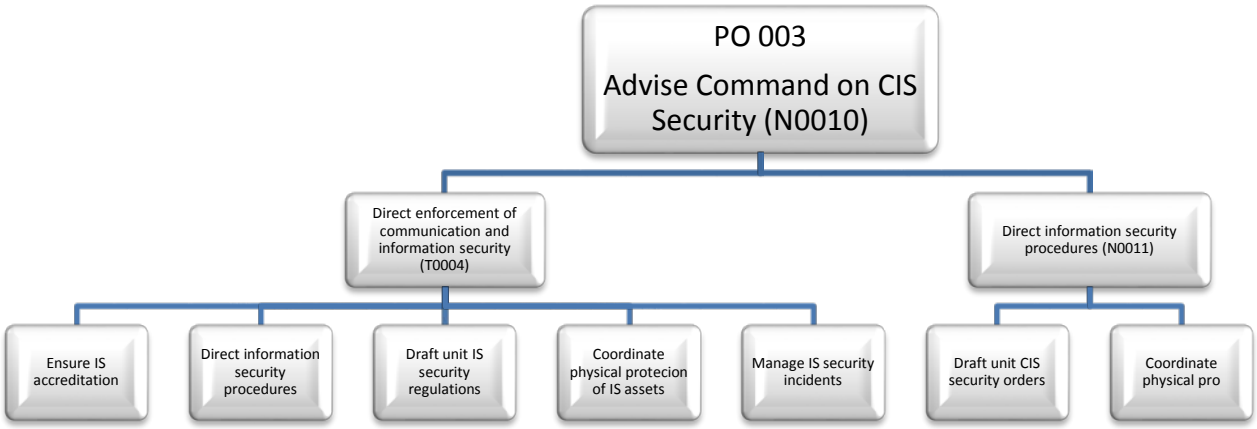
Other Considerations (if applicable):

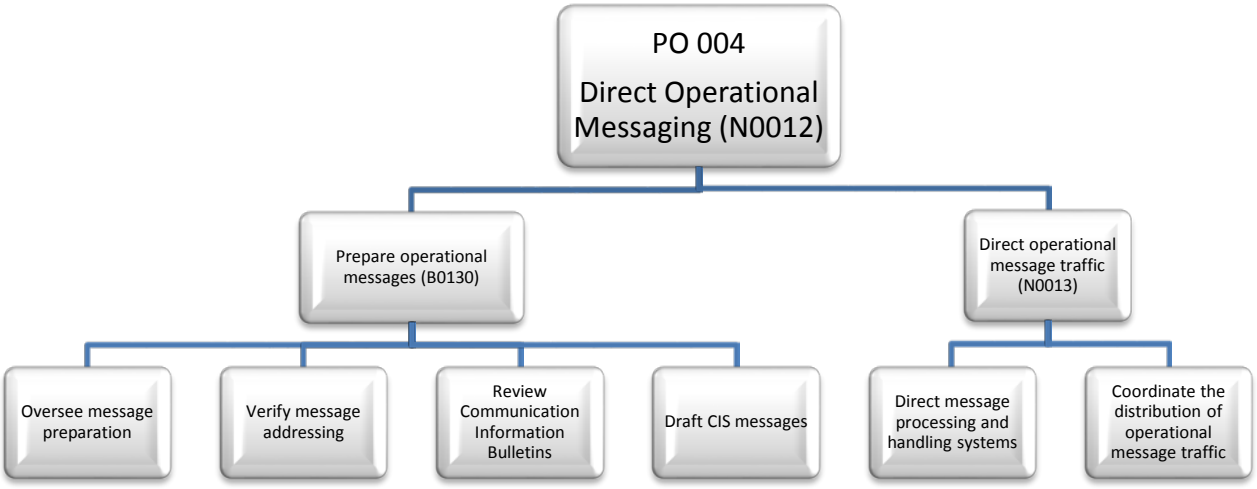
The CISO course and Information Warfare Officer (IWO) Course are scheduled for amalgamation as a part of the Information Warfare Program Review commencing in August and finalizing in October. It is expected that this QSP and the IWO QSP will amalgamate into a new IWO course which combines the CISO and “IMD” functions into one Director Level Course for Officers. The CISO and IWO QSPs should then not be expected to be completed before Oct 2017.

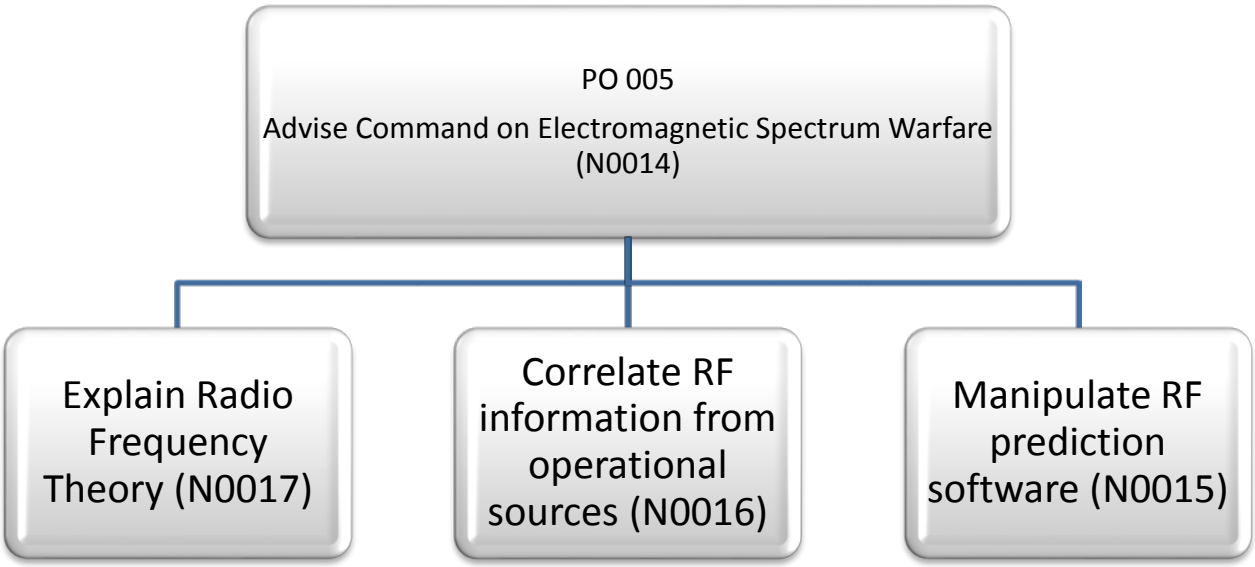
Annex D – QSP Working Group Scalars

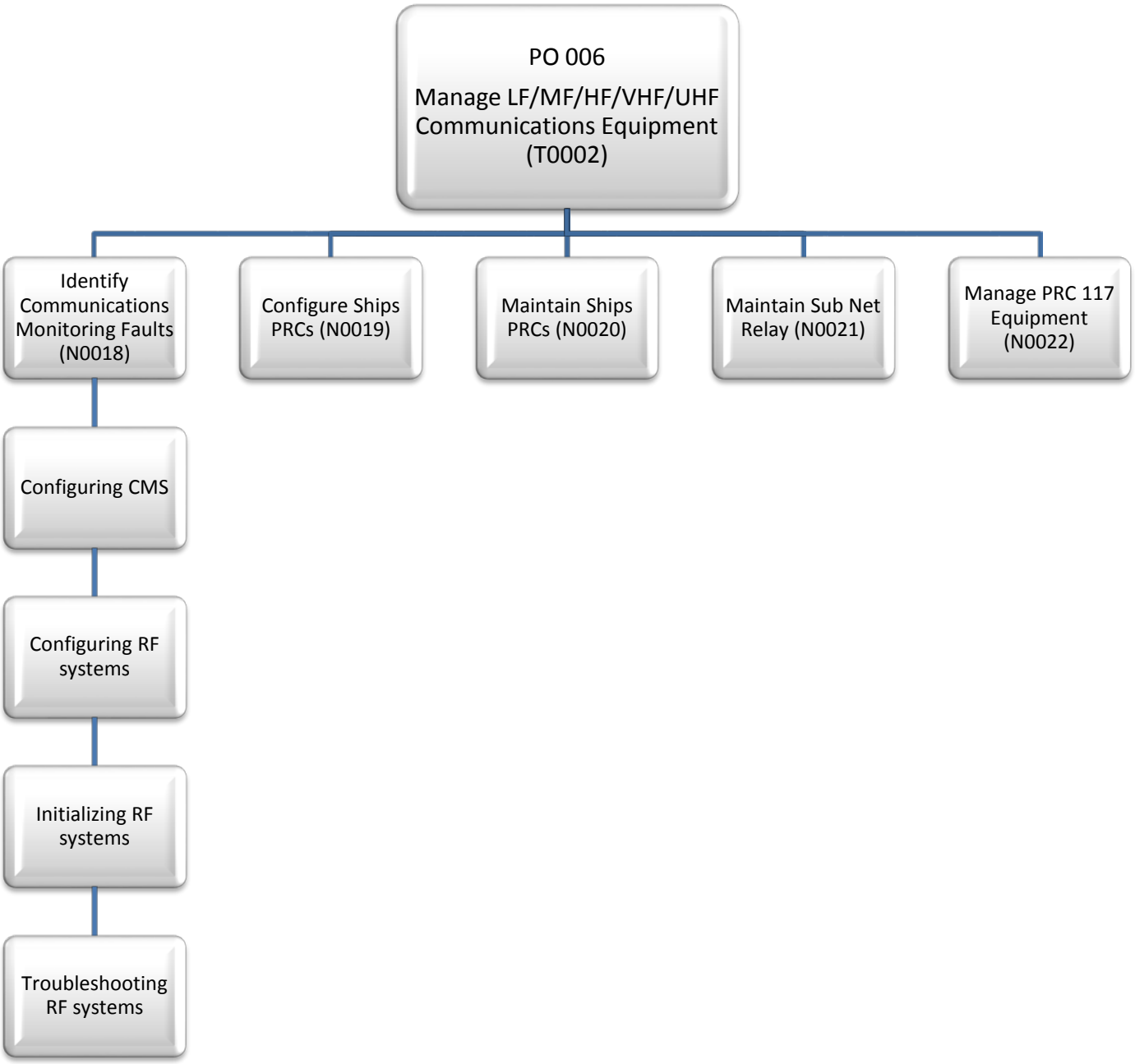


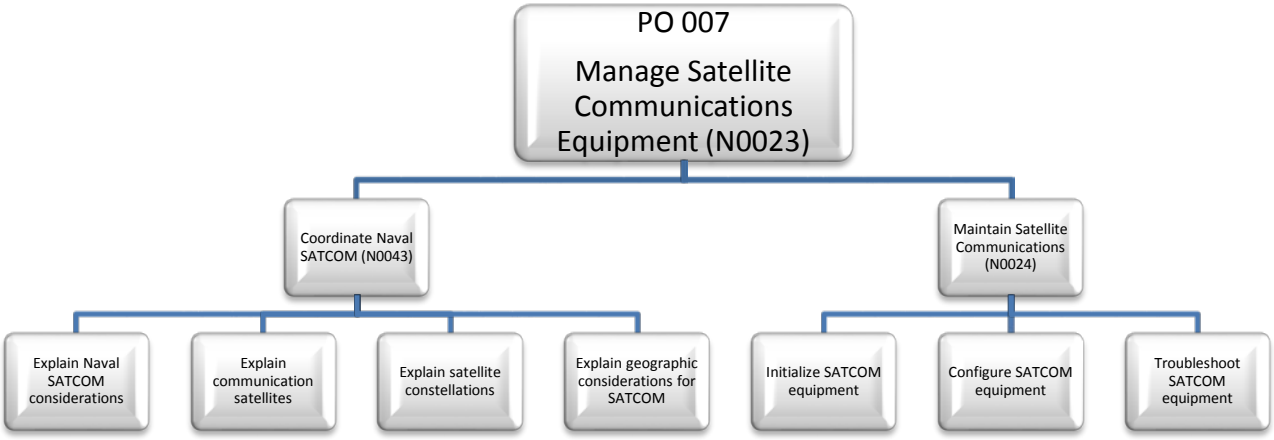

PO 002
Draft Communications Improvement Documents (N0008)
Recommend CIS Improvements (N0009)
Prepare Evaluation Operational Reports (B0140)
Identify systems performance deficienceies, malfuncitons or failures
Submit an IMR
Draft an SOCD
Draft a UCR
Draft CIS input to evaluation reports
Draft CIS input to routine reports
Facilitate the testing of CIS resources by external agencies

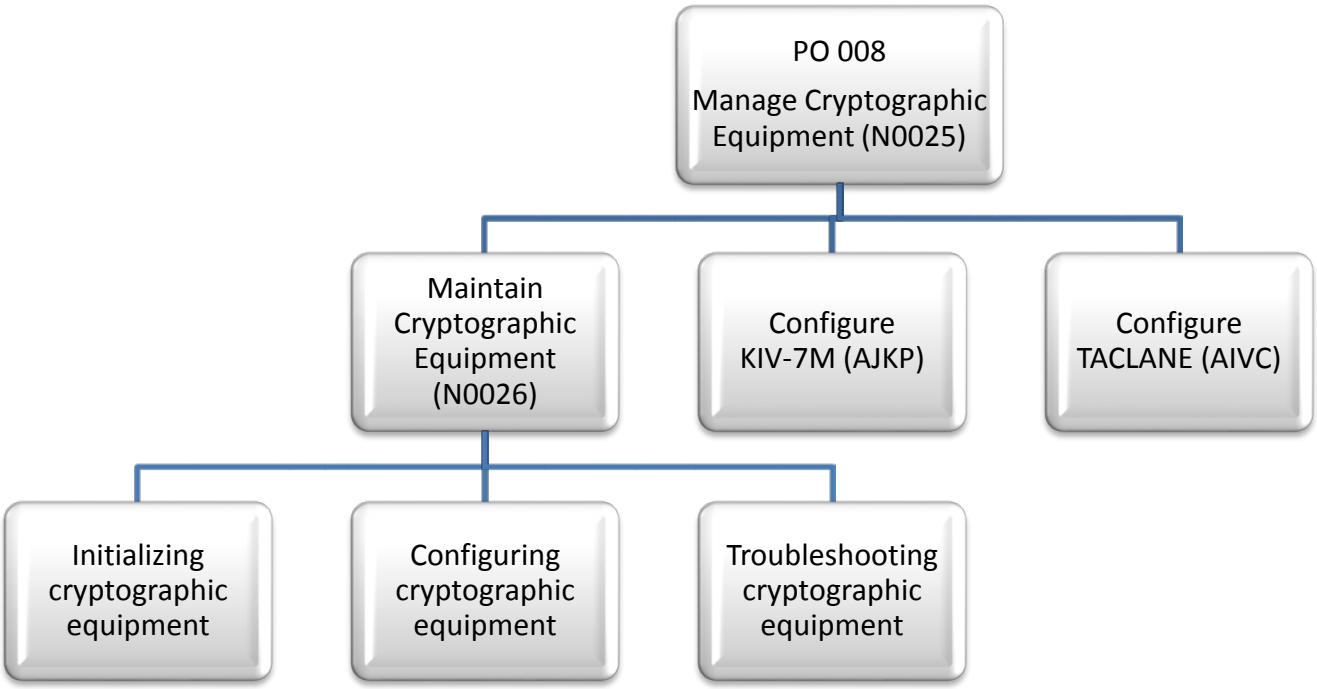

PO 003
Advise Command on CIS Security (N0010)
Direct enforcement of communication and information security (T0004)
Direct information security procedures (N0011)
Ensure IS accreditation
Direct information security procedures
Draft unit IS security regulations
Coordinate physical protecion of IS assets
Manage IS security incidents
Draft unit CIS security orders
Coordinate physical pro

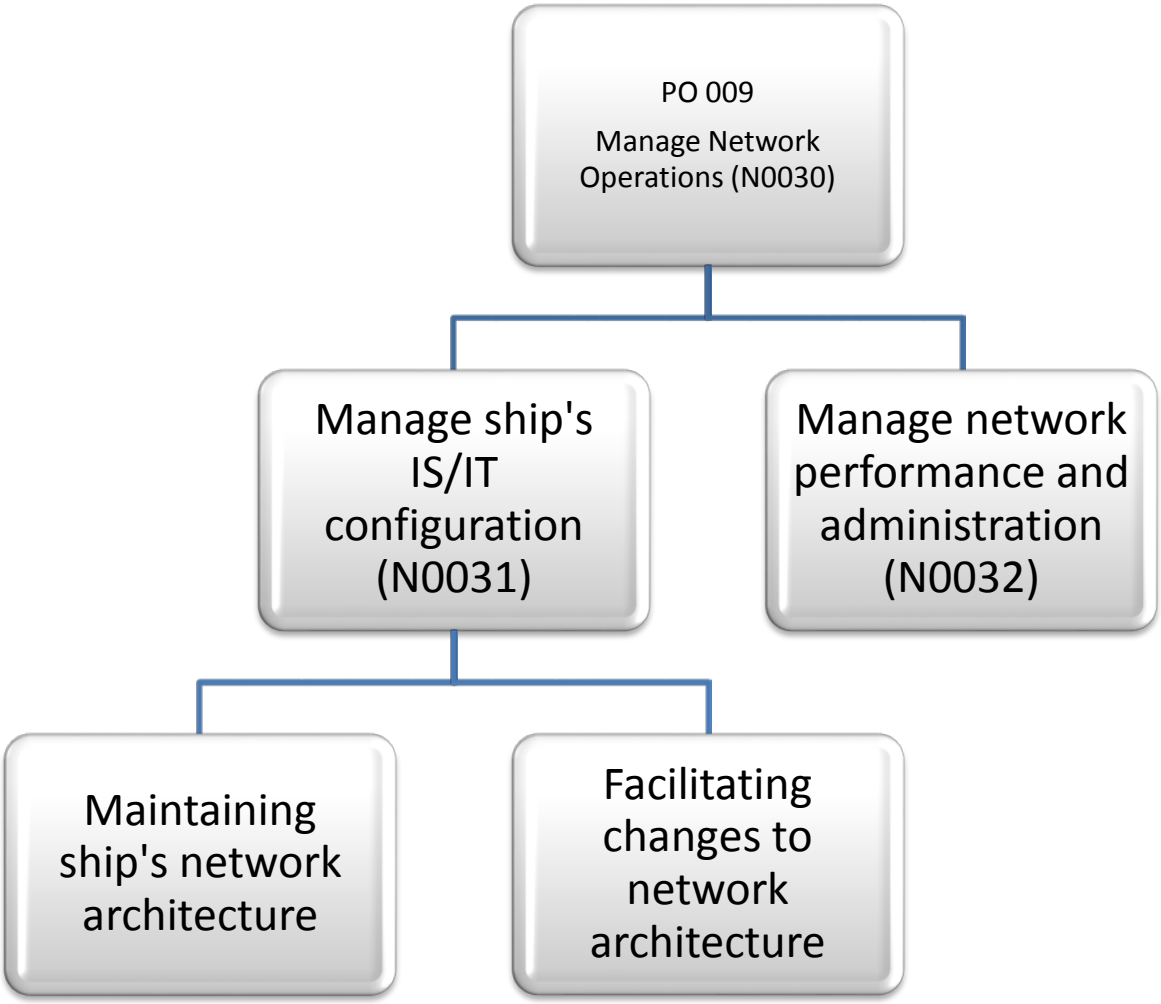

PO 004
Direct Operational Messaging (N0012)
Prepare operational messages (B0130)
Direct operational message traffic (N0013)
Oversee message preparation
Verify message addressing
Review Communication Information Bulletins
Draft CIS messages
Direct message processing and handling systems
Coordinate the distribution of operational message traffic

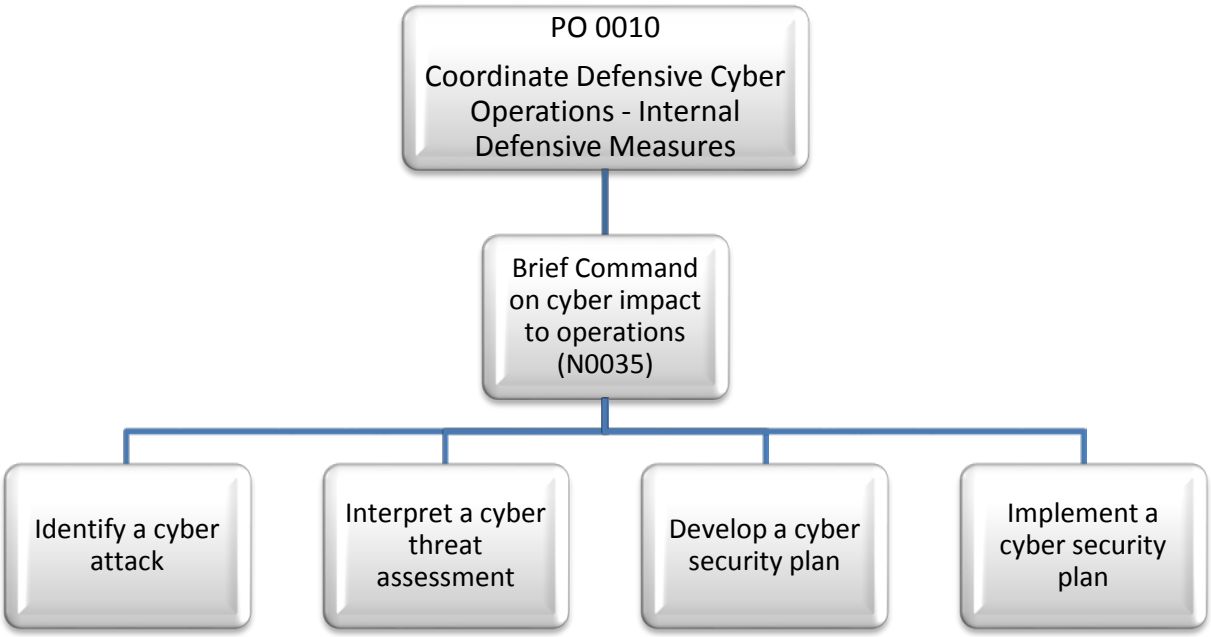

PO 005
Advise Command on Electromagnetic Spectrum Warfare (N0014)
Explain Radio Frequency Theory (N0017)
Correlate RF information from operational sources (N0016)
Manipulate RF prediction software (N0015)

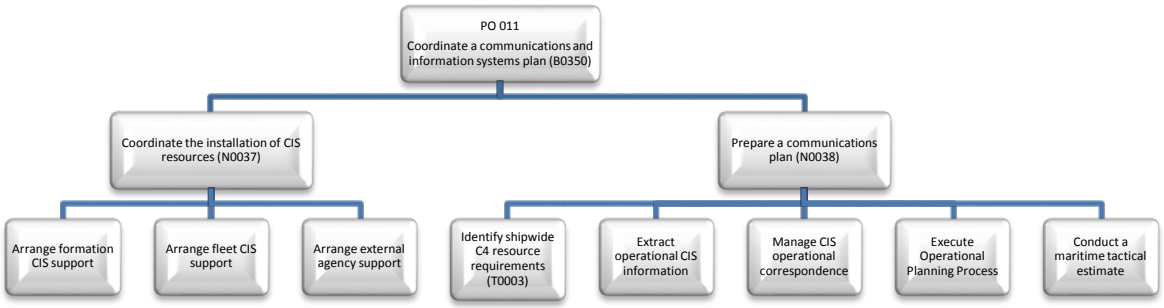

PO 006
Manage LF/MF/HF/VHF/UHF Communications Equipment (T0002)
Identify Communications Monitoring Faults (N0018)
Configure Ships PRCs (N0019)
Maintain Ships PRCs (N0020)
Maintain Sub Net Relay (N0021)
Manage PRC 117 Equipment (N0022)
Configuring CMS
Configuring RF systems
Initializing RF systems
Troubleshooting RF systems

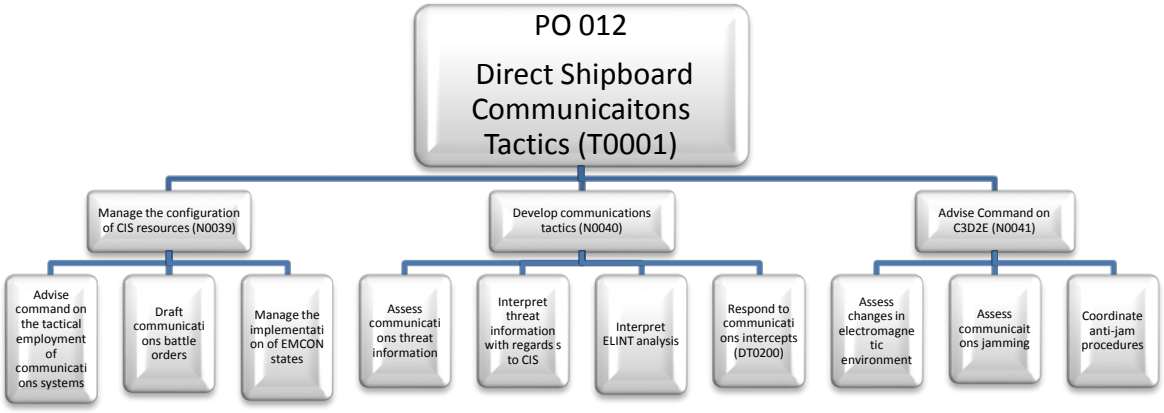

PO 007
Manage Satellite Communications Equipment (N0023)
Coordinate Naval SATCOM (N0043)
Maintain Satellite Communications (N0024)
Explain Naval SATCOM considerations
Explain communication satellites
Explain satellite constellations
Explain geographic considerations for SATCOM
Initialize SATCOM equipment
Configure SATCOM equipment
Troubleshoot SATCOM equipment


PO 008
Manage Cryptographic Equipment (N0025)
Maintain Cryptographic Equipment (N0026)
Configure KIV-7M (AJKP)
Configure TACLANE (AIVC)
Initializing cryptographic equipment
Configuring cryptographic equipment
Troubleshooting cryptographic equipment


PO 009
Manage Network Operations (N0030)
Manage ship's IS/IT configuration (N0031)
Manage network performance and administration (N0032)
Maintaining ship's network architecture
Facilitating changes to network architecture


PO 0010
Coordinate Defensive Cyber Operations - Internal Defensive Measures
Brief Command on cyber impact to operations (N0035)
Identify a cyber attack
Interpret a cyber threat assessment
Develop a cyber security plan
Implement a cyber security plan


PO 011
Coordinate a communications and information systems plan (B0350)
Coordinate the installation of CIS resources (N0037)
Prepare a communications plan (N0038)
Arrange formation CIS support
Arrange fleet CIS support
Arrange external agency support
Identify shipwide C4 resource requirements (T0003)
Extract operational CIS information
Manage CIS operational correspondence
Execute Operational Planning Process
Conduct a maritime tactical estimate


PO 012
Direct Shipboard Communicaitons Tactics (T0001)
Manage the configuration of CIS resources (N0039)
Develop communications tactics (N0040)
Advise Command on C3D2E (N0041)
Advise command on the tactical employment of communications systems
Draft communications battle orders
Manage the implementation of EMCON states
Assess communications threat information
Interpret threat information with regards s to CIS
Interpret ELINT analysis
Respond to communications intercepts (DT0200)
Assess changes in electromagnetic environment
Assess communicaitons jamming
Coordinate anti-jam procedures

Annex E – OJPR - NIL

Not Required.

Annex F – Acronyms or Abbreviations

Acronym or Abbreviation	Term or Definition
AEHF	Advanced Extremely High Frequency
AHWCS	Advanced Harpoon Weapon Control System
AO	Area of Operation
AOPS	Arctic and Offshore Patrol Ship
AOR	Area of Responsibility
AREPS	Advanced Refractive Effects Propagation System
ARMOR	
ASK	Amplitude Shift Keying
ATT	Actual Training Time
AUSCANZUKUS	Australia Canada New Zealand United Kingdom United States
BAT	Base Administrative Time
BCA	Broadcast Authority
BCOA	Broadcast Control and Operating Authority
BCS	Broadcast Control Station
BIS	Base Information Services
BW	Bandwidth
C&A	Certification and Accreditation
C/NPTG	Commander Naval Personnel and Training Group
C2	Command and Control
C2D2E	Command and Control in a Denied or Degraded Environment
C4	Command Control Communications and Computers
CANCOMSECADMINGEN	Canadian Communications Security Administrative General
CANCOMSECGEN	Canadian Communications Security General
CAPs	Capabilities
CCD	Canadian Cryptographic Doctrine
CCIR	Commanders Critical Information Requirements
CCMS	Communications Control Monitoring System
CCR	Communications Control Room
CCS	Communications Control System
CFCD	Canadian Forces Controlled Document
CFCSS	Canadian Forces Crypto Support Squadron
CFCUS	Canadian Forces Cryptographic Support Unit
CFITES	Canadian Forces Individual Training and Education System
CFMWC	Canadian Forces Maritime Warfare Centre
CFNOC	Canadian Forces Network Operations Centre
CHOP	Change of Operational Command
CIB	Communication Information Bulletins
CIS	Communication and Information Systems
CISN	Communications Information Systems and Networks
CISO	Communication and Information Systems Officer
CMS	Communications Monitoring System
CO NTDC (A)	Commanding Officer Naval Training Development Center Atlantic
COMMCEN	Communications Center
COMFAC	
COMPLAN	Communications Plan

COMSEC	Communication Security
COMSPOT	
COMSTAT	
CPF	Canadian Patrol Frigate
CPF	Canadian Patrol Frigate
CRR	Combat Readiness Requirement
CRYPTO	Cryptographic
CSC	Canadian Surface Combatant
CSD	Classified Standard Document
CSEO	Combat Systems Engineering Officer
CSNI	Consolidated Secret Network Infrastructure
CSRF	Common Source Route File
CW	Continuous Wave
DAMA	Demand Access Multiple-Access
DET	Detachment
DFSM	Director Frequency Spectrum Management
DLP	Data Link Processor
DLPS	Data Link Processor System
DMOC	Distributed Mission Operations Centre
DMSS	
DWAN	Defense Wide Area Network
EC	Enabling Check
EFR	Equipment Failure Report
EM	Electromagnetic
EMCON	Emissions Control
EMSEC	Emission Security
EO	Enabling Objective
FBB	Fleet Broad Band
FCA	Formation Consecutive Authority
FCA	Formation COMSEC Authority
FMF	Fleet Maintenance Facility
FSK	Frequency Shift Keying
GEO	Geostationary Equatorial Orbit
GPS	Global Positioning System
GSO	Geosynchronous Orbit
HEO	Highly-Elliptical Orbit
HERF	Hazardous Electromagnetic Effects on Fuel
HERO	Hazardous Electromagnetic Effects on Ordinance
HERP	Hazardous Electromagnetic Effects on Personnel
HF	High Frequency
HMC	Her Majesties Canadian
HMC	Her Majesty's Canadian
HODs	Head of Departments
HSDC	High Speed Data Connectivity
IAW	In Accordance With
IER	Information Exchange Requirements
IMDO	Information Management Director Officer
IMR	Information Management Request
INFOCON	Information Control

INNWOAT	International Maritime Satellite
INMT	Initialization and Network Management Tool
IS	Information System
ISA	Information System Administrator
ISSI	Information System Security Incident
ISSIR	Information System Security Incident Report
ISSO	Information System Security Officer
IT	Information Technology
ITU	International Telecommunications Union
IW	Information Warfare
JIIFC	Joint Information and Intelligence Fusion Center
KEK	Key Encryption Key
KEYMAT	Keying Material
KMS	Knowledge Management System
LCMM	Life Cycle Material Manager
LE Det	Law Enforcement Detachment
LEO	Low Earth Orbit
LIMs	Limitations
LLC	
LOS	Line of Sight
MARTIs	Maritime Tactical Instructions
MCOIN	Maritime Command Operation Information Network
MEO	Medium Earth Orbit
METOC	Meteorological Operations Center
MHDS	Message Handling and Distribution System
MHS	Message Handling System
MIL	Military
MIP	Master Implementation Plan
MITE	Military Individual Training and Education
MPS	Message Processing System
MPS	Message Processing System
MSAT	Mobile Satellite
MTE	Maritime Tactical Environment
MTWAN	Maritime Tactical Wide Area Networking
MULTOTS	Multi-Unit Link Testing and Operational Training System
NATO	North Atlantic Treaty Organization
NATO	North Atlantic Treaty Organization
NavIS	Naval Information System
NAVSET	Navigation Set
NCS	Net Control Station
NDSOD	National Defence Security Orders and Directives
NEO	Naval Extraction Operations
NETE	Naval Engineering Test Establishment
NETSEC	Network Security
NFS (A)	Naval Fleet School Atlantic
NFS (P)	Naval Fleet School Pacific
NMT	Network Management Tool / Navy Multi-band Terminal
NOC	Network Operation Centre
NOPQ	Naval Officer Professional Qualification

NPTG	Naval Personnel and Training Group
NPTORD	Naval Personnel Training Orders
NQual	National Qualification
NRS	Naval Radio Station
NTDC (A)	Naval Training Development Center Atlantic
NWA	Naval Warfare Authority
OCO	Offensive Cyber Operations
OCS	Officer Conducting Serial
OGDs	Other Government Departments
OOW	Officer of the Watch
OPANAL	Operational Analysis
OPEVAL	Operational Evaluation
OPGEN	Operation General
OPORD	Operation Order
OPSEC	Operation Security
OPSIG	Operational Signals
OPSTAT	Operational Status
OPTASK	Operation Tasking
ORS	Operations Room Supervisor
OTAD	Over the Air Distribution
OTAR	Over The Air Radio
OTAT	Over The Air Transmission
PC	Performance Check
PDR	Post Deployment Report
PIP	Pre-Instructional Package
PLA	Plain Language Address
PLAR	Prior Learning Assessment and Recognition
PMSC	Protected Military Satellite Communications
PO	Performance Objective
PRB	Progress Review Board
PRC	Portable Radio Communications
PRC	Personal Radio Communication
PSK	Phase Shift Keying
PSS	Physical Site Survey
PSU	Power Supply Unit
PT	Physical Training
QPSK	Quadrature Phase Shift Keying
QSP	Qualification Standard Plan
QSP	Qualification Standard and Plan
QSPWB	Qualification Standard Plan Working Board
RA	Response Attacks
RADHAZ	Radiation Hazard
RATT	Radio Teletype
RCN	Royal Canadian Navy
RDS	Ready Duty Ship
RF	Radio Frequency
RFC	Request for Change
RFS	Request for Support
RI	Routing Indicator

RQ	Rank Qualification
SA&A	Security Assessment and Accreditation
SATCOM	Satellite Communications
SBC	Single Board Computer
SCA	Supply Control Account
SCOT	Satellite Communications Onboard Terminal
SGT	Satellite Ground Terminal
SKL	Simple Key Loader
SMSS	Strategic Message Switching System
SNC	Senior Naval Communicator
SNC	System Network Controller
SOC	Space Operations Center
SOCD	Statement of Operational Capability Deficiency
SOF	Special Operations Force
SOP	Standard Operating Procedures
SPC	
SS	Specialty Specification
SSOs	Ships Standing Orders
STAT REP	Status Report
STOW	Senior Technician of the Watch
STSCU	Short Term Satellite Communication Upgrade
SWOAD	Ship Without Air Detachment
TA	Training Authority
TACLANE	Tactical Local Area Network
TACNOTEs	Tactical Notes
TAT	Training Administrative Time
TCI	Technical Compliance Inspection
TE	Training Establishment
TEK	Truk Encryption Key
TP	Training Plan
TRA	Threat Risk Analysis
TRANSEC	Transmission Security
TRB	Training Review Board
TRKEK	Transfer Key Encryption Key
TTT	Travel Training Time
UCR	Unsatisfactory Condition Report
UHF	Ultra High Frequency
USMTF	United States Message Traffic Format
VHF	Very High Frequency
VM	Virtual Machine
WADS	Wideband Audio and Data Switch
WGS	Wideband Global SATCOM

Annex G – Modularization and Suggested Timings

Module 1

	Suggested timings	EO	EC*	PC*	PO
Content	**Title**				
PO 001	**Lead NAV COMM Section**				
EO 001.01	Manage NAV COMM Personnel	5	1	1	7
EO 001.02	Coordinate Ship's CIS training	4	1		5
PO 002	**Draft Communications Improvement Documents**				
EO 002.01	Recommend CIS Improvements	5	1	2	8
EO 002.02	Prepare Evaluation Operational Reports	2	1		3
PO 003	**Advise Command on CIS Security**				
EO 003.01	Direct Enforcement of CIS Security	11	3	2	16
PO 004	**Direct Operational Messaging**				
EO 004.01	Prepare Operational Messages	7	1	2	10
EO 004.02	Direct Operational Message Traffic	11	1		12
PO 005	**Advise Command on Electromagnetic Spectrum Warfare**				
EO 005.01	Manipulate RF Prediction Software	31	2	2	35
	Total	**76**	**11**	**9**	**96**

* Include review and debrief time.

Module 2

	Suggested timings	EO	EC*	PC*	PO
Content	**Title**				
PO 006	**Manage LF/MF/HF/VHF/UHF Communications Equipment**				
EO 006.01	Identify Communications Monitoring Faults	25	2	8	35
EO 006.02	Maintain Ship's PRCs	3	2		5
EO 006.03	Maintain SubNet Relay	6	2		8
EO 006.04	Manage PRC 117 Equipment5	7	3		10
PO 007	**Manage SATCOM Equipment**				
EO 007.01	Coordinate Naval SATCOM	8	1	2	11
EO 007.02	Maintain Satellite Communication	20	6		26
PO 008	**Manage Cryptographic Equipment**				
EO 008.01	Maintain Cryptographic Equipment	18	1		19
EO 008.02	KIV-7M (AJKP)	32	5	4	41
EO 008.03	TACLANE (AIVC)	28	4	4	36
	Total	**147**	**26**	**18**	**191**

* Include review and debrief time.

Module 3

	Suggested timings	EO	EC*	PC*	PO
Content	**Title**				
PO 009	**Manage Network Operations**				
EO 009.01	Managing Ship's IS/IT Configuration	25	7	4	36
EO 009.02	Manage Network Performance and Administration	17	3		20
PO 010	**Coordinate Defensive Cyber Operations – Internal Defensive Measures**				
EO 010.01	Brief Command on Cyber Impact to Operations	12	1	2	15
	Total	**54**	**11**	**6**	**71**

* Include review and debrief time.

Module 4

	Suggested timings	EO	EC*	PC*	PO
Content	**Title**				
PO 011	**Coordinate a Communications and Information Systems Plan**				
EO 011.01	Coordinate the Installation of CIS Resources	5.5	0	3	8.5
EO 011.02	Prepare a Communications Plan	20.5	1		21.5
PO 012	**Direct Shipboard Communications Tactics**				
EO 012.01	Developing Communications Tactics	26	1	2	29
EO 012.02	Manage the Configuration of Communications Equipment	21	1		22
EO 012.03	Advise Command on C2D2E	14	1		15
	Total	**87**	**4**	**5**	**96**

* Include review and debrief time.

Module 5

	Suggested timings	EO	EC*	PC*	PO
Content	**Title**				
EO	COMSEC Custodian	84	0	0	84
	Total	**84**			**84**

* Include review and debrief time.

Annex H – Additional Training Resources

In accordance with Chapter 1, paragraph 3, add any extra tables or paragraphs used for training resources in this annex. (**Not to be deleted**)

#	Resource	QTY	Days
1	NATO e-Learning account		
2	Smart Board/ Projector		
3	Instructional Materials (Texts, pubs, logs, Crypto, Training Messages)		
4	CSNI User account form completed/submitted 30 prior to attending course		
5	Pencils, Pens & Stationary		
6	RF Prediction Software		
7	EM Footprint Software		

Guest Lectures

Organization	Periods enquired	Recommended Personnel	Notes
NTDC(A), NFS(A/P) or MARPAC/MARLANT ISSO	1	NAV COMM MOC Advisor NTDC(A) CPO2 or MARPAC ISSO	EO 001.01
MARPAC or MARLANT N6	3	Formation ISSO	EO 003.01

Tours

Location	Periods Required	Recommended Personnel	Notes
Message Handling Facilities NRS / COMMCEN	1	NCS IC	EO 004.02
HMC Ship	7	IFTPR Coordinator	EO 006.01
MULTOTS	2	FMF Cape Scott IC MULTOTS	EO 009.03

Annex I – Costing Information

1. Include information about cost in whichever format is applicable (**Not to be deleted**).

Costing Annex_Template.xlsx

Estimates only, based on CF Cost Factors Manual FY 2012/13
verify that this link is the most updated version
****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: CISO		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	12	$1,000.0		$12,000	$12,000	$60,000.00
R&Q	12	$36.0	88	$38,016	$38,016	$190,080.00
Incidentals	12	$17.5	88	$18,480	$18,480	$92,400.00
Subtotal				**$68,496**	**$68,496**	**$342,480.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	13	$50.0	1	$700	$700	$3,500.00
Classroom (square meters)	30	$13.00	88	…	…	…
Computer requirements	14	$1,000.0	1	$14,000	$0	$14,000.00
IT Equip (Proj, Smart board etc)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Instructional Materials (Printed / Handouts etc)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$26,150**	**$1,150**	**$30,750.00**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Not Applicable						
		subtotal				
	Total Estimated Costs			**$94,646**	**$69,646**	**$373,230.00**

Annex J – Course History

1. This annex is to record course history issues such as explanations of amendments, pertinent findings, trends, resource problems and modifications. Focus should be given to both excellent and questionable issues of the course. Capturing important course history information is a useful managerial tool for amendment considerations by future course developers.

Note: This annex shall NOT be deleted. This annex must be annotated with every update / modification to this QSP. This is in addition to making the appropriate entry in the Record of Amendments table. Detail as to the nature and reasons for amendments is to be provided here.

Past CCISO course training was more concentrated on traditional Radio Frequency (RF) communications and the associated C2 equipment. The last QSPWB was conducted in 2006. The 1999 QSP was updated to reflect current technologies and to change the course title to Communication and Information Systems Officer (CISO). The first course to use the QSP developed in 2006 was taught in 2010 (CISO 0001) and was the first attempt at this type of NWO specialist training since 2002. Post course input from the Course Instructor and students indicated that further improvements needed to be made due to rapid changes in technology and expansion in some areas (IP networking/COMSEC). The decision was made to retain RF theory and delete some information/procedures that were not deemed relevant for employment of a CISO at sea. This QSPWB examined the separation of duties between the CISO and IMD. It has determined that the IMD has been given responsibility for Information, Surveillance and Reconnaissance (ISR) on board therefore this material has been removed from the updated QSP.

26 Jul 2012: The second pilot course, CISO 0002, took place from 31 Oct 11 to 12 Mar 12. There were four students on course and Lt(N) Heckman, CFNOS CISO, was the course officer. Throughout the course the students completed course critiques after each PO, as per CFNOS SSOs. In addition to this, Lt(N) Heckman discussed each lesson plan with the students to gain a better understanding of material relevance, order of TPs etc. This data was recorded in the Pilot Course Evaluation IAW CFNOS SSOs. This document was submitted to standards upon completion of the course by Lt(N) Heckman. Based on the direct feedback from students, guest lecturers and the CFNOS NAV COMM section, proposed changes to the QSP (which was approved in Jul 2012) were noted. Most of the changes were minor, such as TP rewording/addition/deletion for relevance. The largest included a reorganization of POs based on student feedback and instructor concurrence that the material was not taught in an order conducive to building a strong foundation before more complex tasks were required of the students. Additional content from Fleet Staff was added for both the East and West coast, which was a noted issue during the previous course. Lastly, the Space Applications Course (SAC) was removed due to fiscal constraints. The teaching points from the SAC were however added to the space portion of the QSP and the relevant material will be generated in house in order to mitigate this loss. Lt(N) Heckman drafted a revised QSP (2.0) with PO1 Helpard, CFNOS STANDARDS providing guidance and expertise as to the manner in which these changes should be made. In Jul 12, STDS submitted the Pilot Course Evaluation, old QSP, revised QSP and QSP change request documents, which were subsequently approved 26 Jul 12.

5 Dec 2012: After the completion of CISO 0002 the course was deemed to be steady state, however this decision was not unanimous. CISO 0003 was conducted from 2 Aug 2012 to 5 Dec 2012. There were seven students on course and Lt(N) Heckman, CFNOS CISO, was the

course officer. A similar approach as described above was again employed to solicit feedback from students and staff throughout the entirety of the course. Overall the course was successful. It was clear that the revisions made after CISO 0002 were useful and the course flowed much more smoothly in general. West Coast content was delivered using subject matter experts via video teleconference. This proved to be an excellent addition. It should be noted that the effectiveness of this course depends heavily on subject matter expertise, particularly during the radio theory (CFNES SME Sean Kavanaugh), space (DG Space SME LCdr Chris Willmes) and CIS (N6 West SME LCdr Rob Sibbald). The continued success of the CISO course will rest heavily in obtaining the appropriate SMEs to deliver this important material. Finally, Lt(N) Heckman generated another QSP revision of the QSP based on student feedback and personal experiences over four months. The most notable change was the breaking up of PO 002 into several smaller, more manageable POs. In addition to this seven periods of Cyber was added to the course based on the changing threat and CF perspective on the subject. Other improvements included timing adjustment and wording revision based on new technology, nomenclature and organizations throughout the CF. New course material will be added as required and the next serial is set to take place from 25 Mar 2013 to 24 Jul 2013.

29 Aug 13: CISO 0004 was conducted from 25 Mar – 24 Jul 2013. The previous revisions to the course were well received and the flow of the course was greatly enhanced through the splitting up of PO 002 into several smaller, more manageable modules. QSP Revision 2.2 changes were relatively minor. Similar TPs were merged, lectures on UNCLOS and LOAC were added based on Command direction to have these topics included in all D-Level coursing, and timings were further tweaked based on student feedback and instructor observation. In an effort to save money, the course instructor liaised with CFCSS Det East and West to have the West coast students complete the COMSEC phase of the course on the West coast, effectively eliminating two and a half weeks of TD and ensuring that the students can return home sooner. The intention is to make efforts to continue this in future coursing. TD funding was cut to have a space SME join the course so the course instructor will develop new material and lesson plans for this section and will instruct this section himself for CISO 0005 which will take place from 15 Oct 13 – 18 Feb 14.

6 Mar 14: CISO 0005 was conducted from 15 Oct 13 – 18 Feb 14 with 10 students in attendance. QSP V 2.2 was used and little negative feedback was received from this version aside from timing adjustments which will be made for QSP V 2.3. The next CISO course will be CISO 0007 as the MITE system was unable to change the dates for the previously scheduled CISO 0006. QSP V 2.3 changes include the removal of the Ottawa phase and all subject matter expert participation due to fiscal constraints. Intention is to utilize VTC to have some of the Ottawa based organizations continue to be involved in the course while the course officer, Lt(N) Heckman will instruct the material on space and computer networking himself. In addition to these changes, the TDL 100 course was also removed, as the course was changed and will soon be going online. The students and course instructors agreed that the material could be condensed and taught by the course officer. Lastly, QSP V2.3 greatly increased the emphasis on the practical application of course material. Two scenarios were developed for this purpose and students were given an increased number of periods to plan and generate messages and other correspondence related to the overall scenario.

22 Jul 15: Based on the CAAR from CISO 0008 multiple points were addressed in the QSP:

1) Performance Checks were changed from written examinations to oral presentations in order to push the trainees to research the course material in-depth in order to create and present a staff product to an acceptable level. This also addresses shortfalls in briefing ability noted during PC 011 Part II of multiple serials of the CISO course;

2) SOCD and UCR were relocated from PO 011 to PO 001 with similar topics to avoid redundancy;

3) Excess periods were eliminated from multiple enabling objectives based on observations from multiple courses; and

Many of the Enabling Checks were changed to allow more practical exercise and take-home assignments.

Form found in NPTORD 4500-22 Annex B

NPTORD 4500-22
Annex B_QSP Amend

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.01	EO 001.01	To assess trainee's competence when managing NAV COMM personnel .	Practical Assessment	Trainee must achieve _ of _ SATs
EC 001.02	EO 001.02	To assess trainee's competence when coordinating ship's CIS training.	Practical Assessment	Trainee must achieve _ of _ SATs
PC 001	PO 001	To assess trainee's competence when leading the NAV COMM section	Practical Assessment	Trainee must achieve _ of _ SATs
EC 002.01	EO 002.01	To assess trainee's competence when recommending CIS improvements.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 002.02	EO 002.02	To assess trainee's competence when preparing evaluation operational reports	Practical Assessment	Trainee must achieve _ of _ SATs
PC 002	PO 002	To assess trainee's competence when drafting communications improvement documents	Practical Assessment	Trainee must achieve _ of _ SATs
EC 003.01	EO 003.01	To assess trainee's competence when directing enforcement of communications and information security	Practical Assessment	Trainee must achieve _ of _ SATs
EC 003.02	EO 003.02	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
PC 003	PO 003	To assess trainee's competence when advising command on CIS security.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 004.01	EO 004.01	To assess trainee's competence when preparing operational messages	Practical Assessment	Trainee must achieve _ of _ SATs
EC 004.02	EO 004.02	To assess trainee's competence when directing operational message traffic.	Practical Assessment	Trainee must achieve _ of _ SATs
PC 004	PO 004	To assess trainee's competence when directing operational messaging.	Practical Assessment	Trainee must achieve _ of _ SATs

EC 005.01	EO 005.01	To assess trainee's competence when manipulating RF prediction software	Practical Assessment	Trainee must achieve _ of _ SATs
EC 005.02	EO 005.02	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
PC 005	PO 005	To assess trainee's competence when advising command on electromagnetic spectrum warfare.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 006.01	EO 006.01	To assess trainee's competence when identifying communications monitoring faults.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 006.02	EO 006.02	To assess trainee's competence when maintaining ship's PRCs.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 006.03	EO 006.03	To assess trainee's competence when maintaining Subnet Relay	Practical Assessment	Trainee must achieve _ of _ SATs
EC 006.04	EO 006.04	Assessed IAW PRC 117 QSP	IAW PRC 117 QSP	IAW PRC 117 QSP
PC 006	PO 006	To assess trainee's competence when managing LF/MF/HF/VHF/UHF equipment.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 007.01	EO 007.01	To assess trainee's competence when coordinating naval SATCOM.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 007.02	EO 007.02	To assess trainee's competence when maintaining satellite communications.	Practical Assessment	Trainee must achieve _ of _ SATs
PC 007	PO 007	To assess trainee's competence when managing SATCOM equipment.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 008.01	EO 008.01	To assess trainee's competence when maintaining cryptographic equipment.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 008.02	EO 008.02	Assessed IAW KIV-7M QSP.	IAW KIV-7M QSP	IAW KIV-7M QSP

EC 008.03	EO 008.03	Assessed IAW TACLANE (KG-175D) QSP	IAW TACLANE QSP	IAW TACLANE QSP
PC 008	PO 008	To assess trainee's competence when managing cryptographic equipment.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 009.01	EO 009.01	Assessed IAW TDL 100 (ALDF) QSP.	IAW TDL 100 QSP	IAW TDL 100 QSP
EC 009.02	EO 009.02	Assessed IAW TDL 200 (ALDG) QSP…	IAW TDL 200 QSP	IAW TDL 200 QSP
EC 009.03	EO 009.03	To assess trainee's competence when troubleshooting Multi-Link.	Practical Assessment	Trainee must achieve _ of _ SATs
PC 009	PO 009	To assess trainee's competence when managing Multi-Link.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 010.01	EO 010.01	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
EC 010.02	EO 010.02	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
PC 010	PO 010	To assess trainee's competence when managing network operations	Practical Assessment	Trainee must achieve _ of _ SATs
EC 011.01	EO 011.01	To assess trainee's competence when briefing command on cyber impact to operations.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 011.02	EO 011.02	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
EC 011.03	EO 011.03	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
PC 011	PO 011	To assess trainee's competence when coordinating defensive cyber operations – internal defensive measures.	Practical Assessment	Trainee must achieve _ of _ SATs

EC 012.01	EO 012.01	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
EC 012.02	EO 012.02	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
PC 012	PO 012	To assess trainee's competence when coordinating communications and information system plan.	Practical Assessment	Trainee must achieve _ of _ SATs
EC 013.01	EO 013.01	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
EC 013.02	EO 013.02	To assess trainee's competence when…	Practical Assessment	Trainee must achieve _ of _ SATs
PC 013	PC 013	To assess trainee's competence when directing shipboard communications tactics.	Practical Assessment	Trainee must achieve _ of _ SATs
PC 013	PC 013	Assessed IAW COMSEC Custodian QSP	IAW COMSEC Custodian QSP	IAW COMSEC Custodian QSP

Annex L – TEL Development

Information about development of Technology Enabled Learning (TEL) can be found in the embedded document:

Spirals.docx